



AUTOCAD 2009 MANUAL



BAHIR DAR INSTITUTE OF TECHNOLOGY

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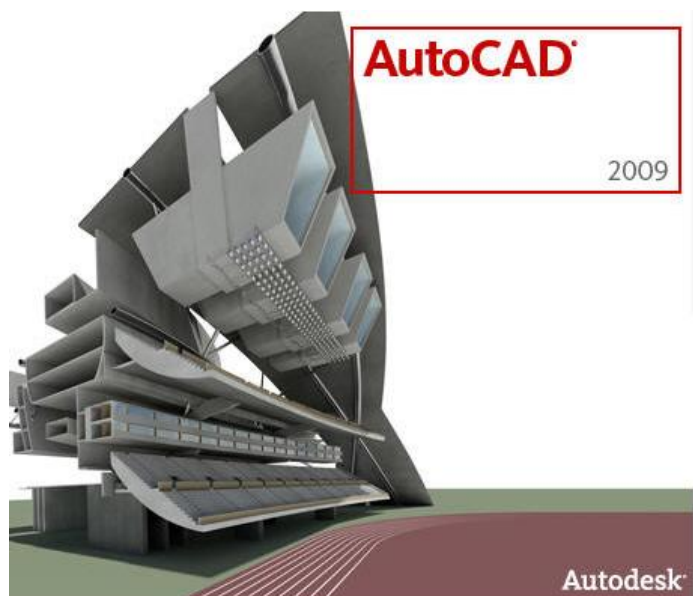


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AUTOCAD 2009

2D DESIGN

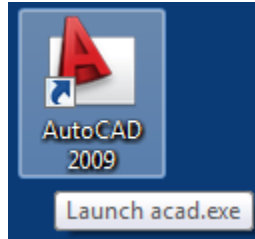
INTRODUCTION

Computer Aided Design (CAD) is the process of doing designs with the aid of computers. This includes the generation of computer models, analysis of design data, and the creation of the necessary drawings. AutoCAD 2009 is computer aided design software developed by Autodesk Inc. The AutoCAD 2009 software is a tool that can be used for design and drafting activities. The two-dimensional and three-dimensional models created in AutoCAD 2009 can be transferred to other computer programs for further analysis and testing. The computer models can also be used in manufacturing equipment such as machining centers, lathes, mills, or rapid prototyping machines to manufacture the product.

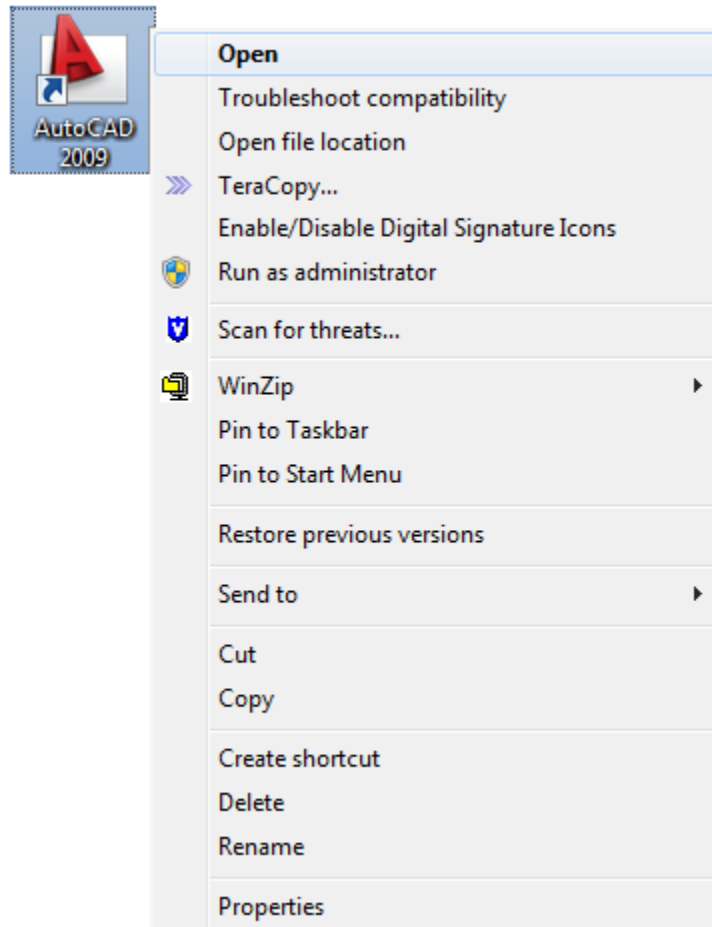
AutoCAD was first introduced to the public in late 1982, and was one of the first CAD software products that were available for personal computers. Since 1984, AutoCAD has established a reputation for being the most widely used PC-based CAD software around the world. By 2007, it was estimated that there were over 6 million AutoCAD users in more than 150 countries worldwide. AutoCAD 2010 is the twenty-third release, with many added features and enhancements, of the original AutoCAD software produced by Autodesk Inc.

Starting AutoCAD 2009

Double Click on **AutoCAD 2009** icon from the desktop.

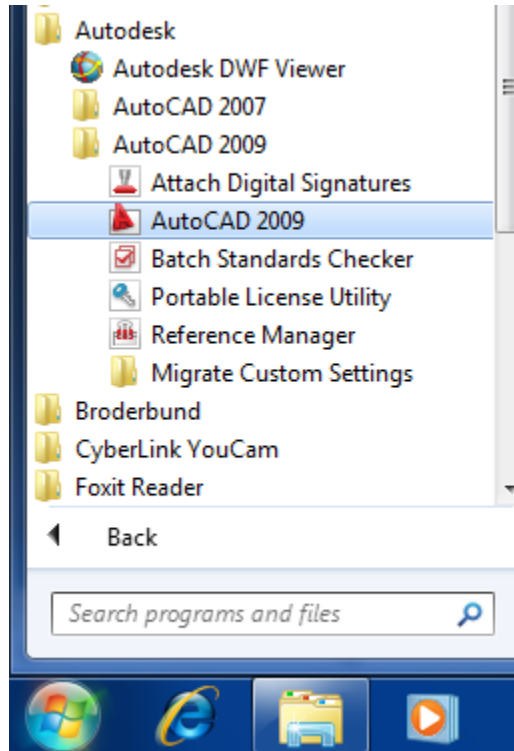


Right click on **AutoCAD 2009** icon then select open.

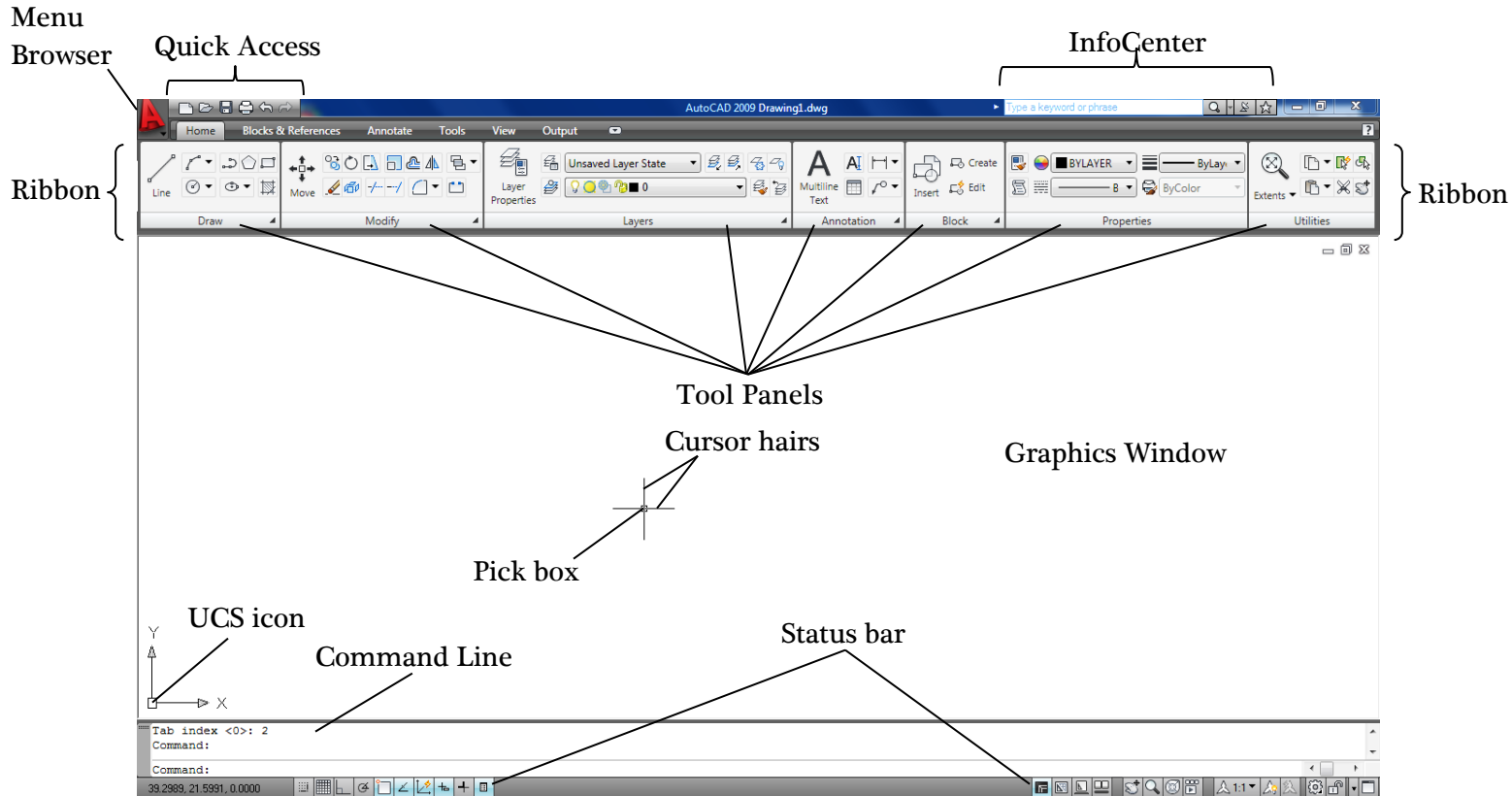


Select the **AutoCAD 2009** option on the Program menu.

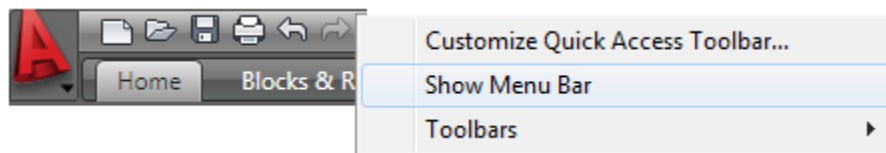
Start ➔ All Programs ➔ Autodesk ➔ AutoCAD 2009(folder) ➔ AutoCAD 2009 (icon)



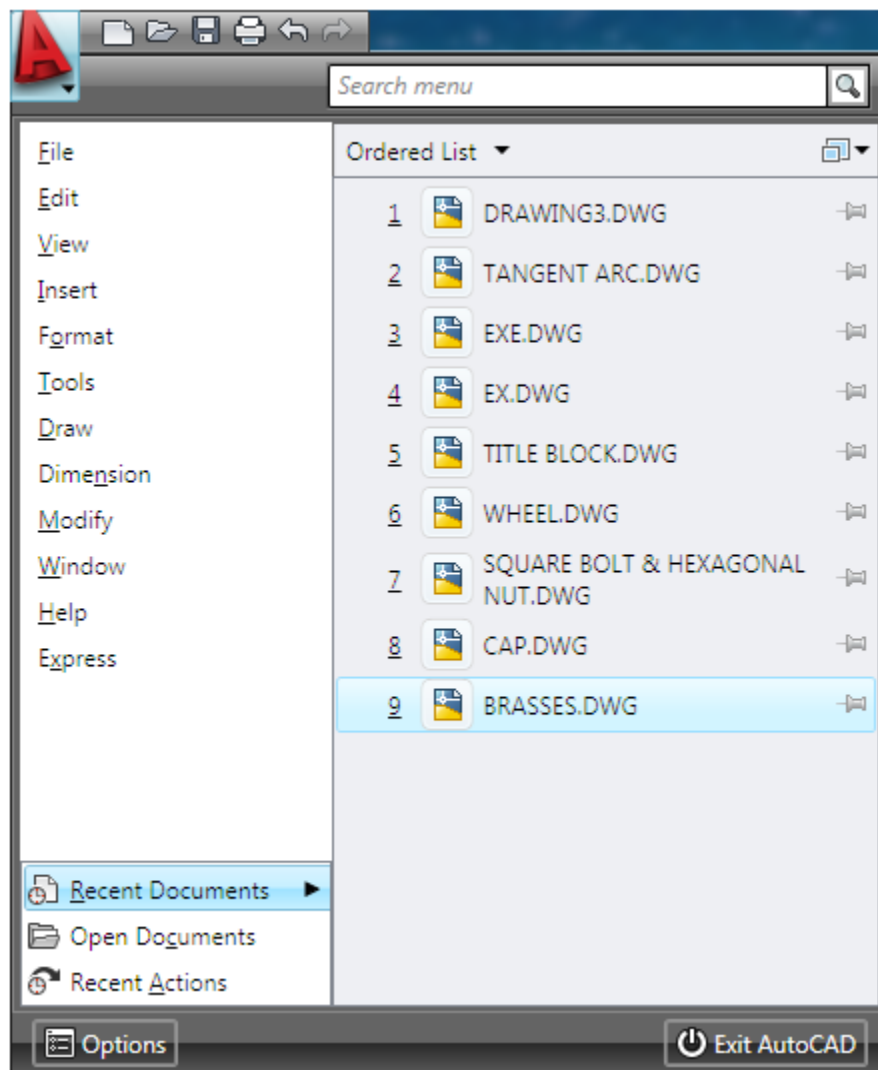
AutoCAD 2009 Screen Layout



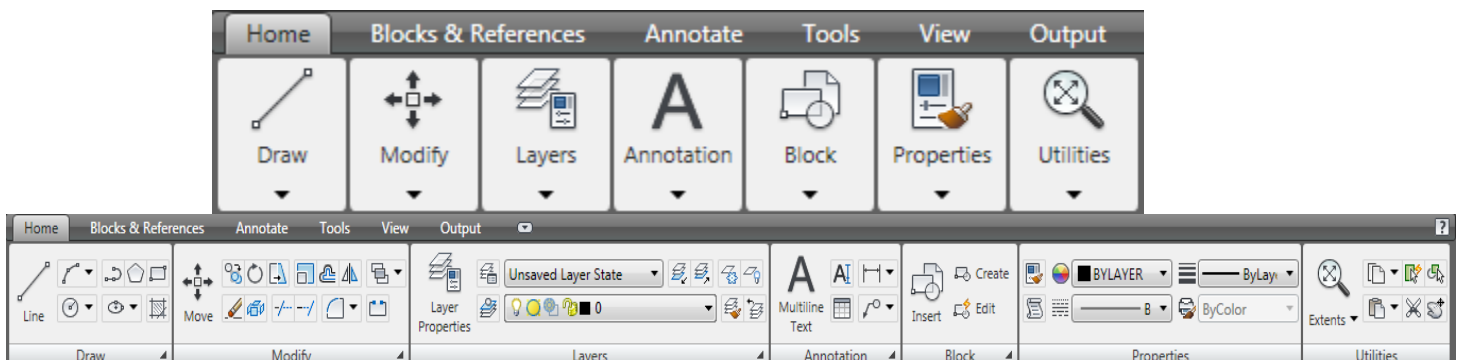
- ☐ Note that AutoCAD automatically assigns generic name, Drawing X, as new drawings are created. In our example, AutoCAD opened the graphics window using the default system units and assigned the drawing name Drawing1.
- ☐ Right click on the Quick Access bar and select Show Menu Bar to display the Menu Bar. The **Menu Bar** provides access to all AutoCAD commands.



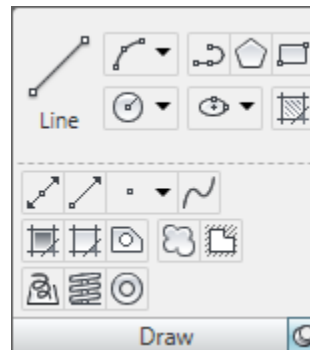
- ❑ **Menu Browser:** at the top of the main window contains access commands, options search menu, preview recently opened drawing files.



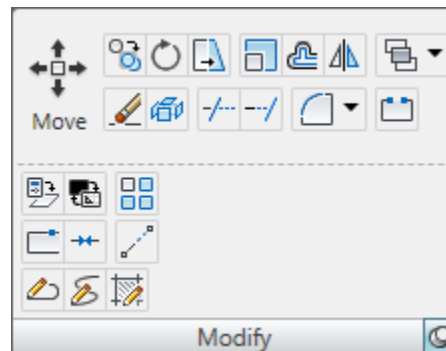
- ❑ **Ribbon:** is a palette that displays task-based buttons and controls in a workspace. The ribbon can be displayed horizontally, vertically, or as a floating palette.



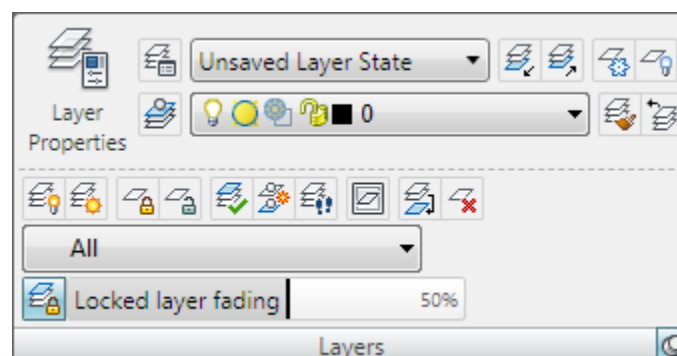
- **Draw:** the draw toolbar panels contain icons for basic draw command such as line, arc, polyline, polygon, rectangle, circle, ellipse, hatch, spline, region and helix.



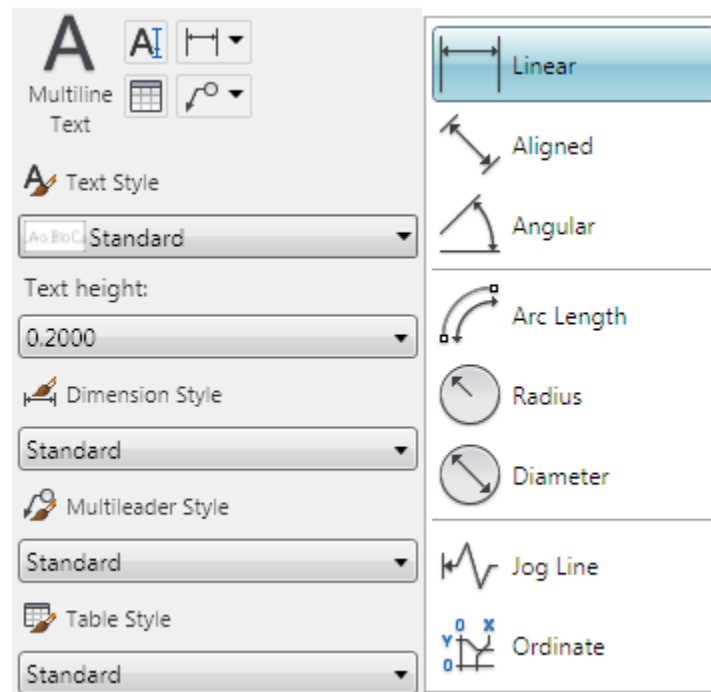
- **Modify:** the modify panels contain icons for modify command like move, copy, rotate, erase, stretch, scale, offset, mirror, trim, array, fillet, chamfer, edit hatch, edit polyline, edit spine and break.



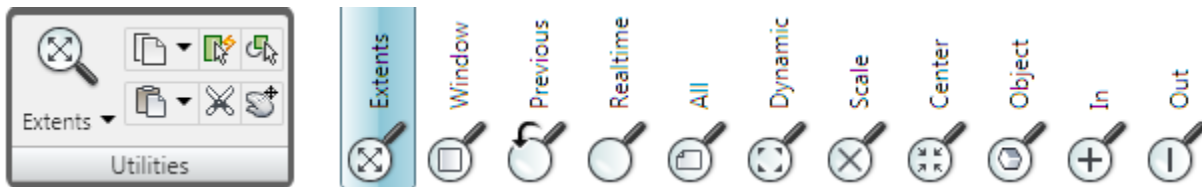
- **Layers:** the layers control toolbar panel contains tools to control the visibility of objects and to assign properties such as color and linetype.



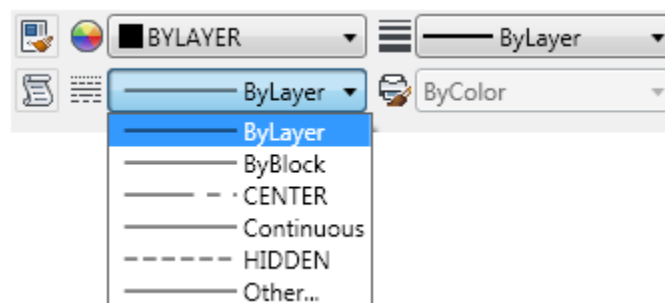
- **Annotation:** the annotation toolbar panel contains tools for creating and editing dimensions and displays text on screen.



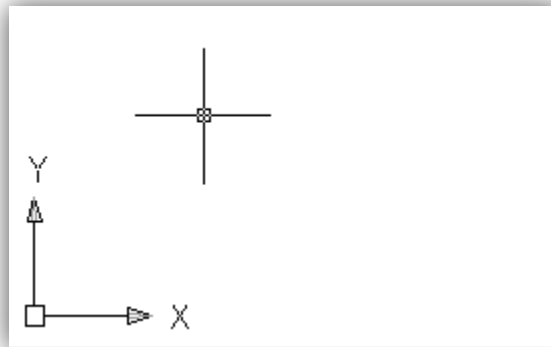
- **Utilities:** contains tools for viewing commands such as zoom and pan.
- ▶ **Zoom:** increases or decreases the apparent size of objects in the current viewport.
 - ▶ **Pan:** moves the view in the current viewport; does not change the location of objects in the drawing, its only changes the view.



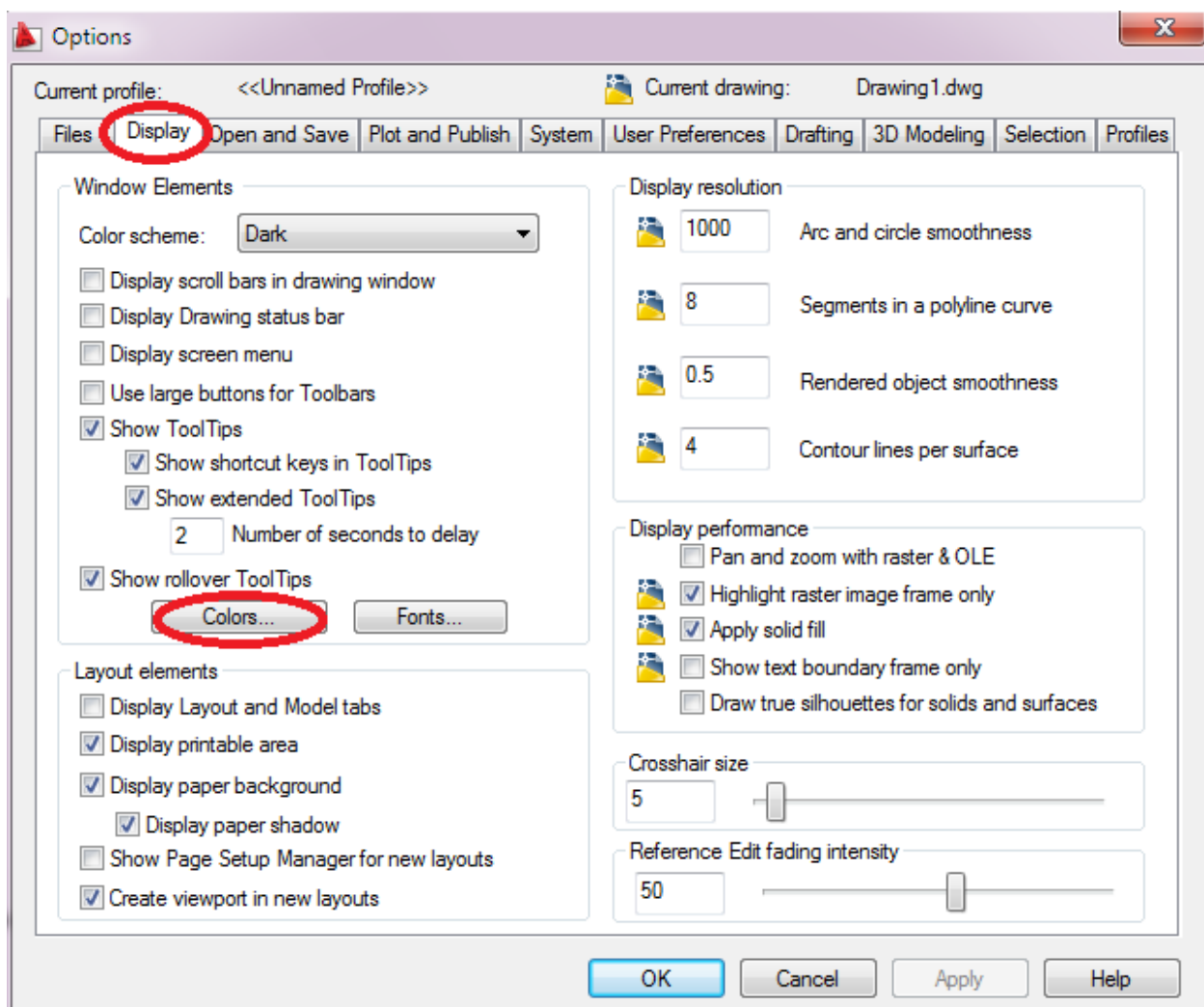
- **Properties:** contains tools linetype, color control, lineweight and plotstyle control.

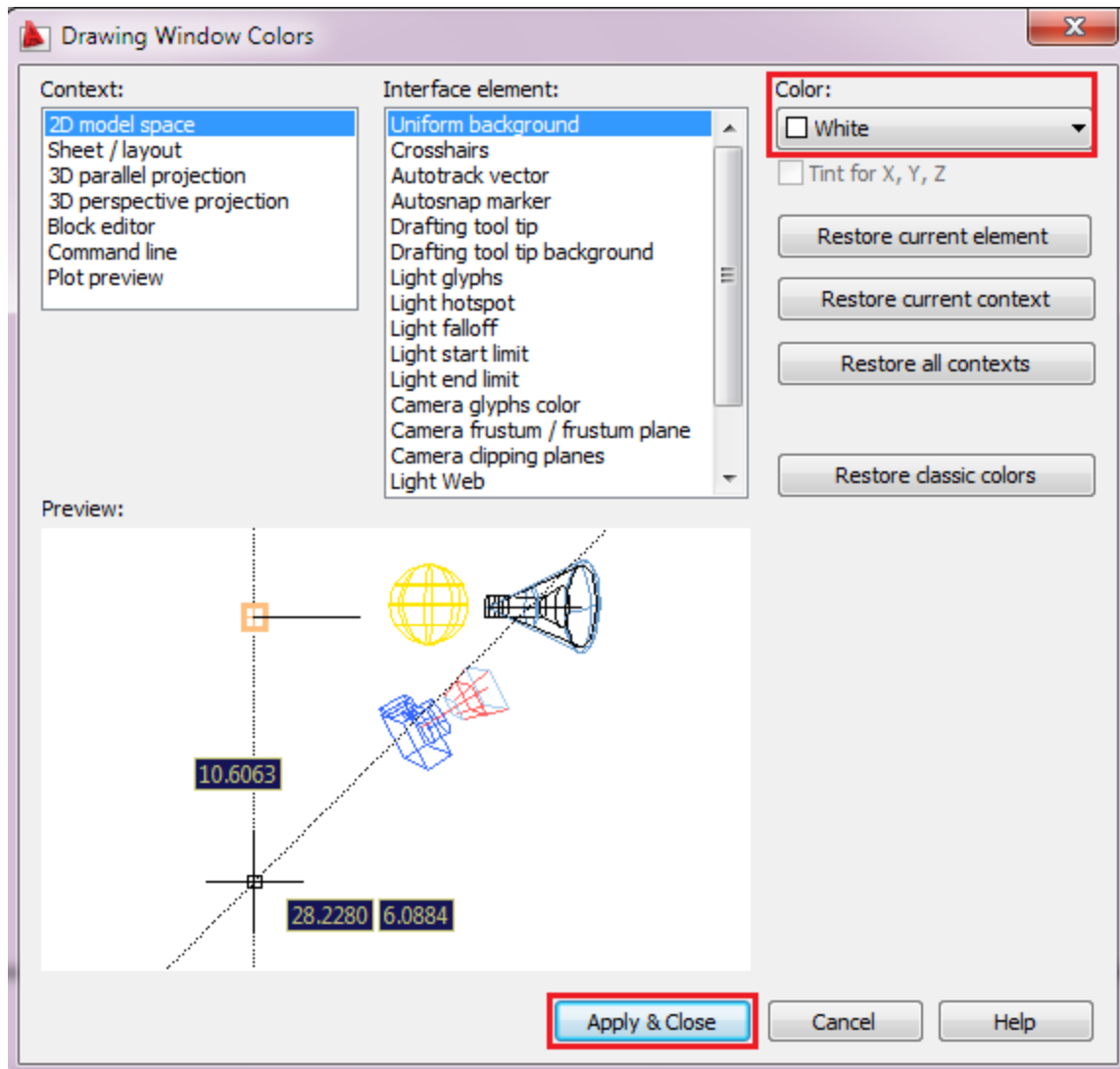


- ❑ **Graphic Window:** is the area where models and drawings are displayed.



- To change the color of graphic window or drawing area
Menu Browser → Options → Display → Color → Apply & Close → Ok



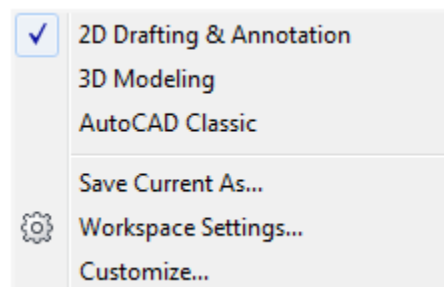


- ❑ **Workspace:** automatically display the ribbon, a special palette with task-specific control panels. To switch workspaces.

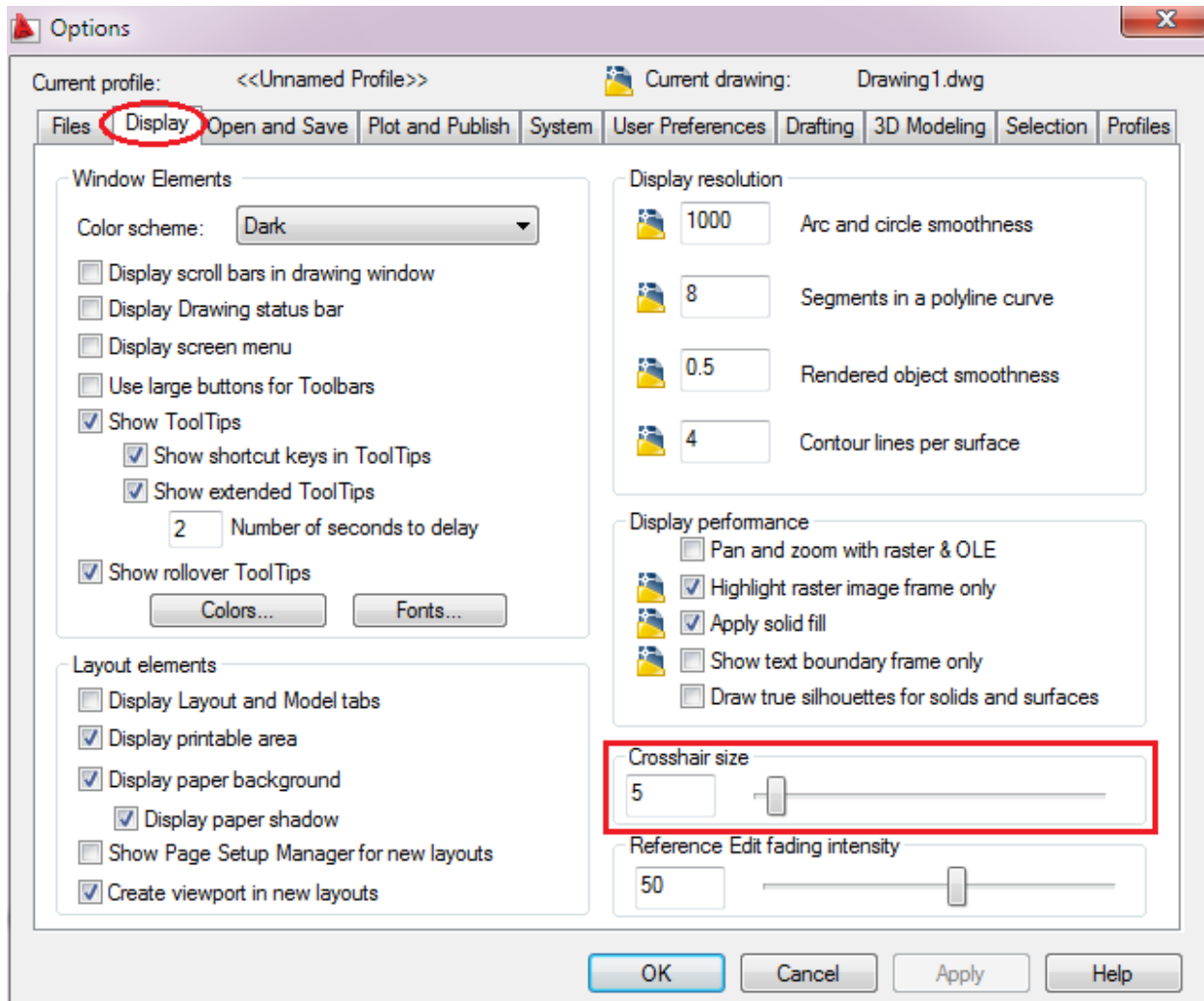
- ▶ Click Workspace Switching icon on the status bar.



- ▶ From the list of workspaces, select on one of the following workspace options.



- ❑ **Cursor or Cross hair:** the range of the crosshair is 1 to 100 percent of the total screen. At 100% the crosshair ends aren't visible. When the size is decreased to 99% or below, the crosshairs have a finite size, and the crosshairs ends are visible when moved to the edge of the graphics area. To control the size of crosshair.
 - ▶ Menu Browser → Options → Display → Crosshair Size → Ok



- ❑ **Status Bar:** the area at the bottom of the application window that contains buttons controlling the mode of operation of the program and displays the coordinates of the cursor location in the drawing area.

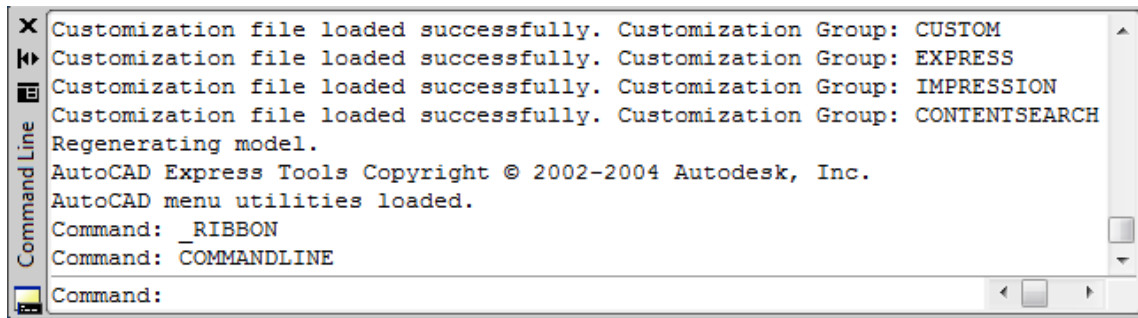


Drawing Coordinates

Model

Layout

- ❑ **Command Line:** used to display commands, system variables, options, messages, and prompts in a dockable and resizable window; enter a command by using keyboard.



- ❑ **Mouse:** AutoCAD uses pointing device like a mouse, a digitizing puck, or a stylus, and it may have more than two buttons.

- **Left Mouse Button**

Left mouse button is pick button used to specify locations, select objects for editing, choose menu options and dialog box buttons and fields.

- **Left Mouse Button**

The operation of the right button on a mouse depends on context. It can be used to end a command in progress, display a shortcut menu, display the Object Snap menu, display the toolbars dialog box.

- **Wheel Mouse**

The wheel mouse has a small wheel between the buttons. It can be used the wheel to zoom and pan in your drawing without using any commands.

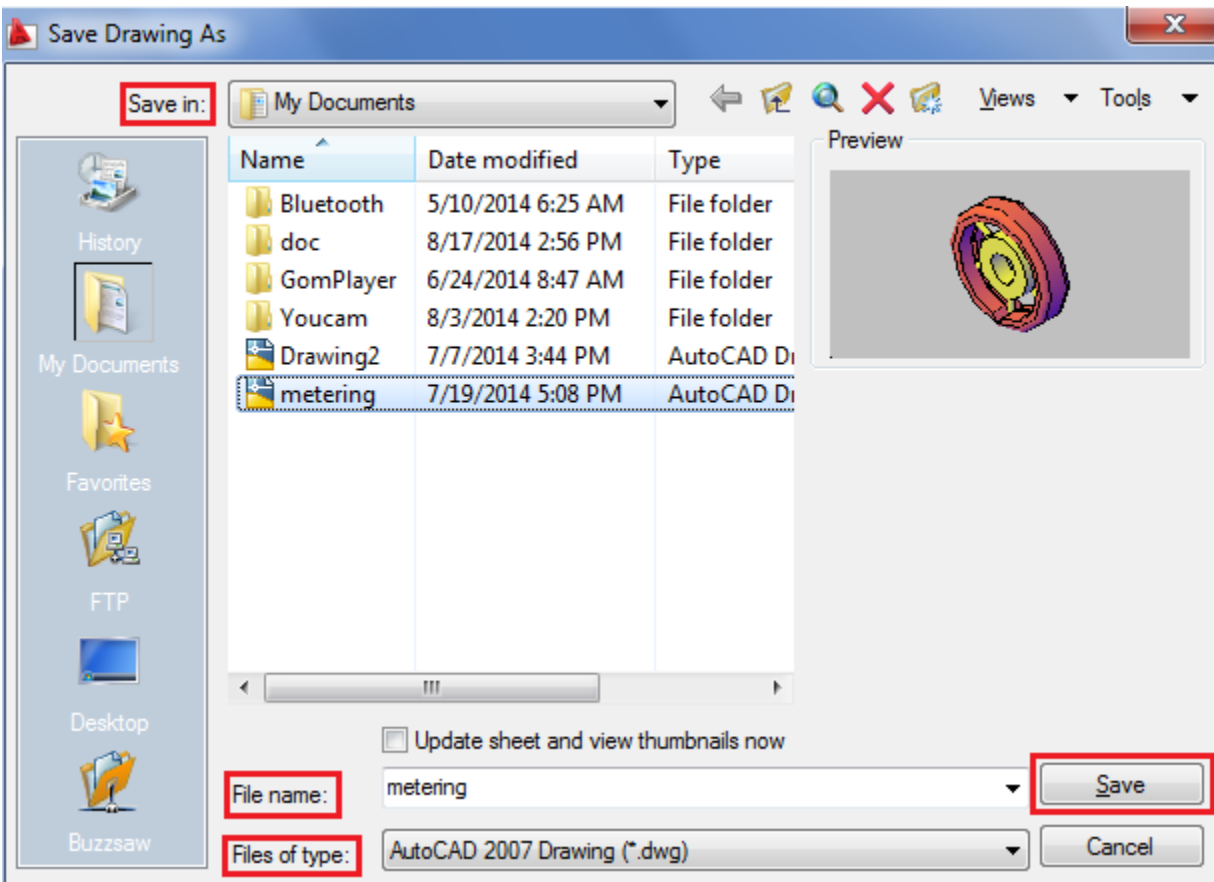
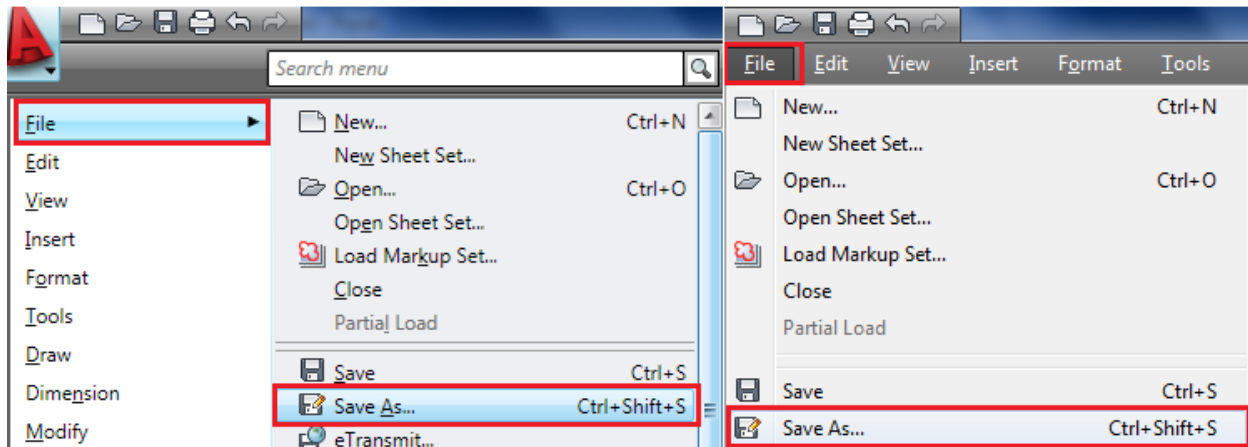


- ❑ **Cancel a Command (Esc):** used to cancel a command that accidentally click and to exist the command. The Esc key is located near the top-left corner of the keyboard.

Save a Drawing

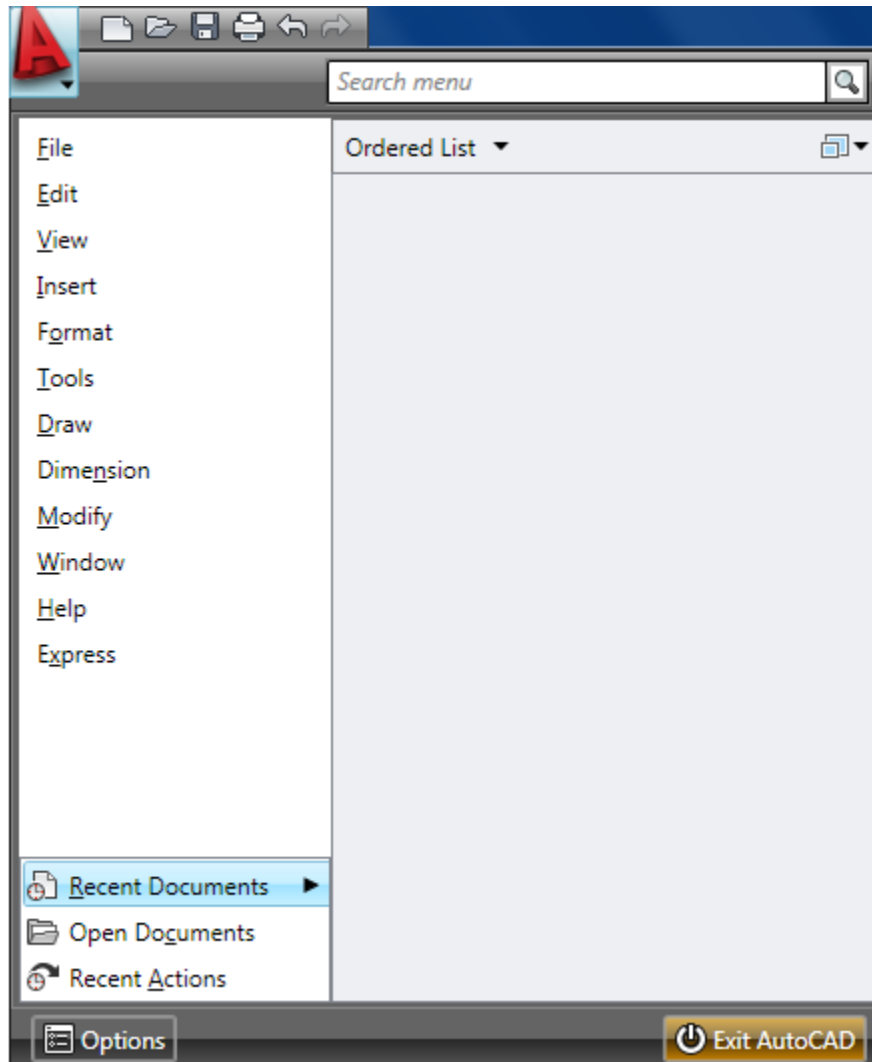
Save the drawing under the current file name or a specified name but never saved the drawing, the Save Drawing As dialog box is displayed. In the Save Drawing As dialog box under File Name, enter the new drawing name; Click Save.

► Menu Browser → File → Save as... or File (Menu Bar) → Save as...



Exiting AutoCAD 2009

To exit AutoCAD 2009, choose Exit AutoCAD in the Menu Browser or Menu Bar; or simply type exit or quit in the command window then press enter or space.



```
AutoCAD menu utilities loaded.  
Command: _RIBBON  
Command: COMMANDLINE  
Command: exit
```

OR

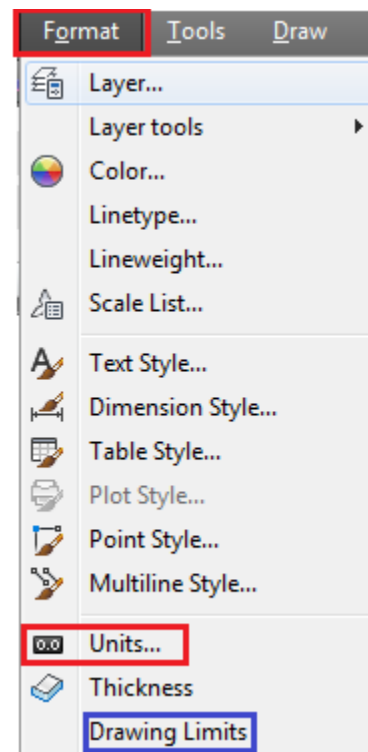
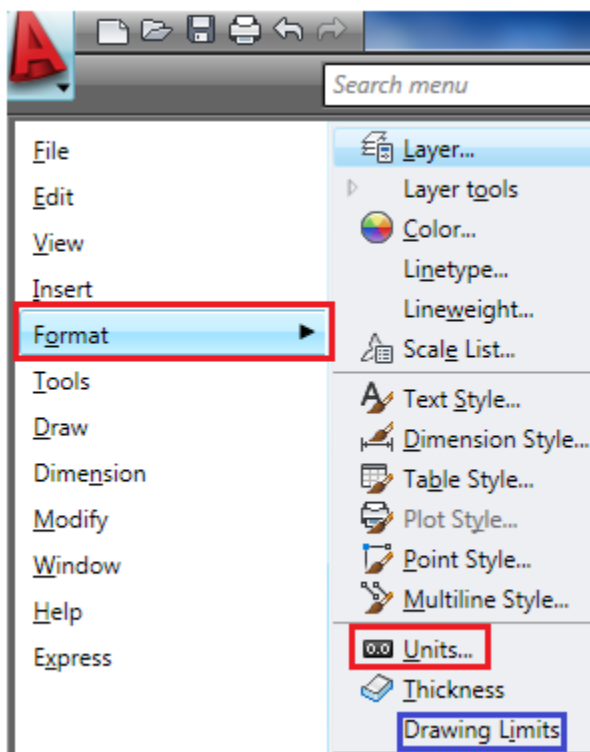
```
AutoCAD menu utilities loaded.  
Command: _RIBBON  
Command: COMMANDLINE  
Command: quit
```

Start a Drawing

There are different ways to start a new drawing including starting from scratch or using a template file. A drawing template file contains standard settings. You can select one of the drawing template files supplied or create your own drawing template files by saving a drawing using a *.dwt* extension.

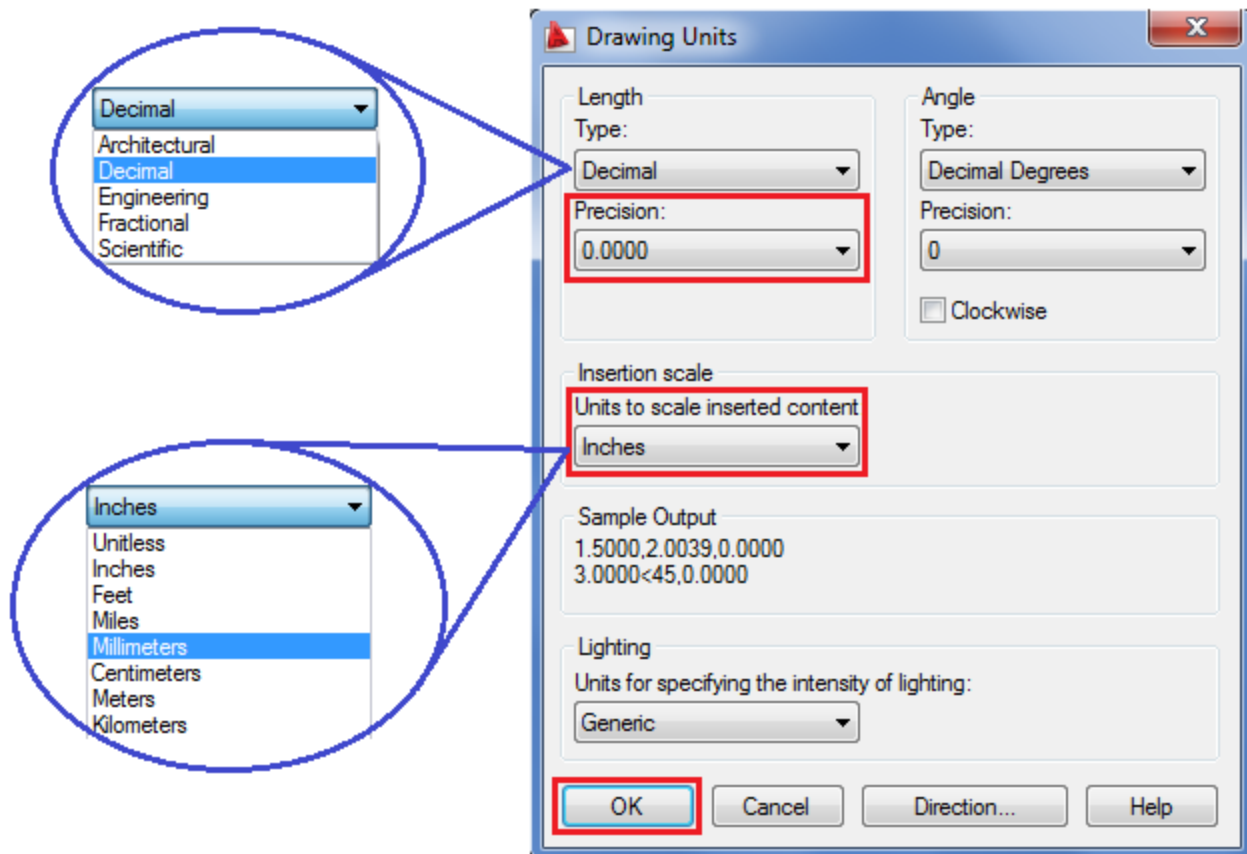
❑ **Drawing Units:** to set the measurement units.

- ▶ Menu Browser → Format → Units... or Format (Menu Bar) → Units...



The Drawing Units dialog box is appearing. In this dialog box set the following:

- ▶ Length option: Type – **Decimal** and Precision – **0**;
- ▶ Insertion scale option: Units to scale inserted content – **Millimeter**
- ▶ Click **OK** to exit the Drawing Units dialog box.



- **Drawing Limits:** The drawing limits are two-dimensional points in the World Coordinate System that represent a lower-left limit and an upper-right limit. It can be used to controls the extents of the grid display in the current Model or Layout tab, and as a plot option that defines an area to be plotted/ printed.
- ▶ Click the Menu Browser → Format → Drawing Limits... **OR**
 - ▶ Click the Format (Menu Bar) → Drawing Limits...(same fig. in Drawing Units)
 - ▶ In the Command line, the limits area set to,
 - ✓ Specify lower left corner (0, 0) then Enter and
 - ✓ Specify upper right corner (420, 297) then Enter.

```

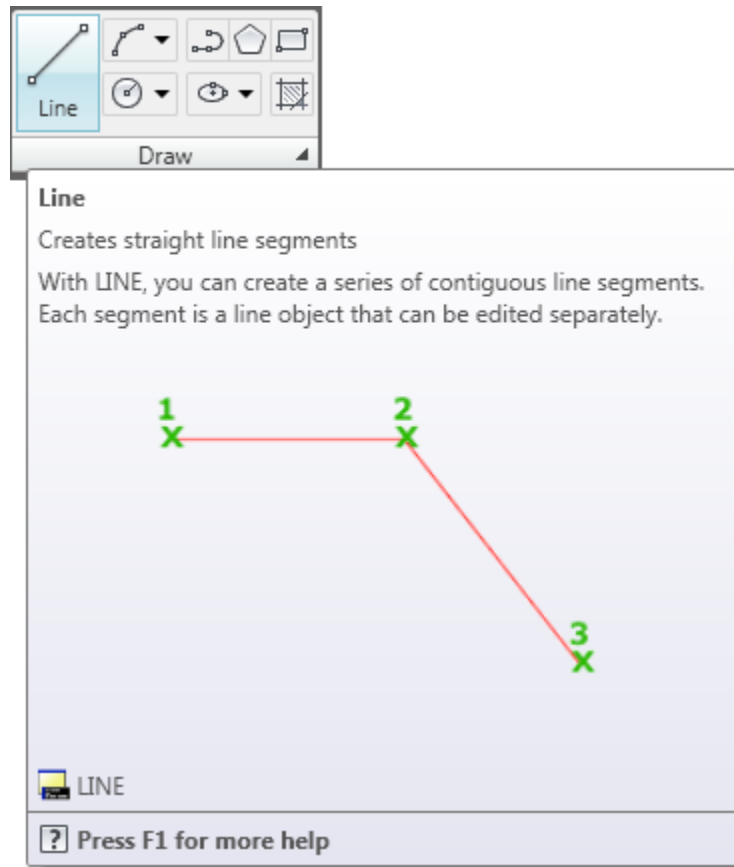
Command: '_limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 420,297
Command:

```

DRAW COMMAND

Line Command  : creates straight line segments.

- ▶ Select **Line** from draw panel **OR**
- ▶ In command line write **line** or **l** then Enter



☐ Drawing with the Line Tool:

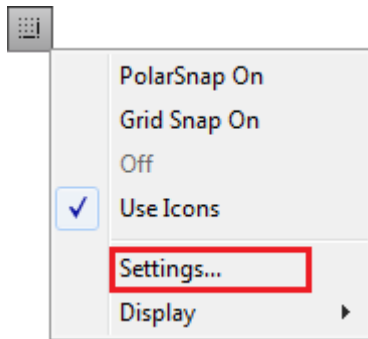
Example:

- ▶ Set **Drawing Units** – Millimeters; **Drawing Limits** – (420, 297); **Grid** – 10; **Snap** -10.

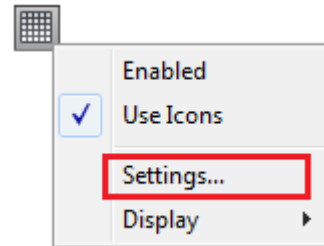
- ☐ **Snap**  **Snap Mode** (F9): restricts the movement of the crosshairs to the given intervals. It used to specifying precise points with the arrow keys or the pointing device.

- ☐ **Grid**  **Grid Display** (F7): rectangular pattern of dots or lines that extends over the area and used to align objects and visualize the distances between them.
- ☐ To specify the snap and grid limits.

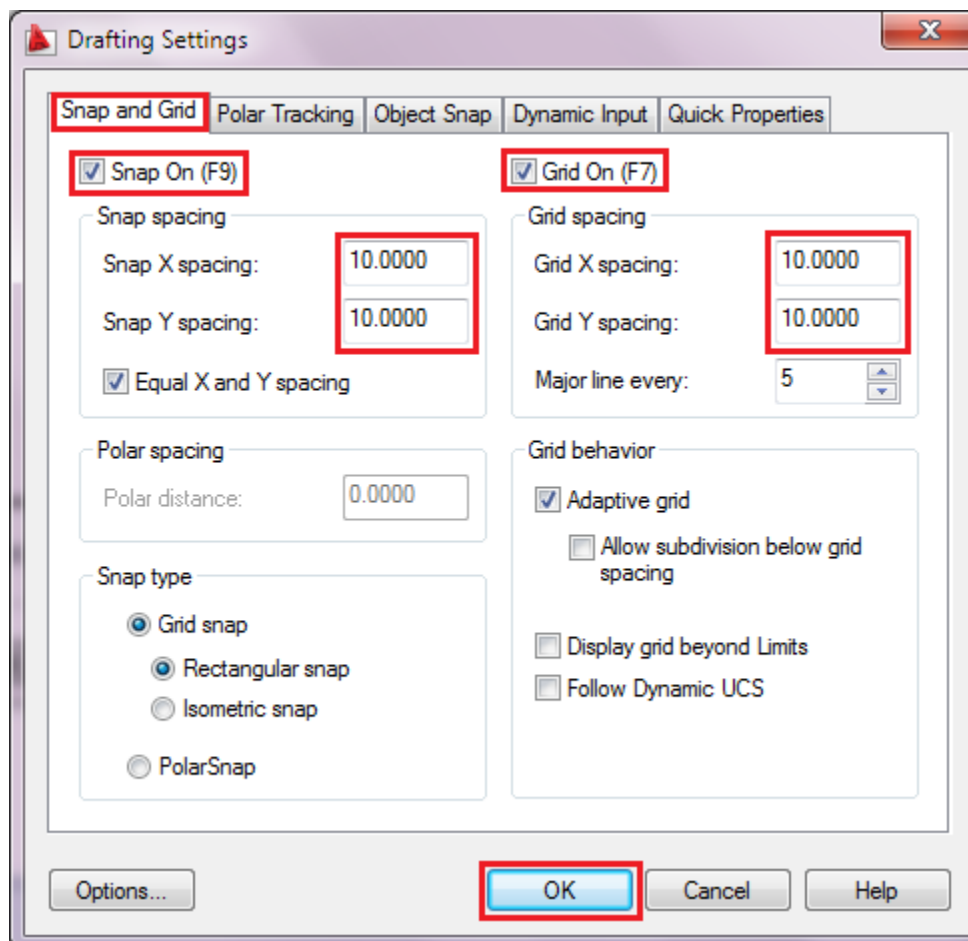
- ▶ Right click the Snap Mode/ Grid Display button on the Status Bar then
- ▶ Select **Setting...**



OR

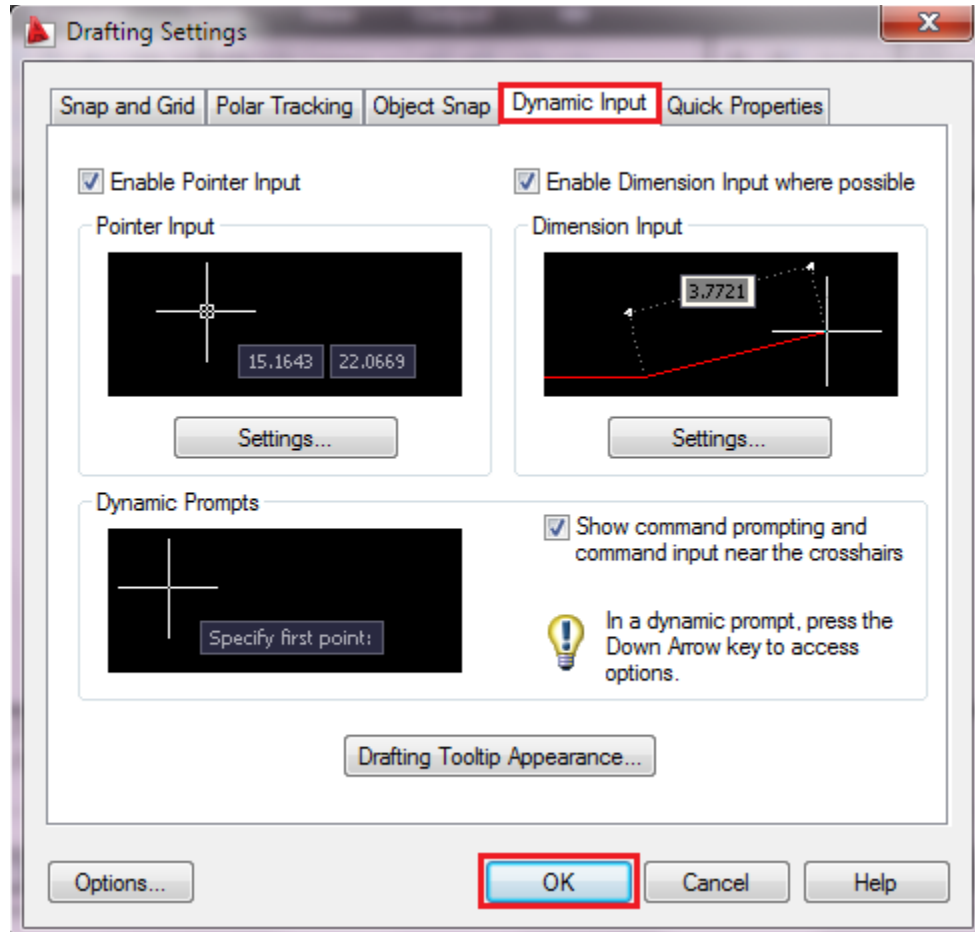


- ▶ **Drafting Settings** dialog box appearing then select Snap and Grid enter 10 in Snap spacing and Grid Spacing.
- ▶ Click **OK**.

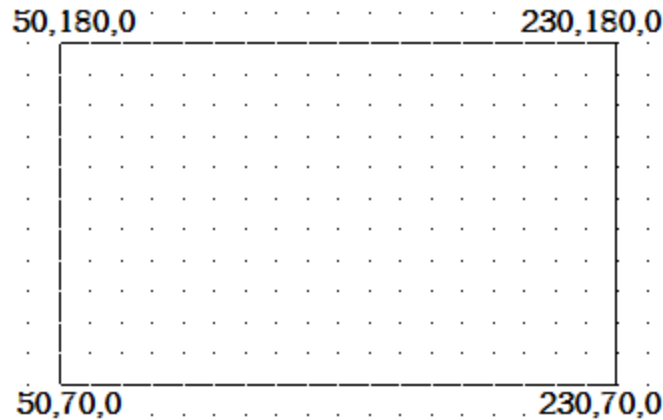


- **Zoom All** : used to display the drawing extents and the grid limits. **View (Menu Bar) → Zoom → All** or you can select from Utilities panel.

-  **Dynamic Input** (F12): provides a command interface near the cursor to help you keep your focus in the drafting area. When Dynamic Input is on, tooltips display information near the cursor that is dynamically updated as the cursor moves. When a command is active, the tooltips provide a place for user entry.
- ▶ **Right click the Dynamics Input button on the Status Bar → Settings... → Drafting Settings dialog box appearing → OK**



- Move the mouse around the drawing area. The cursors pick box will jump from point to point at 10 unit intervals. The position of the pick box will show as coordinate numbers in the **Drawing Coordinates** (on the status bar left-hand end).
 - ▶ Move the mouse until the coordinate numbers show **50, 70, 0** and click.
 - ▶ Move the mouse until the coordinate numbers show **230, 70, 0** and click.
 - ▶ Move the mouse until the coordinate numbers show **230, 180, 0** and click.
 - ▶ Move the mouse until the coordinate numbers show **50, 180, 0** and click.
 - ▶ Then **Right Click** and **Enter** OR simply press **Enter**. The line rectangle appears in drawing area.

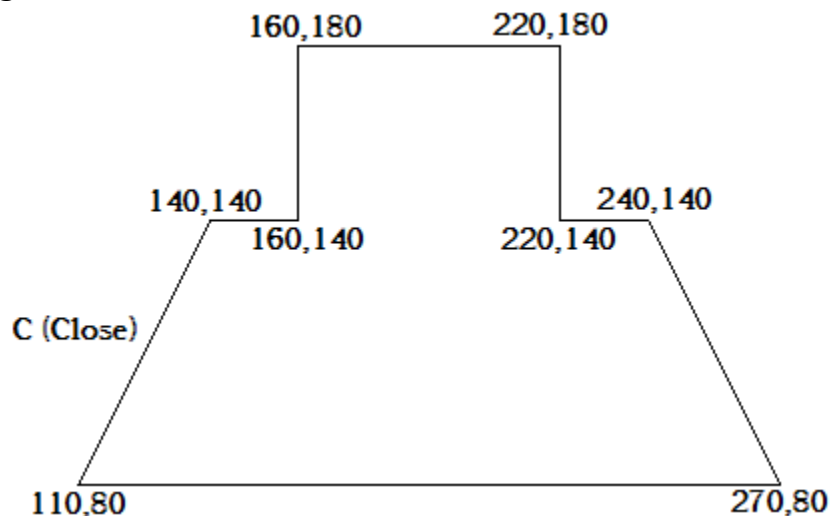


- **Absolute Coordinates (x,y):** the *x, y* coordinates of each corner of the outlines is entered at the command line as required.

Example: select line tool icon and enter the coordinate numbers on command line.

- ▶ Command: `_line` specify the first point: **110,80** then press enter or space.
- ▶ Specify the next point or [Undo]: **240,80** then **Enter**
- ▶ Specify the next point or [Undo]: **240,140** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **220,140** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **220,180** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **160,180** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **160,140** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **140,140** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **c (Close)** then **Enter**

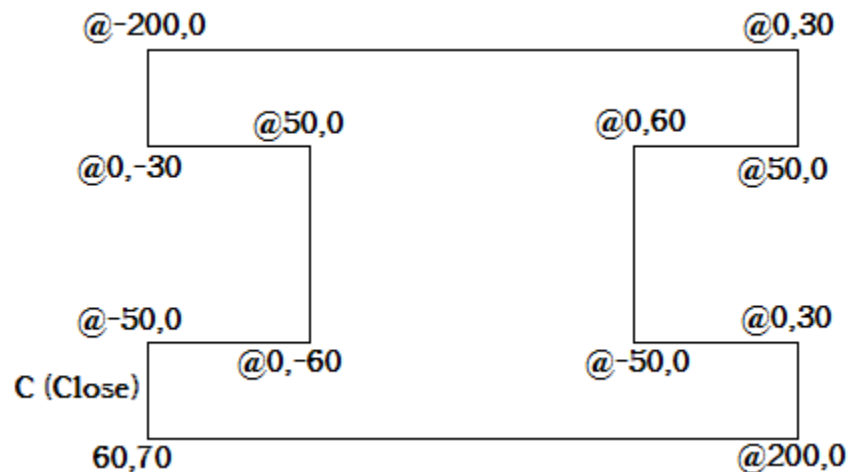
NOTE: If dynamic input (F12) is on, you must type the # sign before entering absolute coordinates (e.g.#110,80).



- **Relative Coordinates (@x,y):** coordinate points are entered relative to the previous entry. In relative coordinate, the @ symbol is entered before each set of coordinates with the following rules in mind: **x** entry is +ve to the right and -ve to the left; **y** entry is +ve to upwards and -ve to downwards.

Example: click on line tool icon and enter figures as follow:

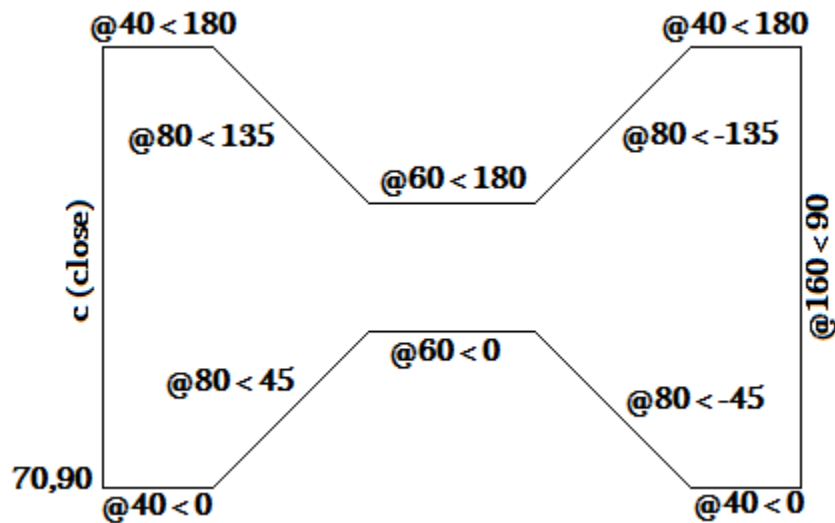
- ▶ Command: _line specify the first point: **60,70** then press enter.
- ▶ Specify the next point or [Undo]: **@200,0** then **Enter**
- ▶ Specify the next point or [Undo]: **@0,30** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@-50,0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@0,60** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@50,0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@0,30** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@-200,0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@0, -30** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@50,0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@0, -60** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@-50,0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **c (Close)** then **Enter**



- **Polar coordinates (@Distance<Angle):** to locate the points when creating the object with the distance and angle relative to the previous entry. Angles increase in the counterclockwise direction and decrease in the clockwise direction. To specify a clockwise direction, enter a negative value for the angle.

Example: click on line tool icon and enter figures as follow:

- ▶ Command: `_line` specify the first point: **70,90** then press enter.
- ▶ Specify the next point or [Undo]: **@40 < 0** then **Enter**
- ▶ Specify the next point or [Undo]: **@80 < 45** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@60 < 0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@80 < -45** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@40 < 0** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@160 < 90** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@40 < 180** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@80 < -135** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@60 < 180** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@80 < 135** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **@40 < 180** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **c (Close)** then **Enter**

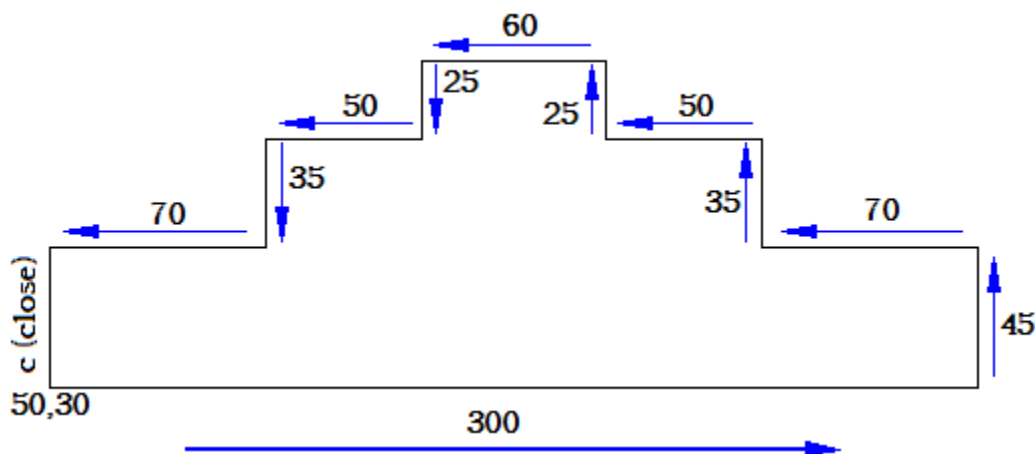


- ☐ **Tracking Method:** another method of constructing accuracy drawing by using line command. It used to specify a point by offsetting vertically and horizontally from a series of temporary points. The direction of the offset is indicated by the rubber-band line. Drag the rubber-band line in any direction and enter a number at the keyboard, followed by press Enter.

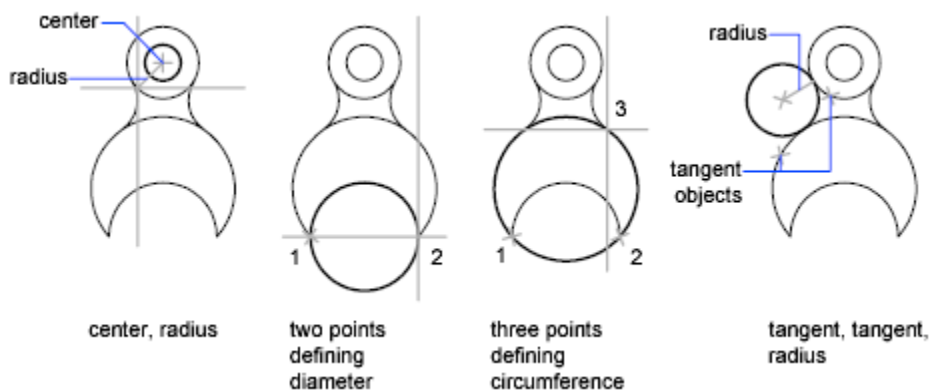
Example: first on **Ortho Mode** from the status bar and select line tool icon.

- ☐ **Ortho**  (F8): constrains cursor movement to the horizontal or vertical direction.

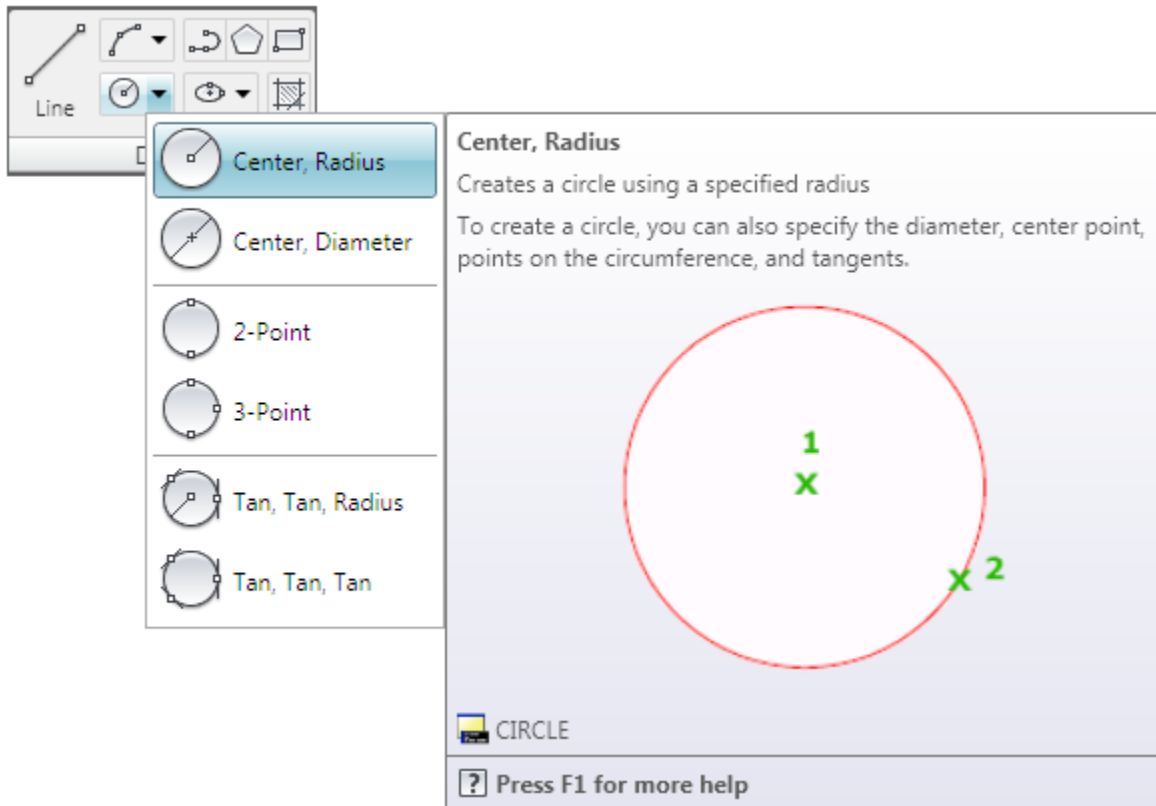
- ▶ Command: `_line` specify the first point: **50,30** then press **Enter/Space**.
- ▶ Drag to right; Specify the next point or [Undo]: **300** then **Enter**
- ▶ Drag to up; Specify the next point or [Undo]: **45** then **Enter**
- ▶ Drag to left; Specify the next point or [Close/Undo]: **70** then **Enter**
- ▶ Drag to up; Specify the next point or [Close/Undo]: **35** then **Enter**
- ▶ Drag to left; Specify the next point or [Close/Undo]: **50** then **Enter**
- ▶ Drag to up; Specify the next point or [Close/Undo]: **25** then **Enter**
- ▶ Drag to left; Specify the next point or [Close/Undo]: **60** then **Enter**
- ▶ Drag to down; Specify the next point or [Close/Undo]: **25** then **Enter**
- ▶ Drag to left; Specify the next point or [Close/Undo]: **50** then **Enter**
- ▶ Drag to down; Specify the next point or [Close/Undo]: **35** then **Enter**
- ▶ Drag to left; Specify the next point or [Close/Undo]: **70** then **Enter**
- ▶ Specify the next point or [Close/Undo]: **c (close)** then press **Enter**



Circle Command : to create circles, you can specify various combinations of center, radius, diameter, points on the circumference, and points on other objects.



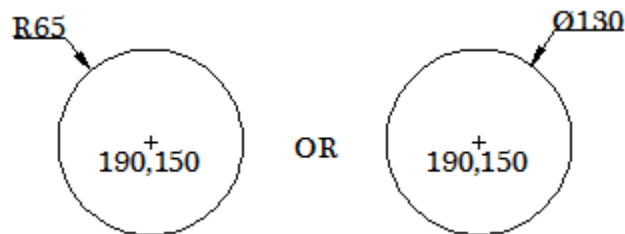
- ▶ Select **Circle** from draw panel **OR**
- ▶ In command line write **circle** or **c** then Enter



□ Drawing with the Circle Tool:

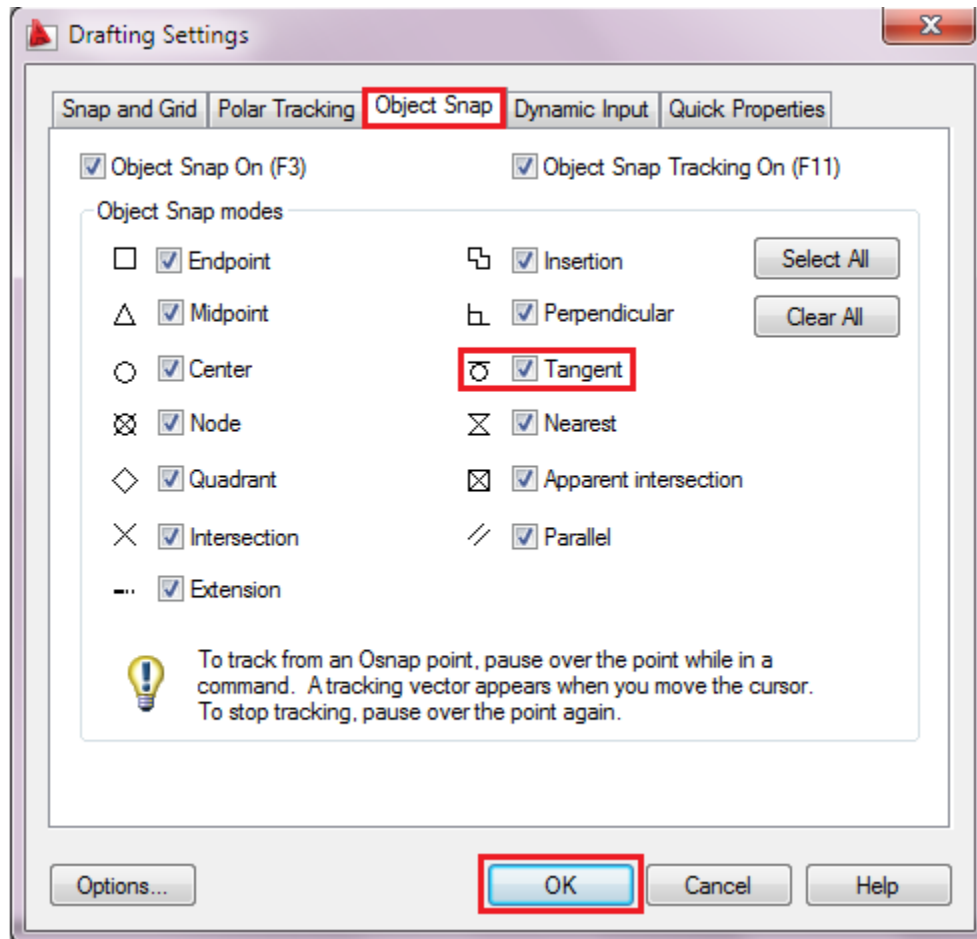
Example: circle, **Center Radius** or **Center Diameter** click on circle tool icon and follow the step below:

- ▶ Command: `_circle` Specify center point for circle or [3P / 2P / Ttr (tan tan radius)]: **190,150**
- ▶ Specify radius of circle or [Diameter]: **65** then **Enter**. **OR**
- ▶ Specify radius of circle or [Diameter]: **d** then **Enter** (If diameter is given)
- ▶ Specify diameter of circle: **130** then **Enter**.

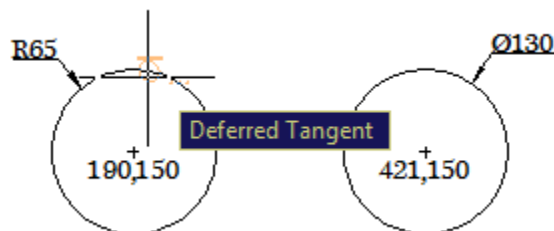


Example: circle, **Tan Tan Radius** first on **Object Snap** from the status bar then click on circle tool icon and follow the step below:

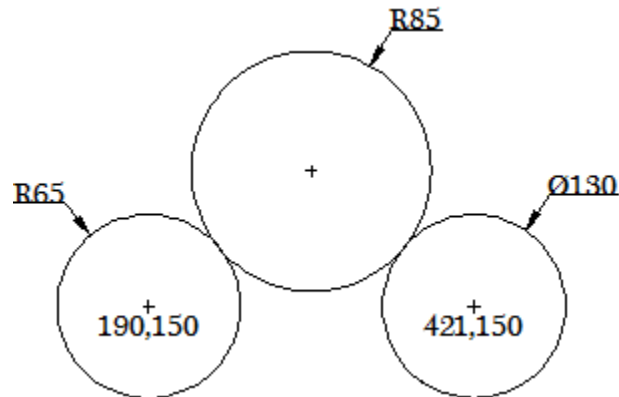
- ❑ **Object Snap (OSNAP)**  (F3): used to specify precise point at locations on objects.
- To specify object snap:
- ▶ **Right click the Object Snap** button on the **Status Bar** then
 - ▶ Select **Setting...**
 - ▶ **Drafting Settings** dialog box appearing then pick object snap modes that you need.
 - ▶ Click **OK**.



- ▶ Command: `_circle` Specify center point for circle or [3P / 2P / Ttr (tan tan radius)]: **t** then **Enter**.

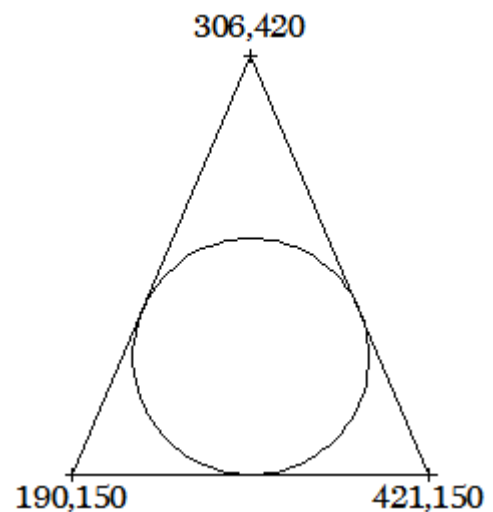
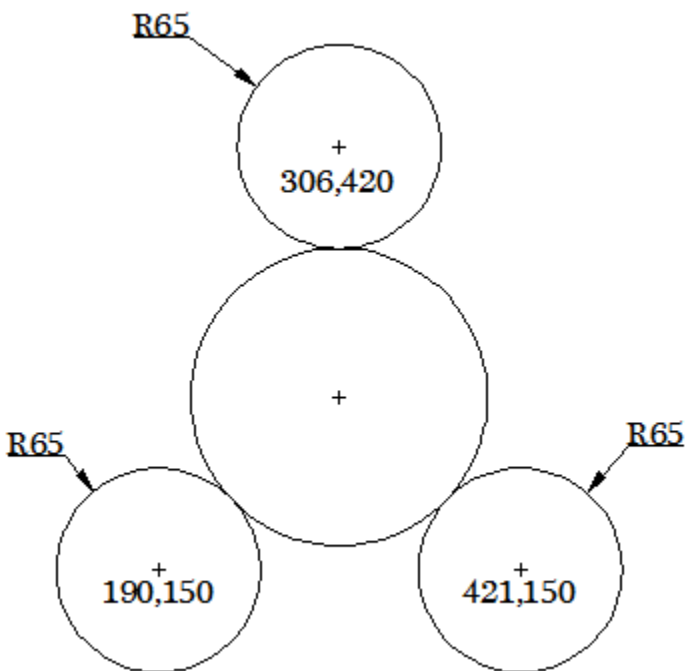


- ▶ Specify point on object for first tangent of circle: **Pick** or **click**
- ▶ Specify point on the object for second of circle: **Pick**
- ▶ Specify radius of circle: **85** then **Enter**



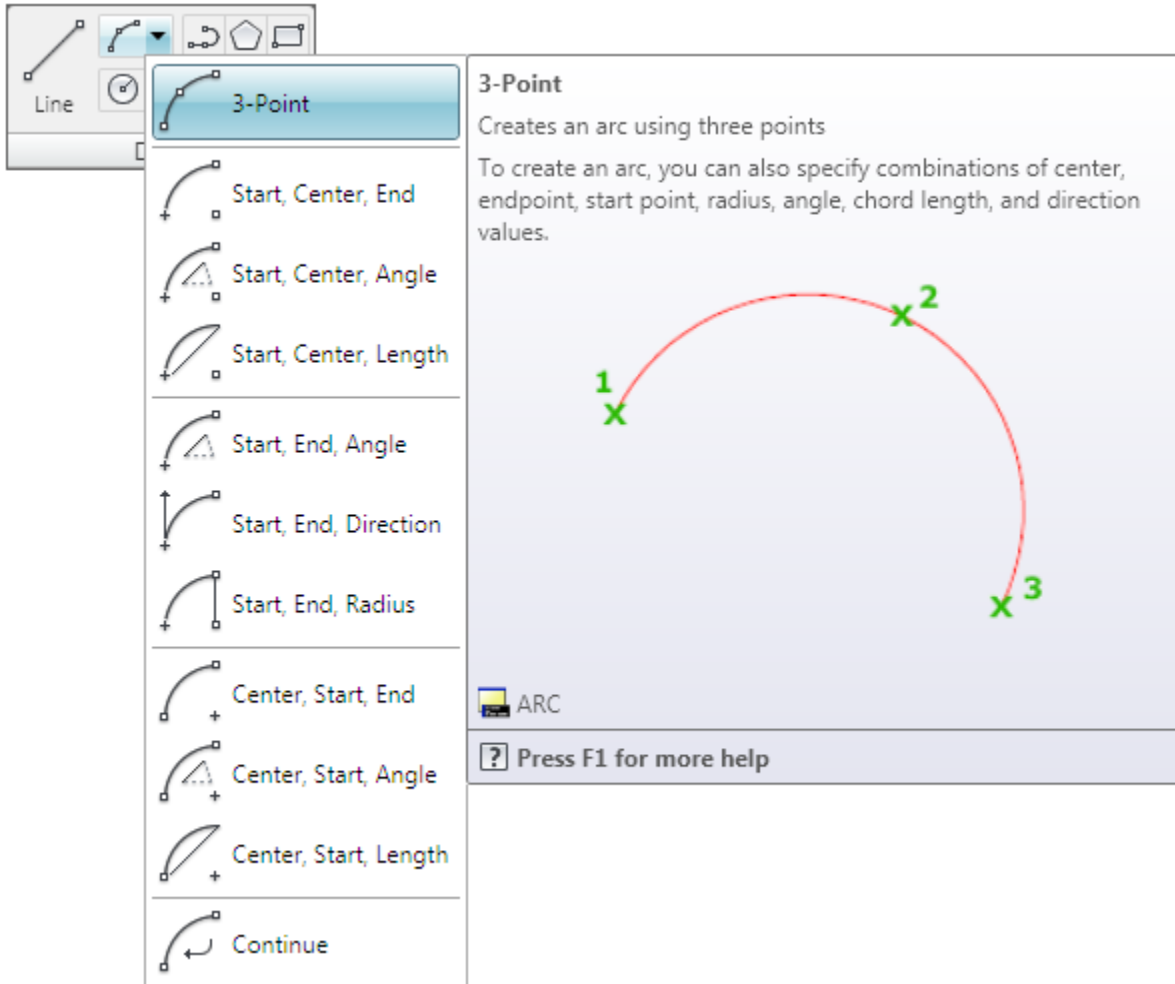
Example: circle, **Tan Tan Tan** click on circle (tan tan tan)  tool icon and follow the step below:

- ▶ Command: `_circle` Specify center point for circle or [3P / 2P / Ttr (tan tan radius)]:
`_3p` specify first point on circle: **_tan** to **Pick**.
- ▶ Specify second point on circle: **_tan** to **Pick**.
- ▶ Specify third point on circle: **_tan** to **Pick**.



Arc Command : to create an arc, you can specify various combinations of center, endpoint, start point, radius, angle, chord length, and direction values. Arcs are drawn counterclockwise from the start point to the endpoint.

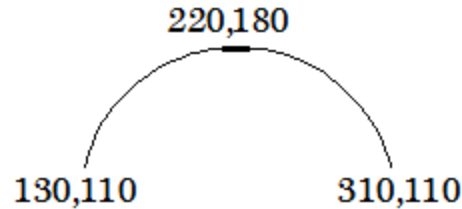
- ▶ Select **Arc** from draw panel **OR**
- ▶ In command line write **arc** or **a** then Enter



□ Drawing with the Arc Tool:

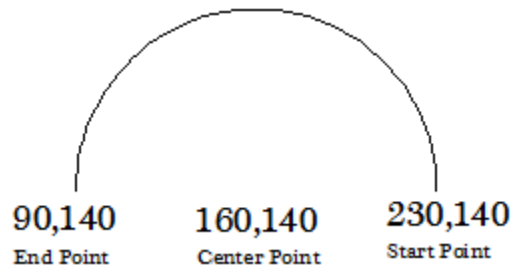
Example: arc, **3-Point** click on arc tool icon and follow the step below:

- ▶ Command: **_arc** Specify start point of arc or [Center]: **130,110** then **Enter**
- ▶ Specify second point of arc or [Center / End]: **220,180** then **Enter**
- ▶ Specify end point of arc: **310,110** then **Enter**



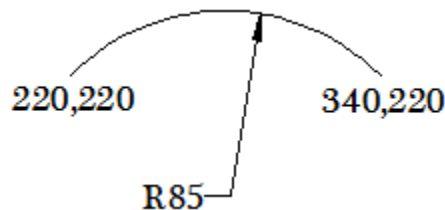
Example: arc, **Start Center End** click on arc tool icon and follow the step below:

- ▶ Command: _arc Specify start point of arc or [Center]: **230,140** then **Enter**
- ▶ Specify second point of arc or [Center / End]: _c Specify center point of arc: **160,140** then **Enter**
- ▶ Specify end point of arc or [Angle / chord Length]: **90,140** then **Enter**



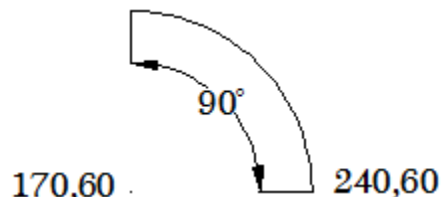
Example: arc, **Start End Radius** click on arc tool icon and follow the step below:

- ▶ Command: _arc Specify start point of arc or [Center]: **340,220** then **Enter**
- ▶ Specify end point of arc: **220,220** then **Enter**
- ▶ Specify center point of arc or [Angle / Direction / Radius]: _r Specify radius of arc: **85** then **Enter**



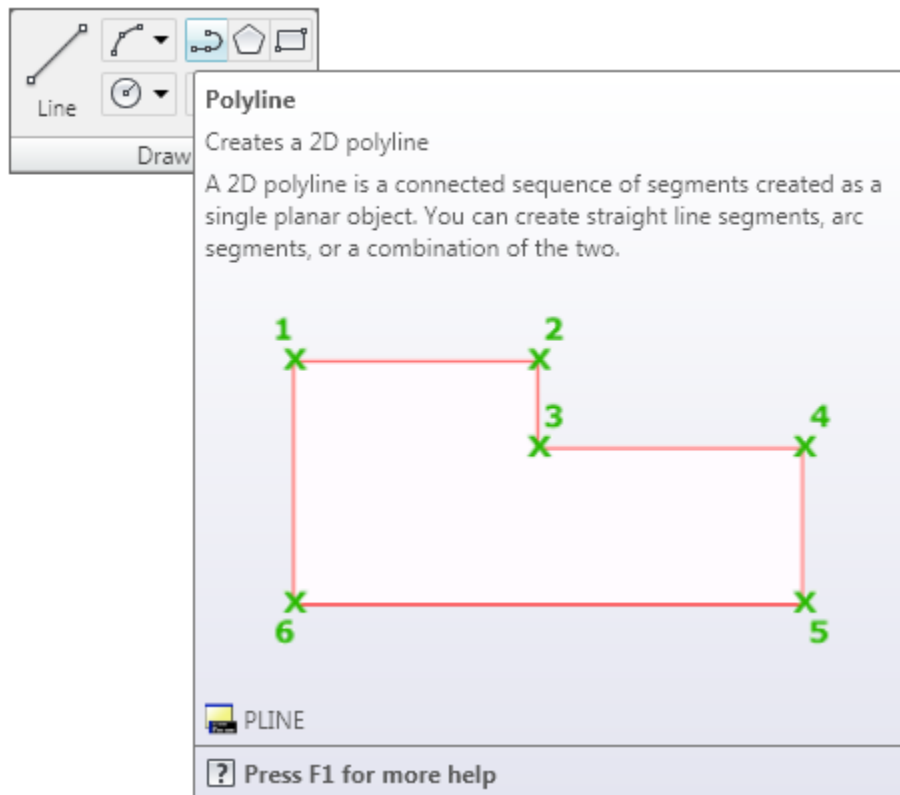
Example: arc, **Center Start Angel** click on arc tool icon and follow the step below:

- ▶ Command: _arc Specify start point of arc or [Center]: _c Specify center point of arc: **170,60** then **Enter**
- ▶ Specify start point of arc: **240,60** then **Enter**
- ▶ Specify end point of arc or [Angle / chord Length]: _a Specify included angle: **90** then **Enter**



Polyline Command  : a connected sequence of segments created as a single object. You can create straight line segments with different thickness, arc segments, or a combination of the two and arrows.

- ▶ Select **Polyline** from draw panel **OR**
- ▶ In command line write **pline** or **pl** then Enter

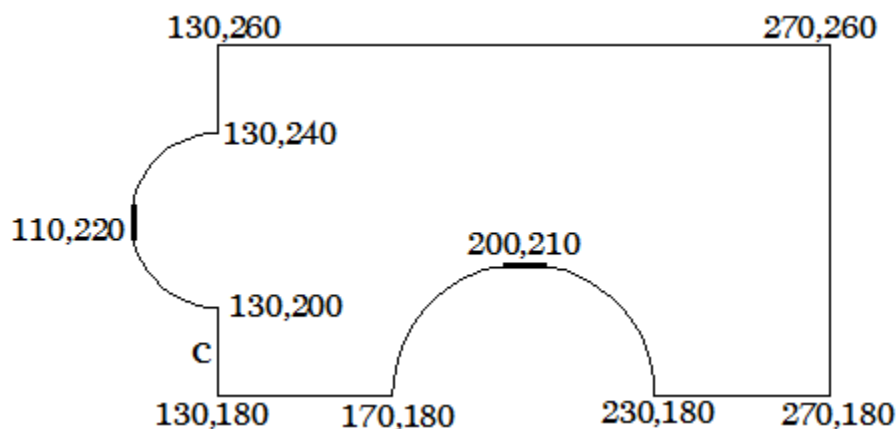


□ Drawing with the Polyline Tool:

Example: click on polyline tool icon and follow the step below:

- ▶ Specify start point: **130,180** then **Enter**
- ▶ Specify next point or [Arc / Halfwidth / Length / Undo / Width]: **170,180** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **a** (arc) then Enter
- ▶ Specify endpoint of arc or [Angle / CEnter / Close / Direction / Halfwidth / Line / Radius / Second pt / Undo / Width]: **s** (Second pt) then **Enter**
- ▶ Specify second point on arc: **200,210** then **Enter**

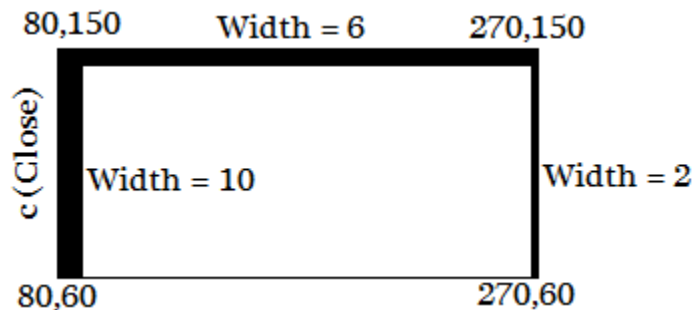
- ▶ Specify end point of arc: **230,180** then **Enter**
- ▶ Specify endpoint of arc or [Angle / CEnter / Close / Direction / Halfwidth / Line / Radius / Second pt / Undo / Width]: **l** (Line) then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **270,180** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **270,260** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **130,260** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **130,240** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **a** (arc) then **Enter**
- ▶ Specify endpoint of arc or [Angle / CEnter / Close / Direction / Halfwidth / Line / Radius / Second pt / Undo / Width]: **s** then **Enter**
- ▶ Specify second point on arc: **110,220** then **Enter**
- ▶ Specify end point of arc: **130,200** then **Enter**
- ▶ Specify endpoint of arc or [Angle / CEnter / Close / Direction / Halfwidth / Line / Radius / Second pt / Undo / Width]: **l** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **c** (Close) then **Enter**



Example: click on polyline tool icon and follow the step below:

- ▶ Specify start point: **80,60** then **Enter**
- ▶ Specify next point or [Arc / Halfwidth / Length / Undo / Width]: **270,60** then **Enter**

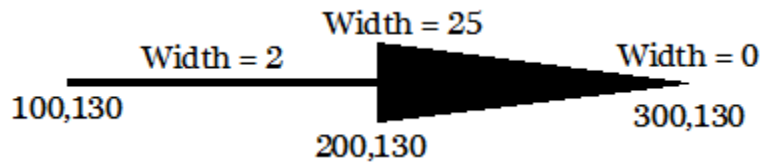
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **w** (width) then **Enter**
- ▶ Specify starting width <0>: **2** then **Enter**
- ▶ Specify ending width <2>: **Enter** (the same width)
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **270,150** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **w** then **Enter**
- ▶ Specify starting width <2>: **6** then **Enter**
- ▶ Specify ending width <6>: **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **80,150** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **w** then **Enter**
- ▶ Specify starting width <6>: **10** then **Enter**
- ▶ Specify ending width <10>: **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **c** (**Close**) then **Enter**



Example: click on polyline tool icon and follow the step below:

- ▶ Specify start point: **100,130** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **w** then **Enter**
- ▶ Specify starting width <0>: **2** then **Enter**
- ▶ Specify ending width <2>: **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **200,130** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **w** then **Enter**

- ▶ Specify starting width <2>: **25** then **Enter**
- ▶ Specify ending width <25>: **0** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **300,130** then **Enter**
- ▶ Specify next point or [Arc / Close / Halfwidth / Length / Undo / Width]: **Enter**



Ellipse Command : the shape of an ellipse is determined by two axes that define its length and width. The longer axis is called the major axis, and the shorter one is the minor axis.

- ▶ Select **Ellipse** from draw panel **OR**
- ▶ In command line write **ellipse** or **el** then Enter



Draw

- Center
- Axis, End**
- Elliptical Arc

Axis, End

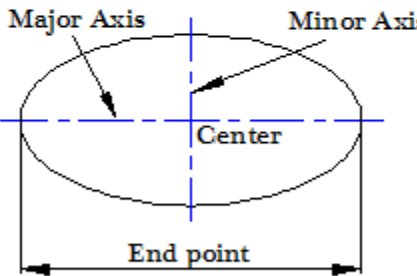
Creates an ellipse or an elliptical arc

The first two points of the ellipse determine the location and length of the first axis. The third point determines the distance between the center of the ellipse and the end point of the second axis.



ELLIPSE

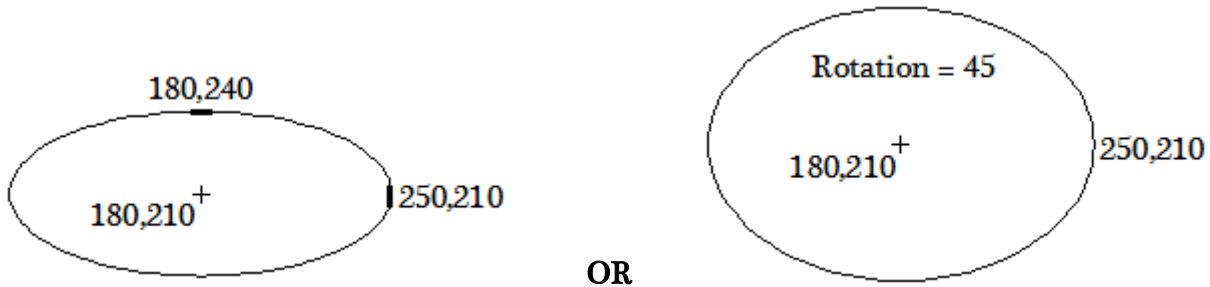
Press F1 for more help



□ Drawing with the Ellipse Tool:

Example: ellipse, **Center** click on ellipse tool icon and follow the step below:

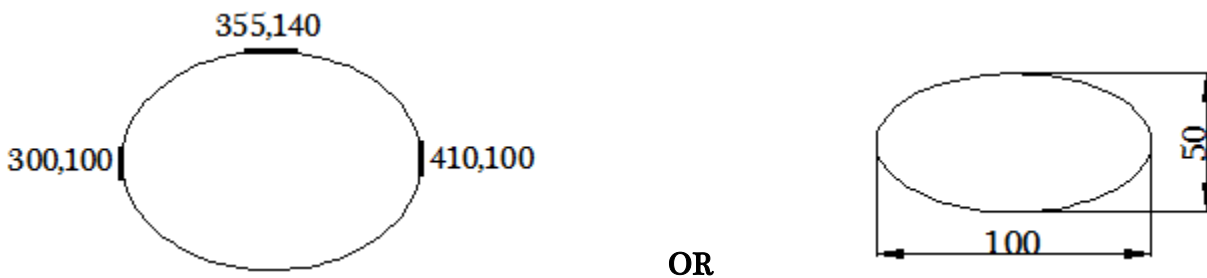
- ▶ Specify center of ellipse: **180,210** then **Enter**
- ▶ Specify endpoint of axis: **250,210** then **Enter**
- ▶ Specify distance to other axis or [Rotation]: **180,240** then **Enter**



- ▶ Specify center of ellipse: **180,210** then **Enter**
- ▶ Specify endpoint of axis: **250,210** then **Enter**
- ▶ Specify distance to other axis or [Rotation]: **r** then **Enter**
- ▶ Specify rotation around major axis: **45** then **Enter**

Example: ellipse, **Axis End** click on ellipse tool icon and follow the step below:

- ▶ Specify axis endpoint of ellipse or [Arc / Center]: **300,100** then **Enter**
- ▶ Specify other endpoint of axis: **410,100** then **Enter**
- ▶ Specify distance to other axis or [Rotation]: **355,140** then **Enter**

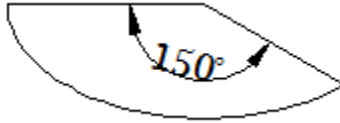


- ▶ Specify axis endpoint of ellipse or [Arc / Center]: **340,210** then **Enter**
- ▶ Drag to right (ortho on); Specify other endpoint of axis: **100** then **Enter**
- ▶ Specify distance to other axis or [Rotation]: **25** then **Enter**

Example: ellipse, **Elliptical Arc** click on ellipse tool icon and follow the step below:

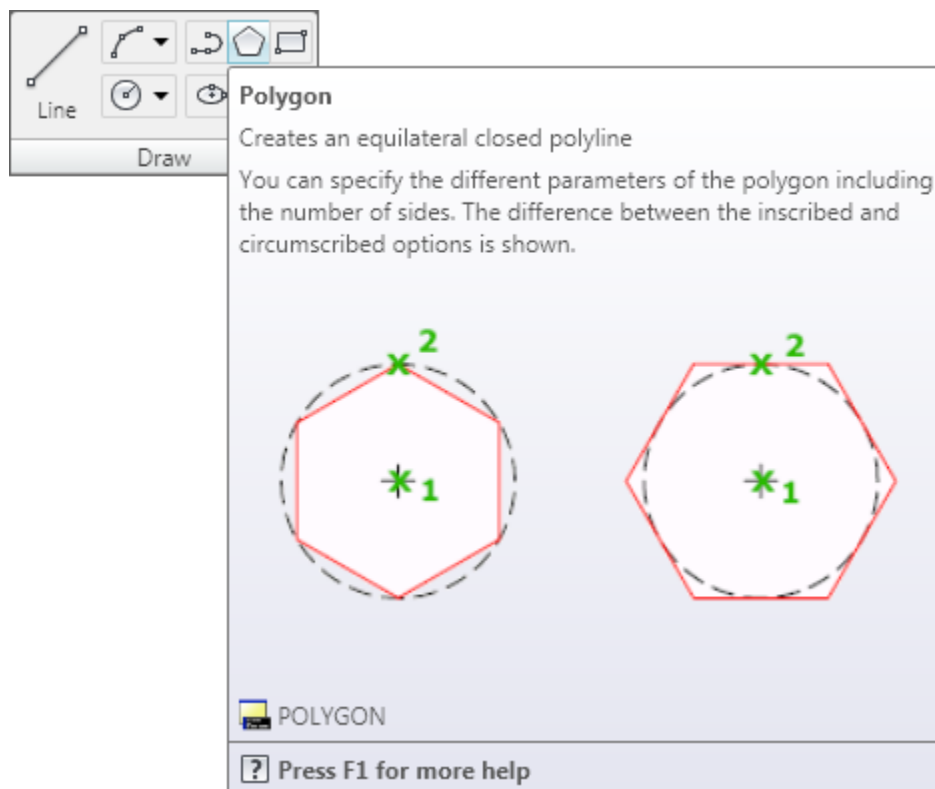
- ▶ Specify axis endpoint of elliptical arc or [Center]: **150,150** then **Enter**
- ▶ Specify other endpoint of axis: **290,150** then **Enter**

- ▶ Specify distance to other axis or [Rotation]: 220,190 then Enter
- ▶ Specify start angle or [Parameter]: 0 then Enter
- ▶ Specify end angle or [Parameter / Included angle]: 150 then Enter



Polygon Command : used to create regular polygon (pentagon, hexagon, heptagon, octagon, so on) quickly.

- ▶ Select **Polygon** from draw panel **OR**
- ▶ In command line write **polygon** or **pol** then Enter

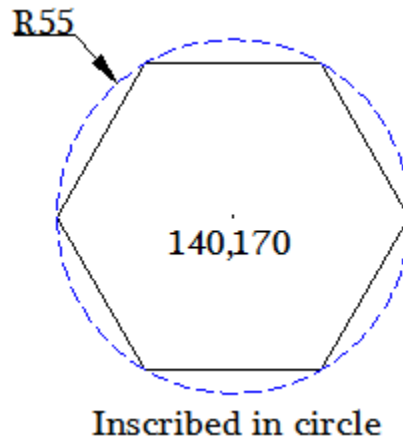


☐ Drawing with the Polygon Tool:

Example: click on polygon tool icon and follow the step below:

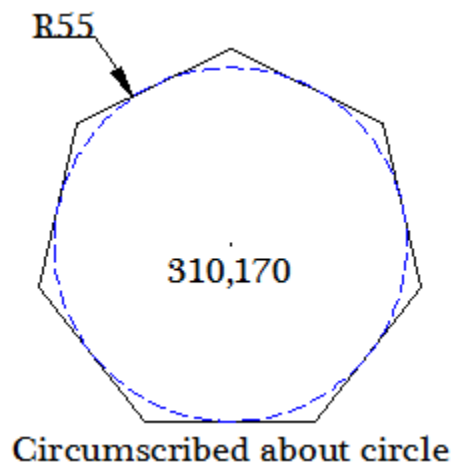
- ▶ Command: `_polygon` Enter number of sides <4>: **6** then **Enter**
- ▶ Specify center of polygon or [Edge]: **140,170** then **Enter**
- ▶ Enter an option [Inscribed in circle / Circumscribed about circle] <I>: **Enter**

- Specify radius of circle: 55 then **Enter**



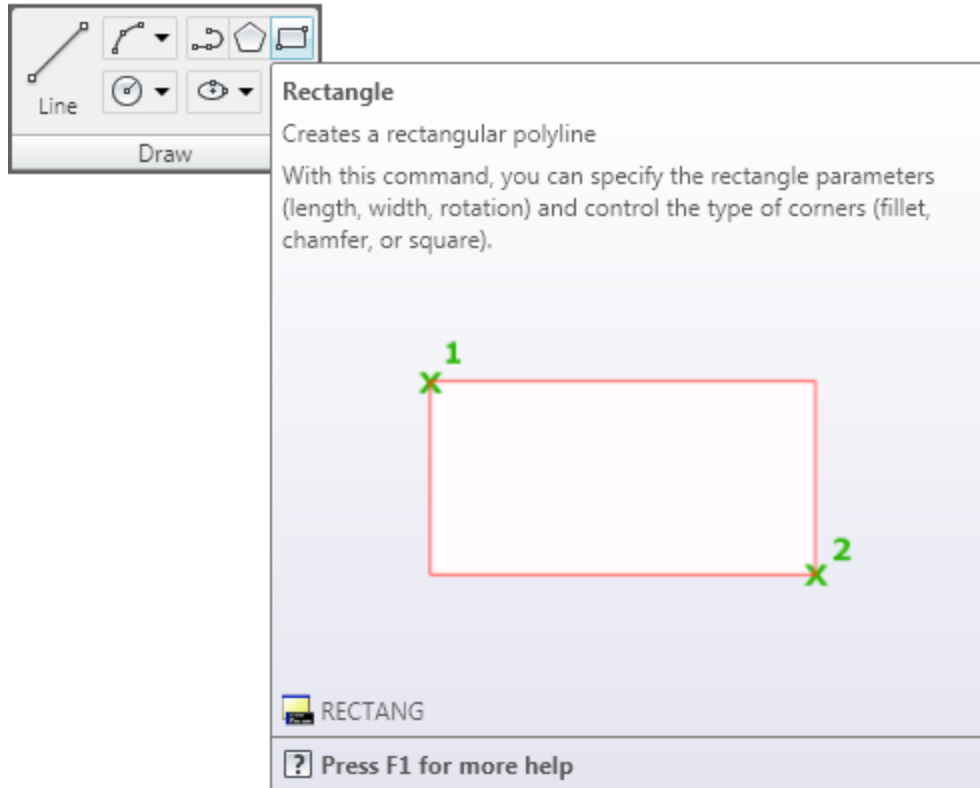
Example: click on polygon tool icon and follow the step below:

- Command: `_polygon` Enter number of sides <4>: 7 then **Enter**
- Specify center of polygon or [Edge]: **310,170** then **Enter**
- Enter an option [Inscribed in circle / Circumscribed about circle] <I>: **c** then **Enter**
- Specify radius of circle: 55 then **Enter**



Rectangle Command : used to create a rectangular polyline.

- Select **Rectangle** from draw panel **OR**
- In command line write **rectangle** or **rec** then Enter



□ Drawing with the Rectangle Tool:

Example: click on rectangle tool icon and follow the step below:

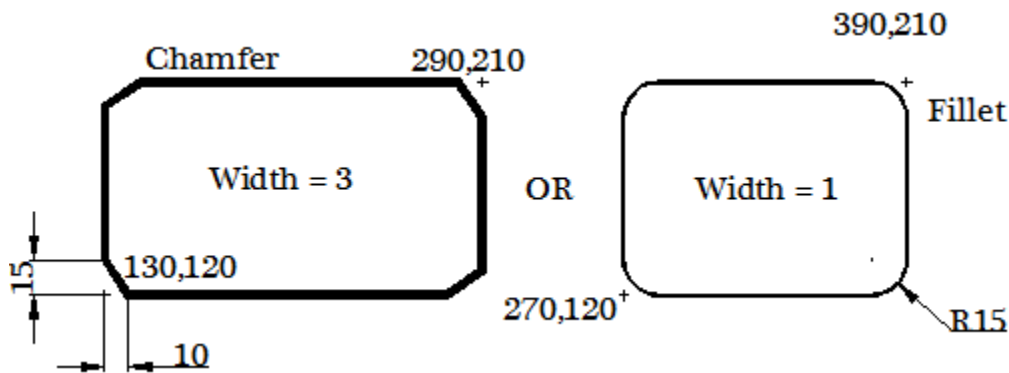
- ▶ Command: `_rectang` Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **90,120** then **Enter**
- ▶ Specify other corner point or [Area / Dimension / Rotation]: **270, 210** then **Enter**



Example: click on rectangle tool icon and follow the step below:

- ▶ Command: `_rectang` Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **w** then **Enter**

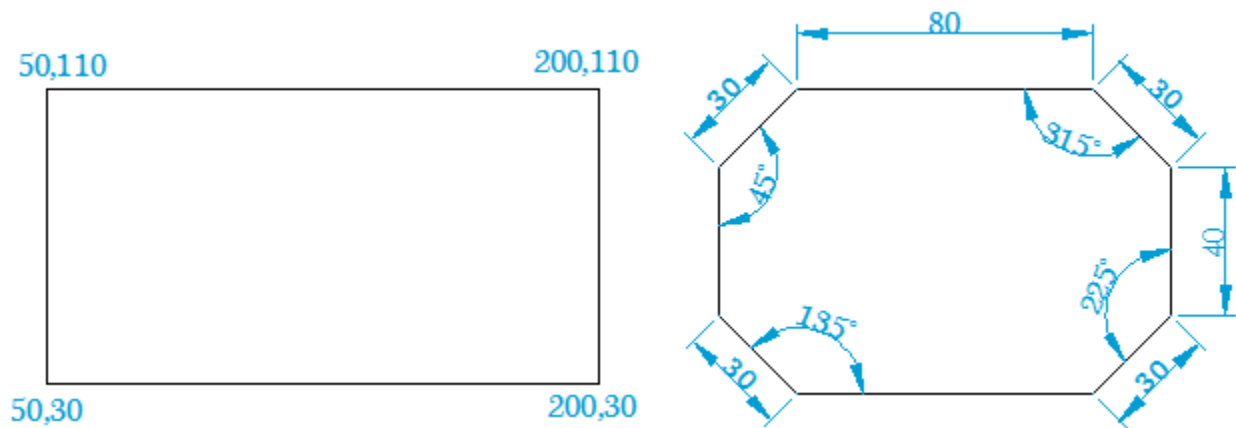
- ▶ Specify line width for rectangles <0>: **3** then **Enter**
- ▶ Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **c** then **Enter**
- ▶ Specify first chamfer distance for rectangles <0>: **15** then **Enter**
- ▶ Specify second chamfer distance for rectangles <15>: **10** then **Enter**
- ▶ Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **130,120** then **Enter**
- ▶ Specify other corner point or [Area / Dimension / Rotation]: **290,210** then **Enter**



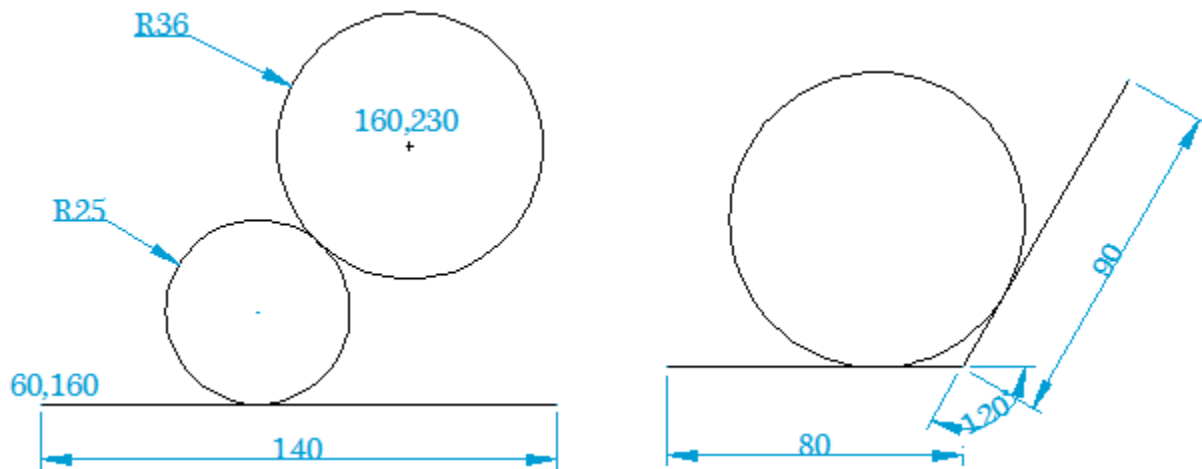
- ▶ Command: `_rectang` Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **w** then **Enter**
- ▶ Specify line width for rectangles <0>: **1** then **Enter**
- ▶ Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **f** then **Enter**
- ▶ Specify fillet radius for rectangles <0>: **15** then **Enter**
- ▶ Specify first corner point or [Chamfer / Elevation / Fillet / Thickness / Width]: **270,120** then **Enter**
- ▶ Specify other corner point or [Area / Dimension / Rotation]: **390,210** then **Enter**

Exercise

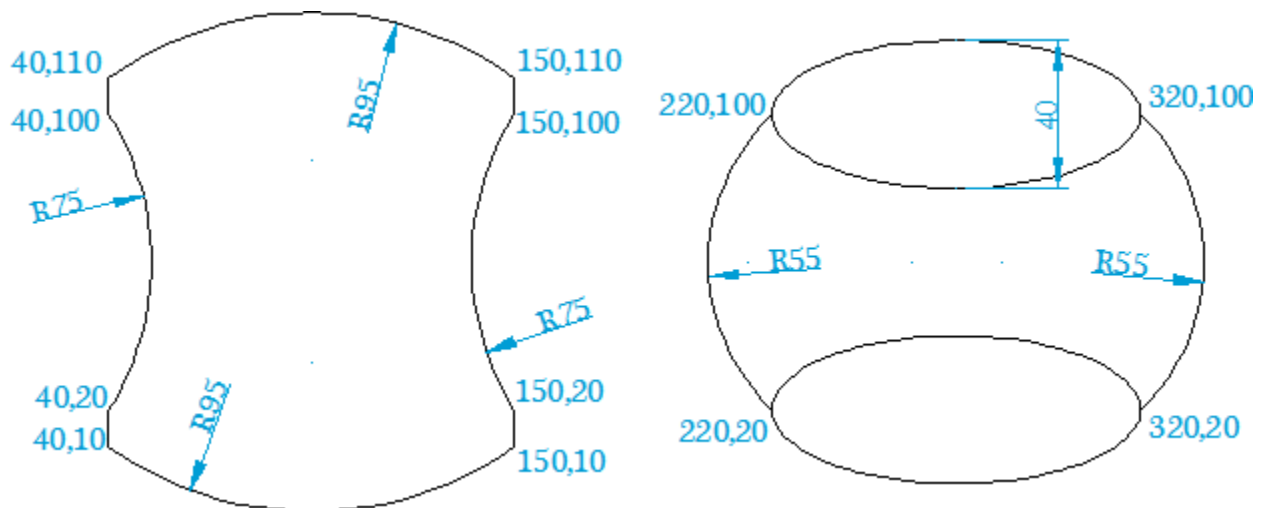
- Using the **Line** tool, construct the outline shown below.



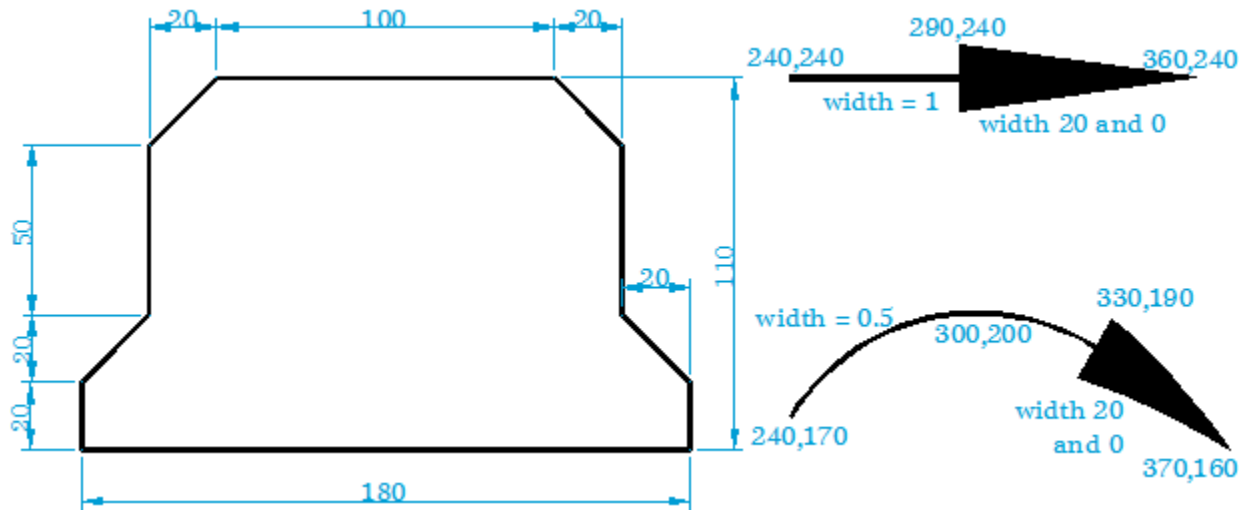
- Using the **Circle** and **Line** tools, construct the outline shown below.



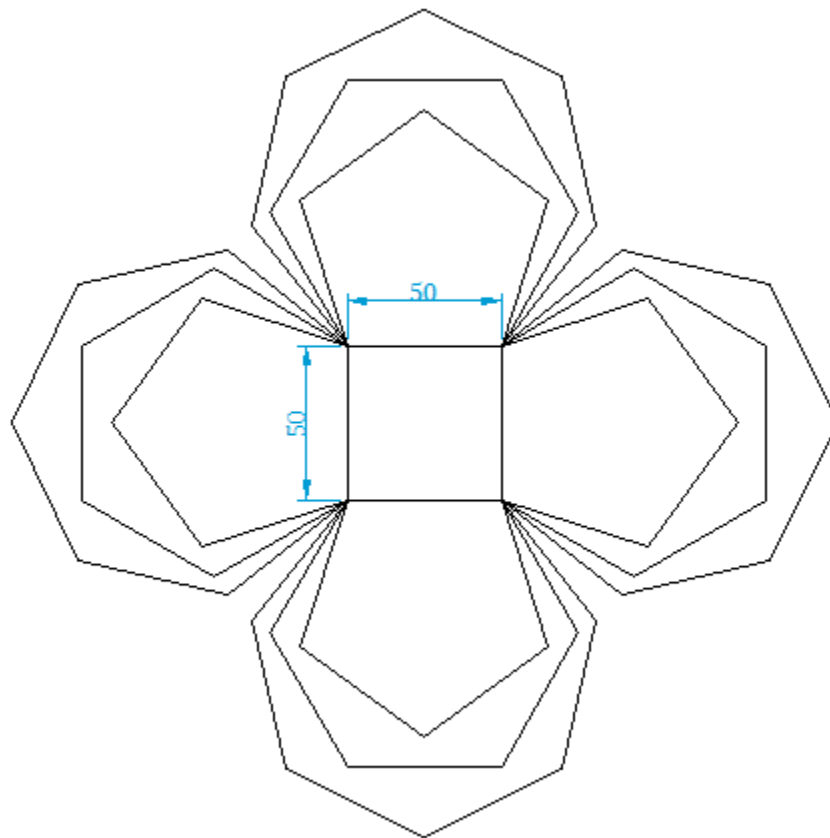
- With the **Line Arc** and **Ellipse** tools, construct the outline shown below.



4. Using the **Polyline** tool, construct the outline given below.



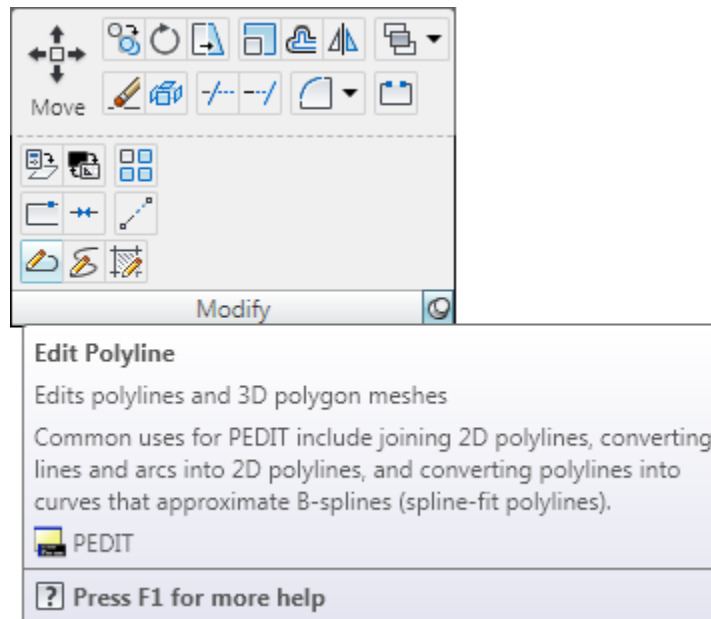
5. Figure shows below a rectangle in the form of a square with polygons (5, 6 and 7) along each edge. Using the **Dimensions** prompt of the **Rectangle** tool construct the square. Then, using the **Edge** prompt of the **Polygon** tool, add the four polygons. Use the **Object Snap endpoint** to ensure the polygons are in their exact positions.



MODIFY COMMAND

Edit Polyline Command : used to converting lines and arcs into 2D polylines; changing the shape of polyline objects; closing and opening of polyline objects by moving, adding or deleting individual vertices.

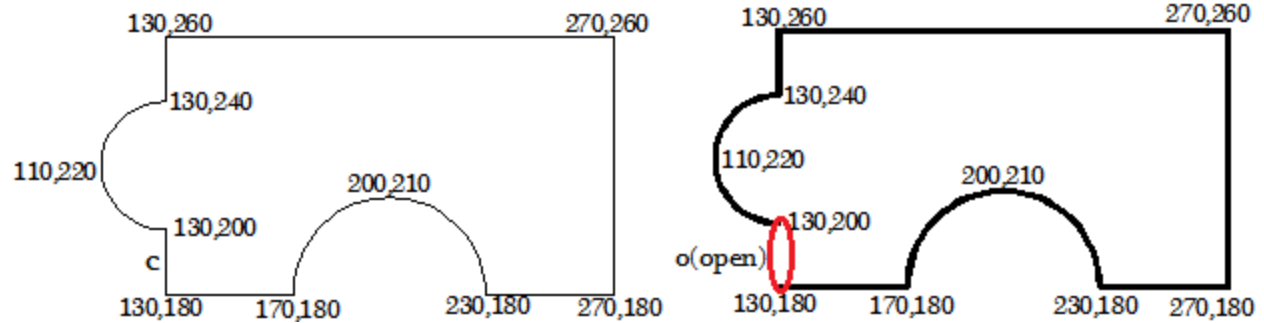
- ▶ Select **Edit Polyline** from modify panel **OR**
- ▶ In command line write **pedit** or **pe** then Enter



□ Edit Polyline Tool:

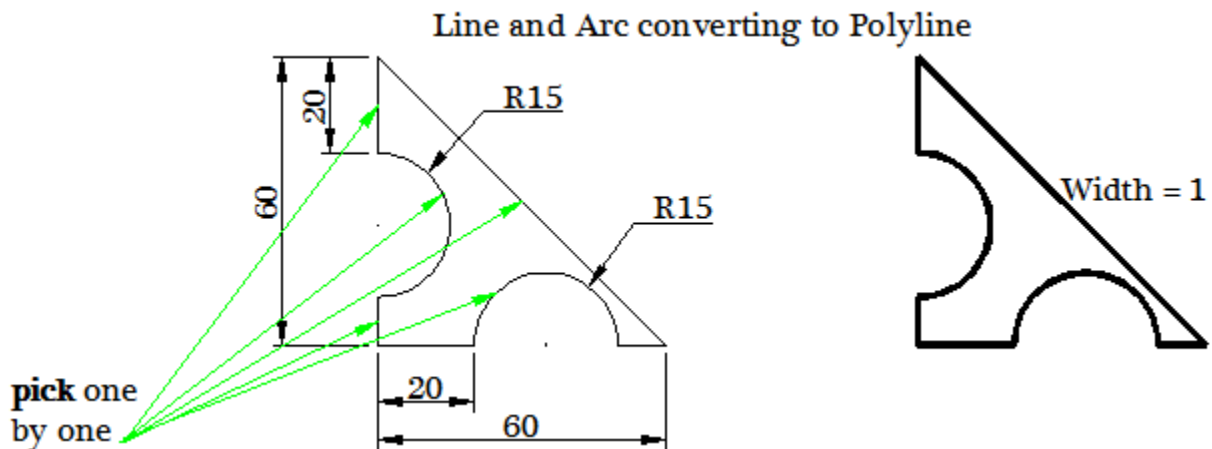
Example: click on edit polyline tool icon and follow the step below:

- ▶ Command: **_pedit** select polyline or [Multiple]: **click (pick)**
- ▶ Enter an option [Open / Join / Width / Edit vertex / Fit / Spline / Decurve / Ltype gen / Undo]: **w** then **Enter**
- ▶ Specify new width for all segments: **1.5** then **Enter**
- ▶ Enter an option [Open / Join / Width / Edit vertex / Fit / Spline / Decurve / Ltype gen / Undo]: **o** then **Enter**
- ▶ Enter an option [Open / Join / Width / Edit vertex / Fit / Spline / Decurve / Ltype gen / Undo]: **Enter**



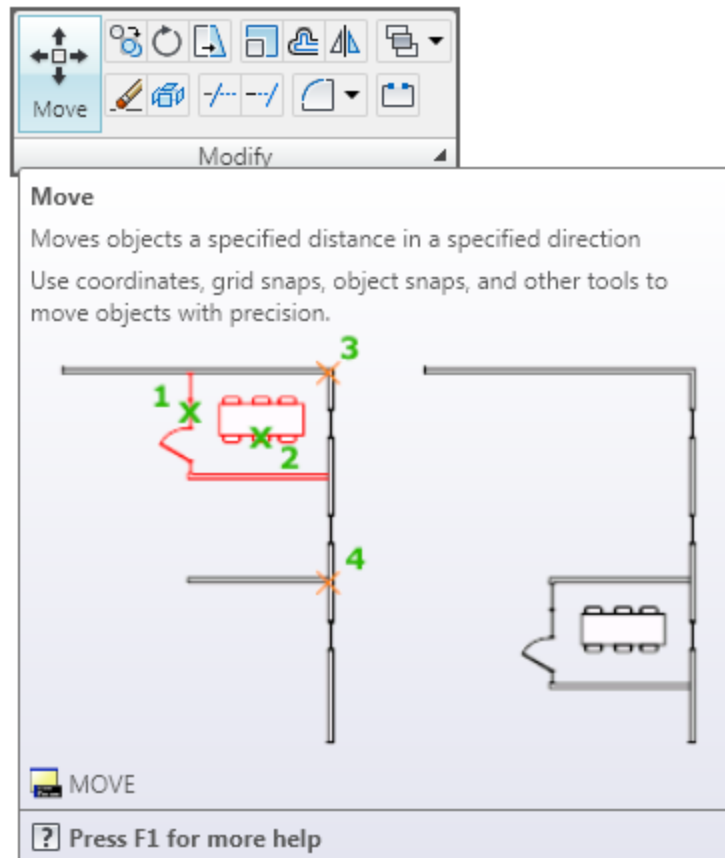
Example: click on edit polyline tool icon and follow the step below:

- ▶ Command: `_pedit` select polyline or [Multiple]: **click (pick)**
- ▶ **Pick** line or arc; object selected is not a polyline Do you want to turn it into one? <Y> press **Enter** (to change polyline)
- ▶ Enter an option [Close / Join / Width / Edit vertex / Fit / Spline / Decurve / Ltype gen / Undo]: **w** then **Enter**
- ▶ Specify new width for all segments: **1** then **Enter**
- ▶ Enter an option [Close / Join / Width / Edit vertex / Fit / Spline / Decurve / Ltype gen / Undo]: **Enter**



Move Command  : used to move objects at a specified distance and direction from the originals.

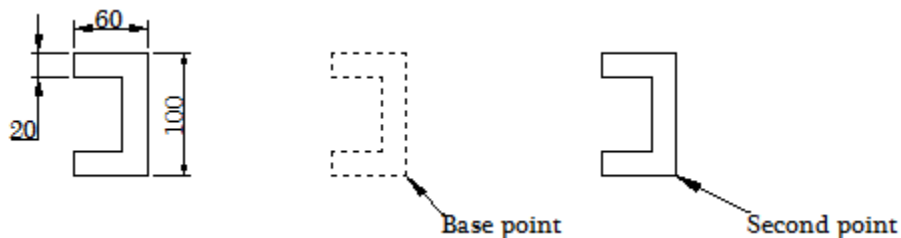
- ▶ Select **Move** from modify panel **OR**
- ▶ In command line write **move** or **m** then Enter



□ Move Tool:

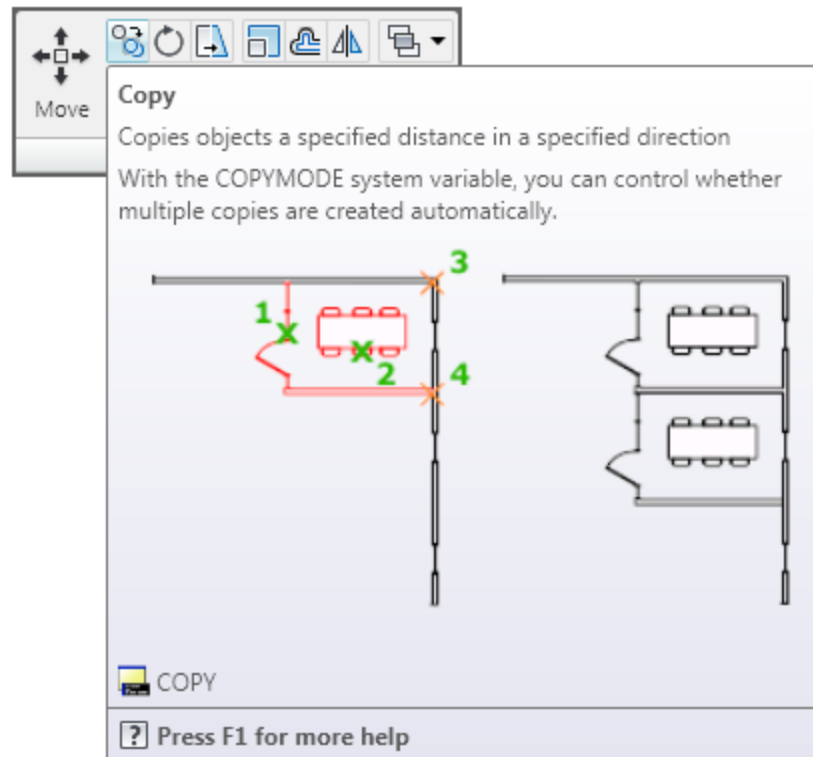
Example: click on move tool icon and follow the step below:

- ▶ Command: `_move` Select objects: **pick**, Specify opposite corner: 8 found and **Enter**
- ▶ Specify base point or [Displacement] <Displacement>: **click (pick)**
- ▶ Specify second point or <use first point as displacement>: **click**



Copy Command  : used to create duplicates of objects at a specified distance and direction from the originals.

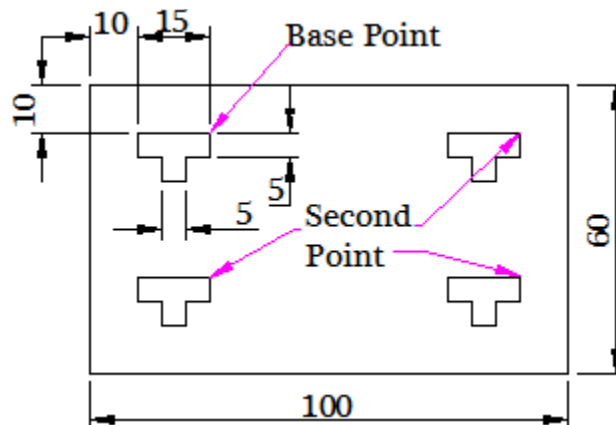
- ▶ Select **Copy** from modify panel **OR**
- ▶ In command line write **copy** or **co** then Enter



□ Copy Tool:

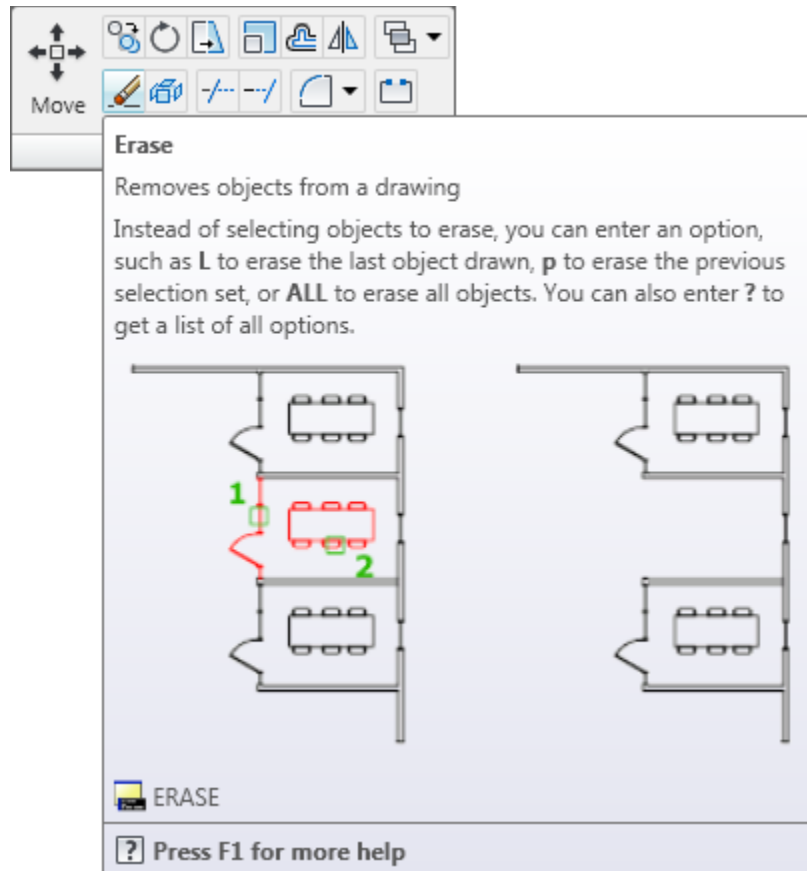
Example: click on copy tool icon and follow the step below:

- ▶ Command: `_copy` select objects: **pick** (1 found, 8 total) then **Enter**
- ▶ Specify base point or [Displacement / mode] <Displacement>: **pick**
- ▶ Specify second point or <use first point as displacement>: **pick**
- ▶ Specify second point or [Exit / Undo] <Exit>: **pick**
- ▶ Specify second point or [Exit / Undo] <Exit>: **pick**
- ▶ Specify second point or [Exit / Undo] <Exit>: **Enter**



Erase Command : used to removed or deleted object or objects that have been incorrectly drawn.

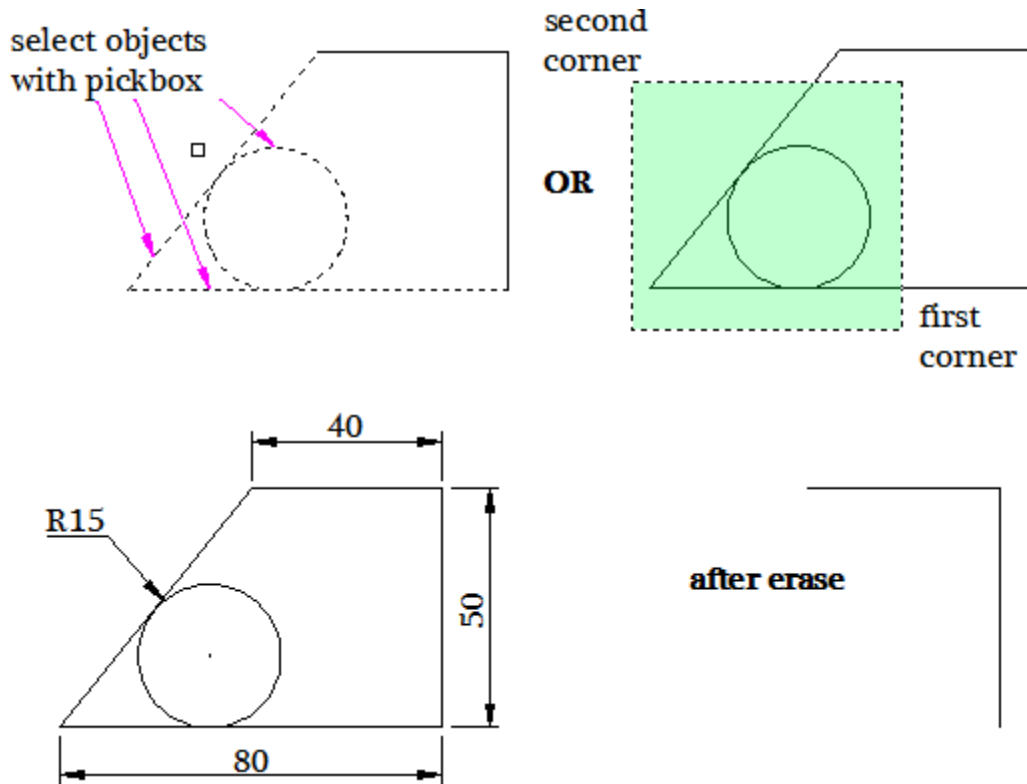
- ▶ Select **Erase** from modify panel **OR**
- ▶ In command line write **erase** or **e** then Enter



□ **Erase Tool:**

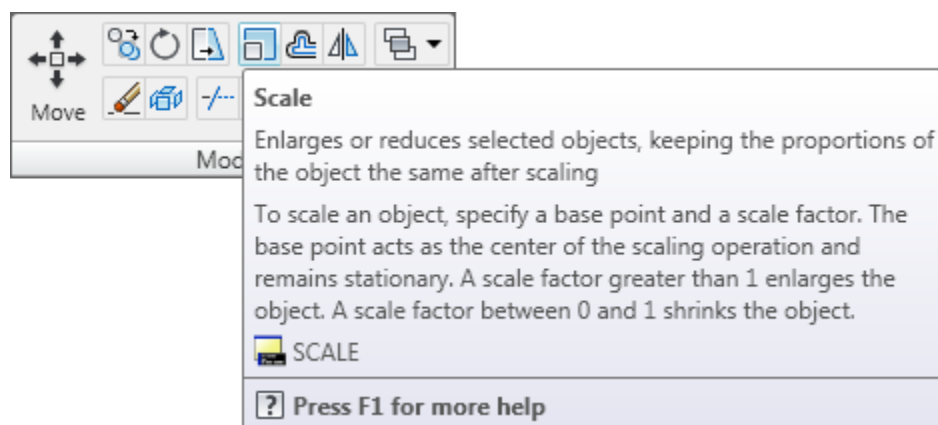
Example: click on erase tool icon and follow the step below:

- ▶ Command: **_erase** Select object: **pick**, 1 found, 3 total
- ▶ Select objects: press **Enter** **OR**
- ▶ Command: **_erase** Select object: **pick** (in first corner)
- ▶ Command: **_erase** Select object: Specify opposite corner: **pick** (in second corner), 3 found
- ▶ Select objects: press **Enter**



Scale Command : used to enlarge or reduce selected objects, keeping the proportions of the object the same after scaling.

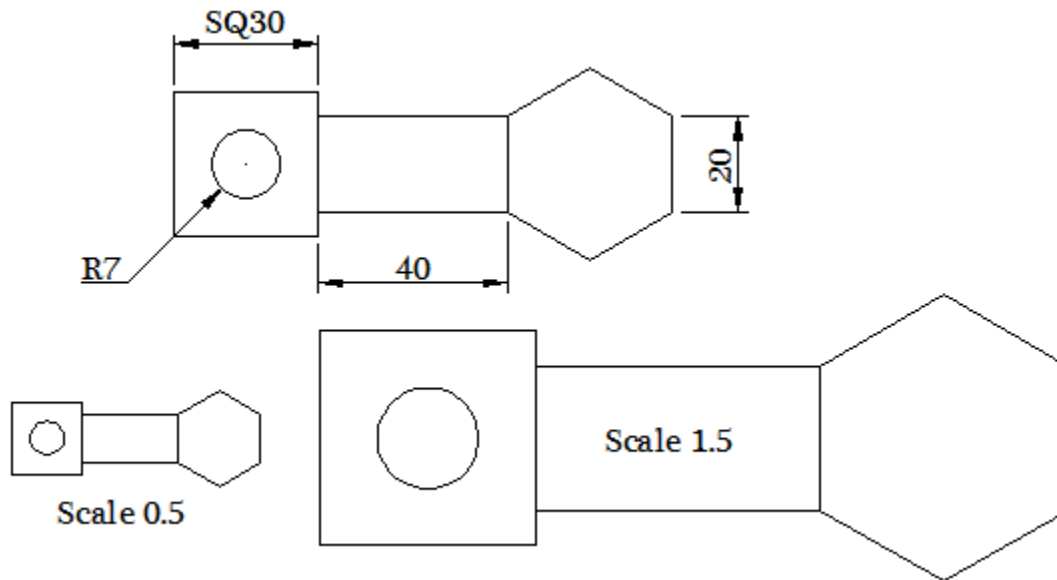
- Select **Scale** from modify panel **OR**
- In command line write **scale** or **sc** then Enter



□ Scale Tool:

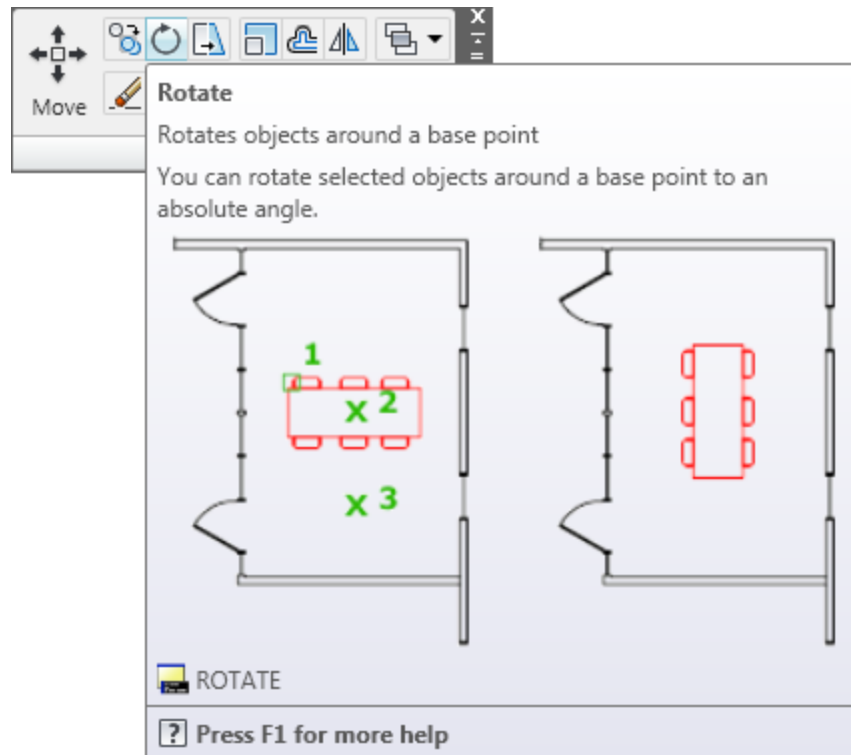
Example: click on scale tool icon and follow the step below:

- ▶ Command: `_scale` Select objects: **pick**
- ▶ Specify opposite corner: **pick** 9 found
- ▶ Select objects: press **Enter**
- ▶ Specify base point: **pick**
- ▶ Specify scale factor or [Copy / Reference] <1>: **0.5** (for reduce) **OR 1.5** (for enlarge) then **Enter**



Rotate Command : you can rotate objects in your drawing around a specified base point. To determine the angle of rotation, you can enter an angle value, drag using the cursor, or specify a reference angle to align to an absolute angle.

- ▶ Select **Rotate** from modify panel **OR**
- ▶ In command line write **rotate** or **ro** then Enter



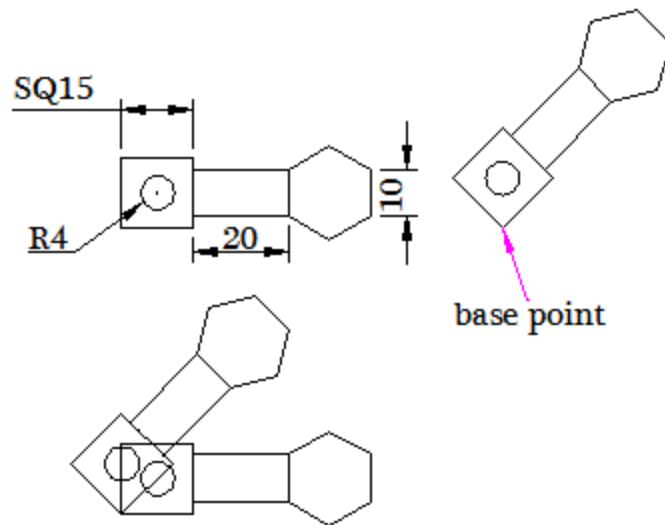
□ Rotate Tool:

Example: click on rotate tool icon and follow the step below:

- ▶ Command: `_rotate` Current positive angle in UCS: ANGDIR=counterclockwise ANGBASE=0 Select objects: **pick**
- ▶ Select objects: Specify opposite corner: **pick** 9 found
- ▶ Select objects: press **Enter**
- ▶ Specify base point: **pick**
- ▶ Specify rotation angle or [copy / Reference] <0>: **45** then **Enter**

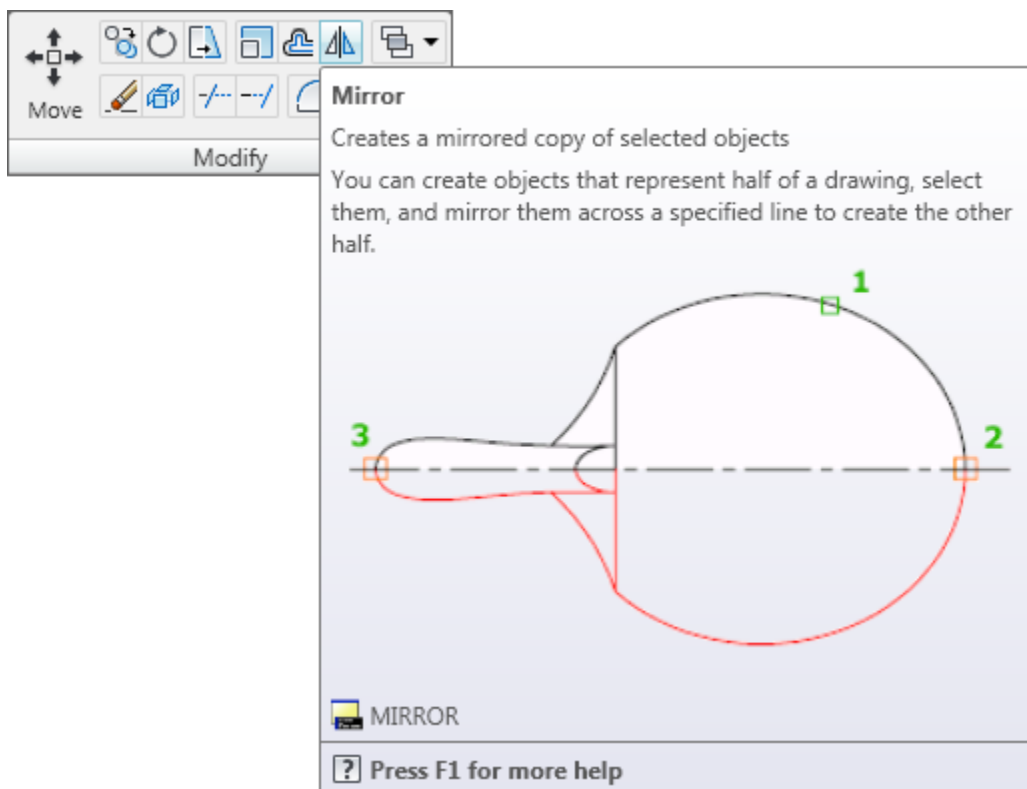
OR

- ▶ Specify rotation angle or [copy / Reference] <0>: **c** then **Enter**
- ▶ Specify rotation angle or [Copy / Reference] <0>: **45** then **Enter**



Mirror Command : used to create symmetrical objects because you can quickly draw half the object and then mirror it instead of drawing the entire object.

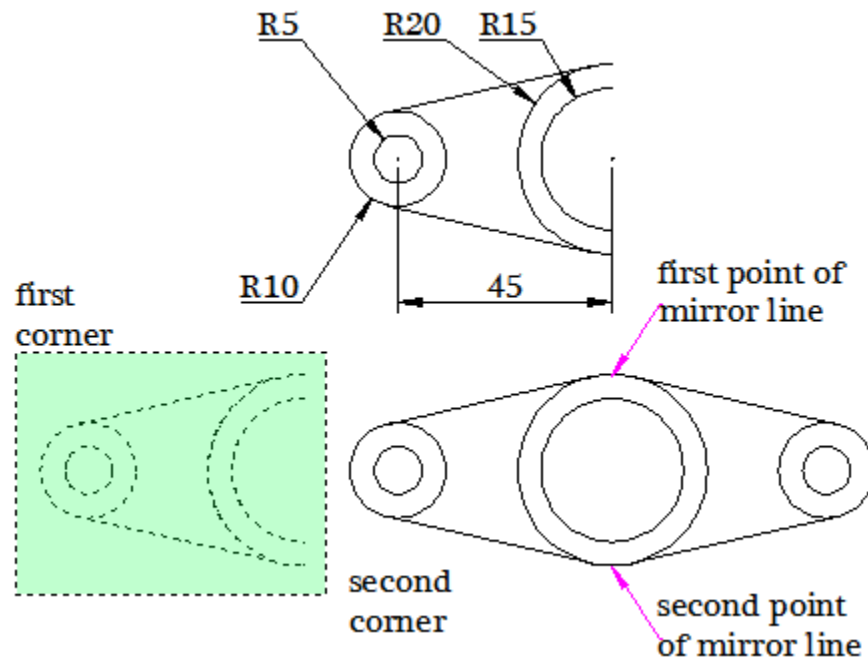
- ▶ Select **Mirror** from modify panel OR
- ▶ In command line write **mirror** or **m** then Enter



□ Mirror Tool:

Example: click on mirror tool icon and follow the step below:

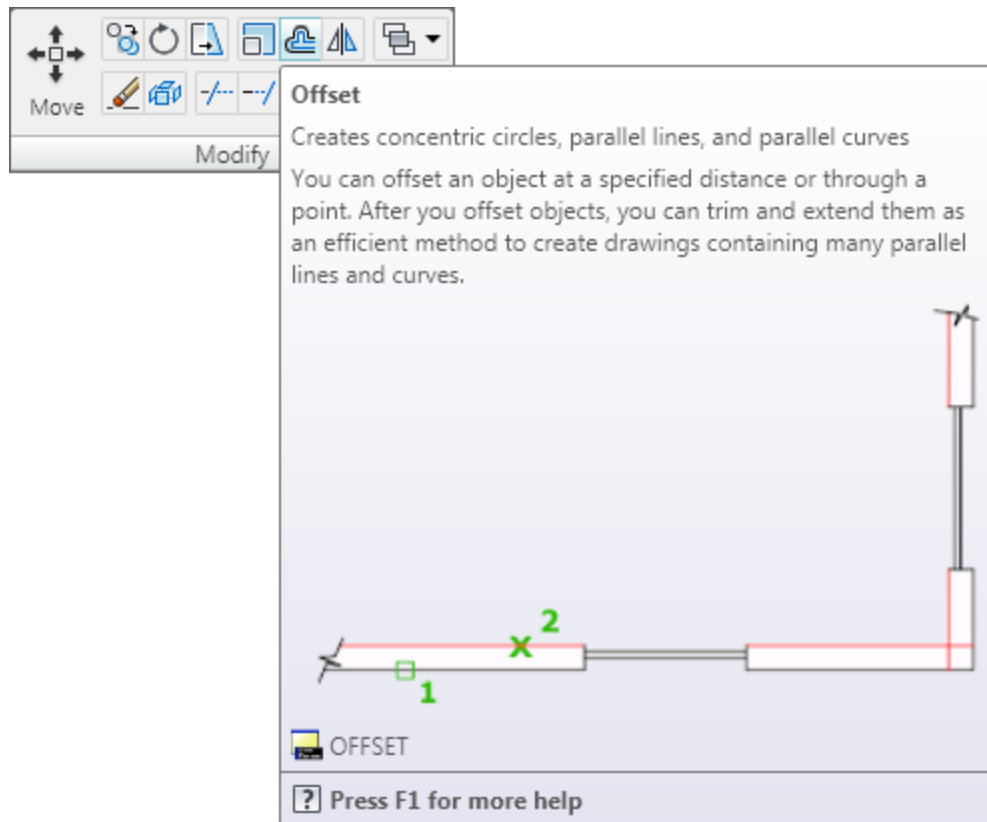
- ▶ Command: `_mirror` Select objects: **pick**
- ▶ Specify opposite corner: **pick** 6 found
- ▶ Select objects: press **Enter**
- ▶ Specify first point of mirror line: **pick**
- ▶ Specify second point of mirror line: **pick**
- ▶ Erase source objects? [Yes /No] <N>: press **Enter**



Offset Command : used to create a new object that shape parallels the shape of the original object. Offsetting a circle or an arc creates a larger or smaller circle or arc, depending on which side you specify for the offset.



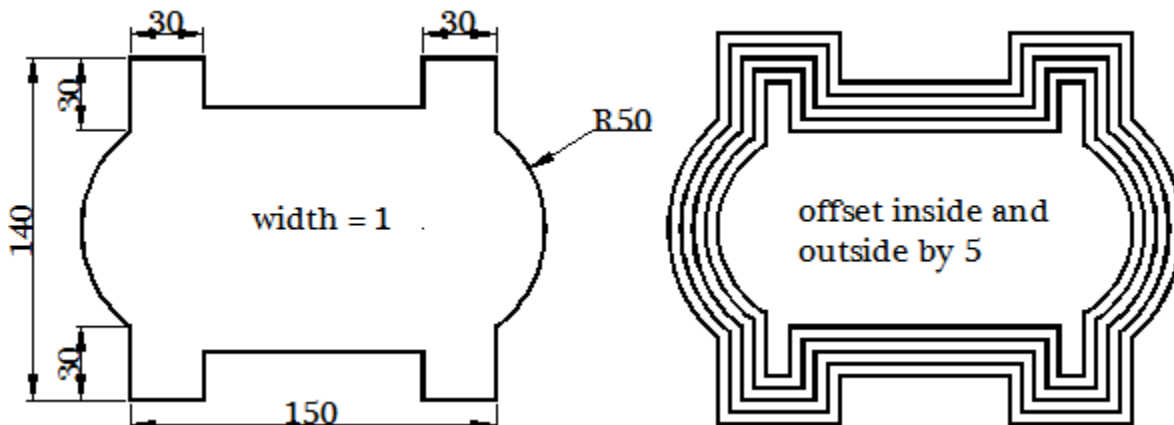
- ▶ Select **Offset** from modify panel **OR**
- ▶ In command line write **offset** or **o** then Enter



□ Offset Tool:

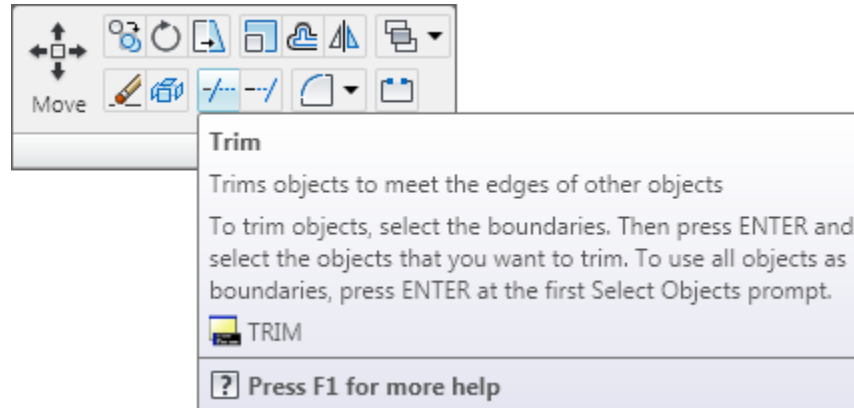
Example: click on offset tool icon and follow the step below:

- ▶ Command: Specify offset distance or [Through / Erase / Layer] <Through>: **5** then **Enter**
- ▶ Select object to offset or [Exit / Undo] <Exit>: **pick (click)**
- ▶ Specify point on side to offset or [Exit / Multiple / Undo] <Exit>: **pick** (inside or outside the object)
- ▶ Select object to offset or [Exit / Undo] <Exit>: **e** then **Enter**



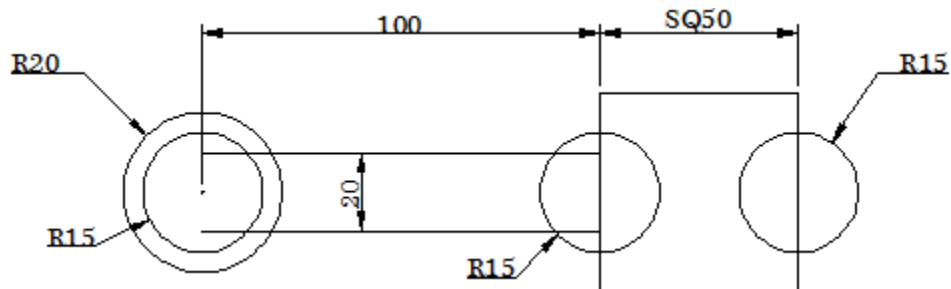
Trim Command : used to precisely fit the end of object at boundary edges defined by other objects.

- ▶ Select **Trim** from modify panel **OR**
- ▶ In command line write **trim** or **tr** then Enter

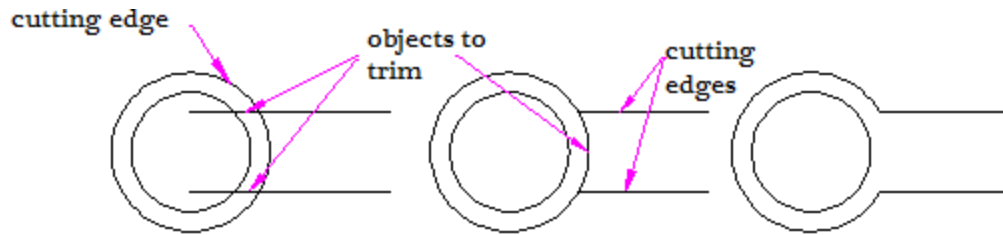


□ Trim Tool:

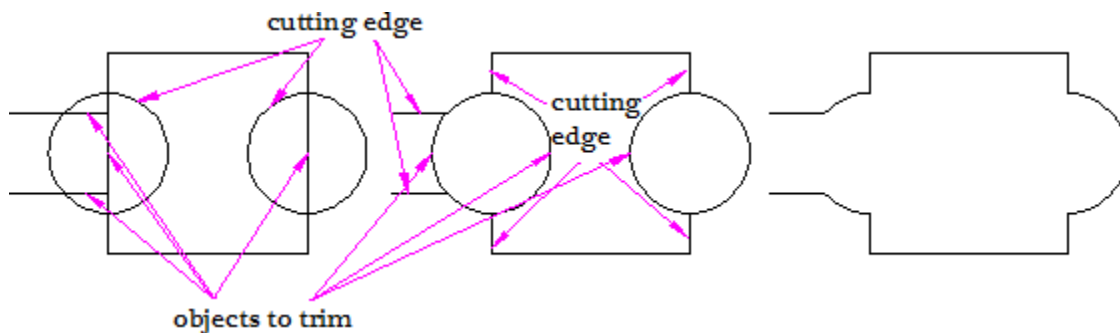
Example: click on trim tool icon and follow the step below:



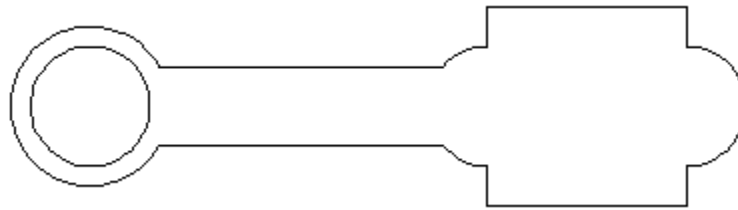
- ▶ Command: `_trim` Current settings: Projection=UCS, Edge=None Select cutting edges...
- ▶ Select objects or <select all>: **pick** (click on circle R20), 1 found
- ▶ Select objects: press **Enter**
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (first object, line to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (second object, line to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **Enter**



- ▶ Command: `_trim` Current settings: Projection=UCS, Edge=None Select cutting edges...
- ▶ Select objects or <select all>: **pick** (click on two circle), 1 found, 2 total
- ▶ Select objects: press **Enter**
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (first object, line to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (second object, line to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (third object, line to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (fourth object, line to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **Enter**
- ▶ Command: `_trim` Current settings: Projection=UCS, Edge=None Select cutting edges...
- ▶ Select objects or <select all>: **pick** (click on cutting edge), 1 found, 6 total
- ▶ Select objects: press **Enter**
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **pick** (objects to trim)
- ▶ Select object to trim or shift – select to extend or [Fence / Crossing / Project / Edge / eRase / Undo]: **Enter**

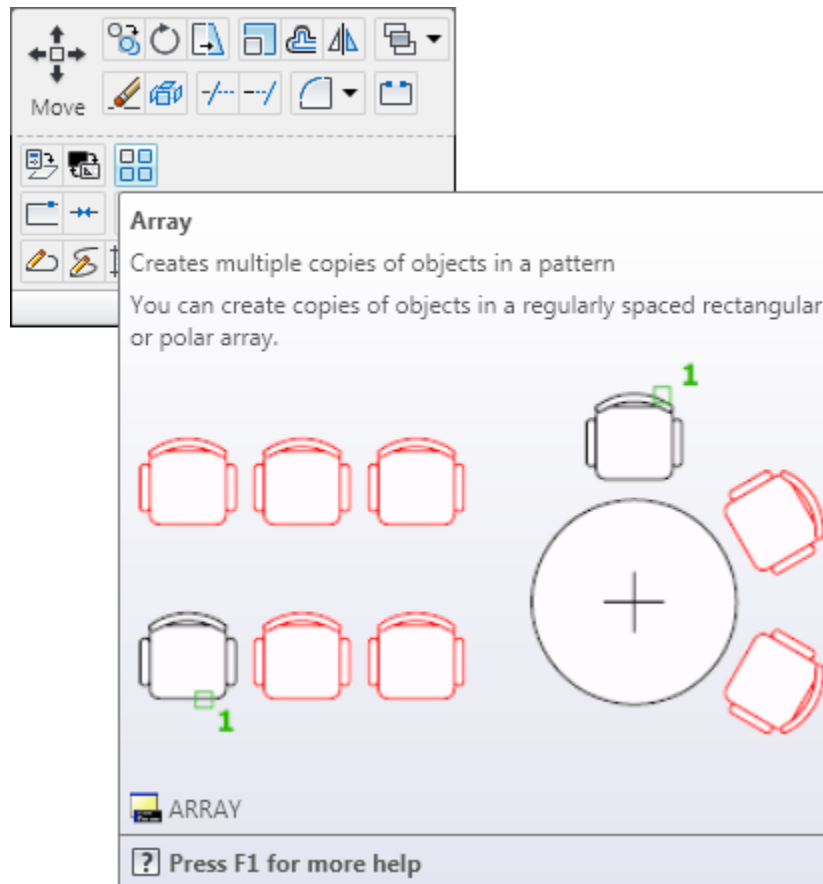


► Result



Array Command : used to create multiple copies of objects in a rectangular or polar (circular) pattern.

- Select **Array** from modify panel **OR**
- In command line write **array** or **ar** then Enter

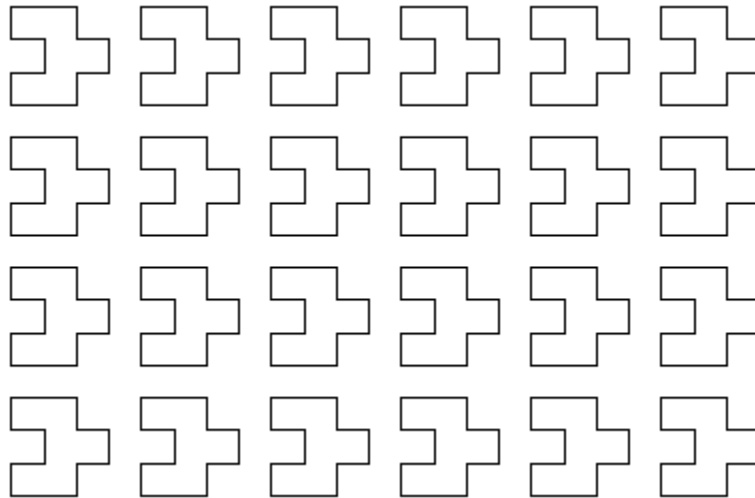
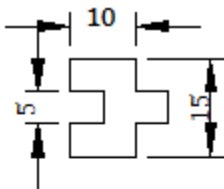
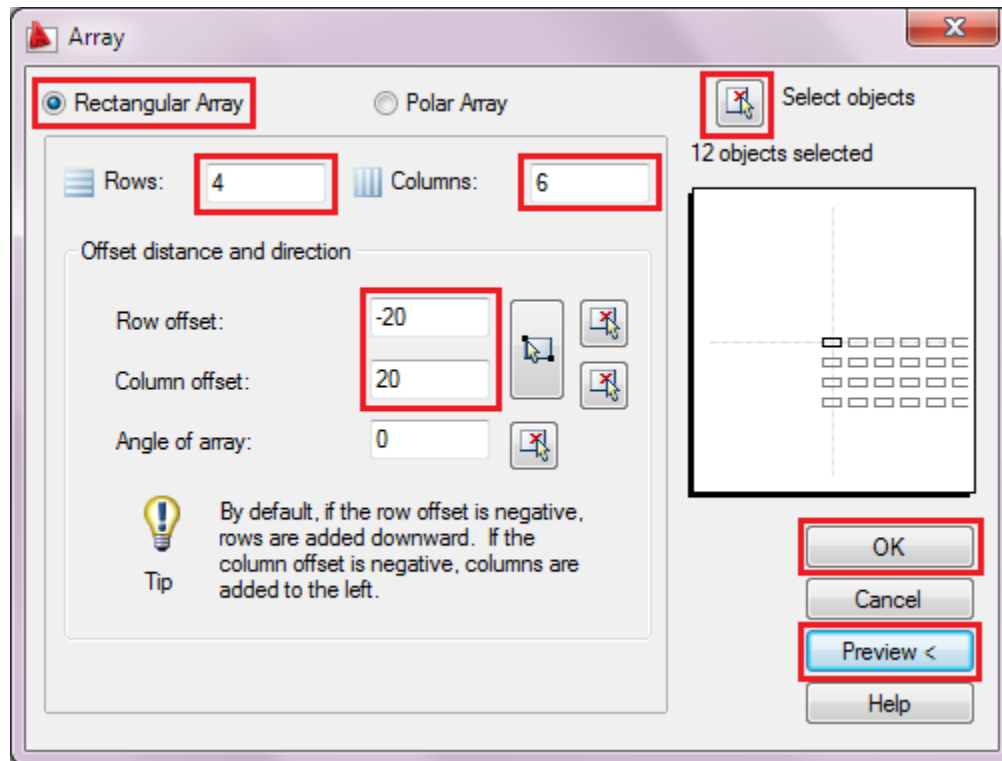


□ **Array Tool:**

Example: Rectangular Array, click on array tool icon and follow the step below:

- Array dialog box appear

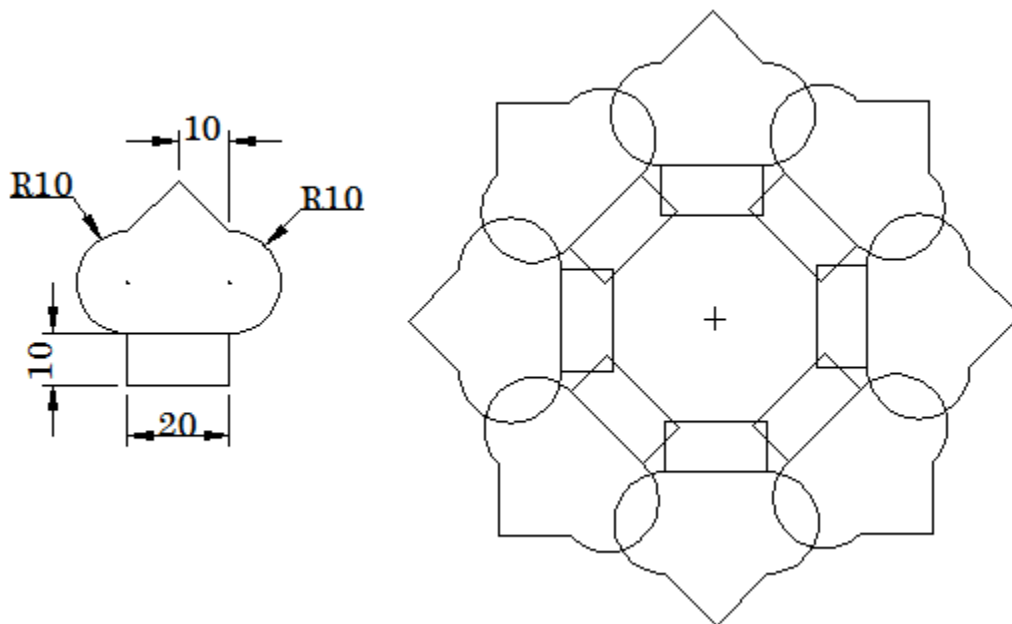
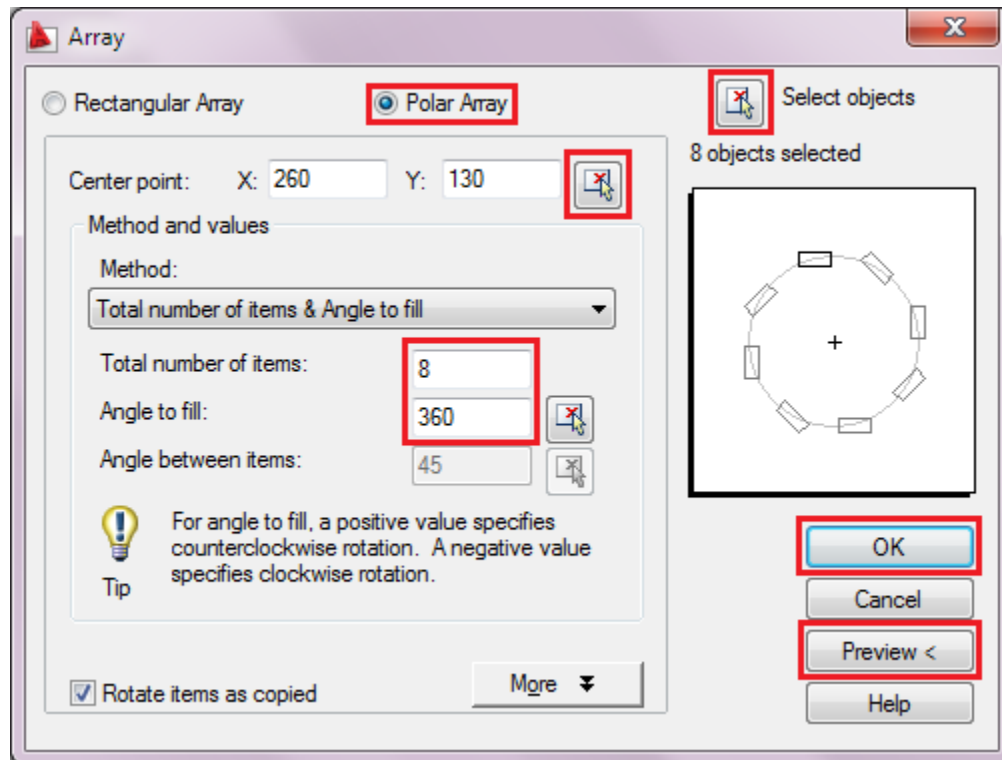
- ▶ Select object  Click then **Enter**
- ▶ Entry the value of Rows: **4**, Columns: **6**, Row offset: – **20** and Column offset: **20** then **OK**



Example: Polar Array, click on array tool icon and follow the step below:

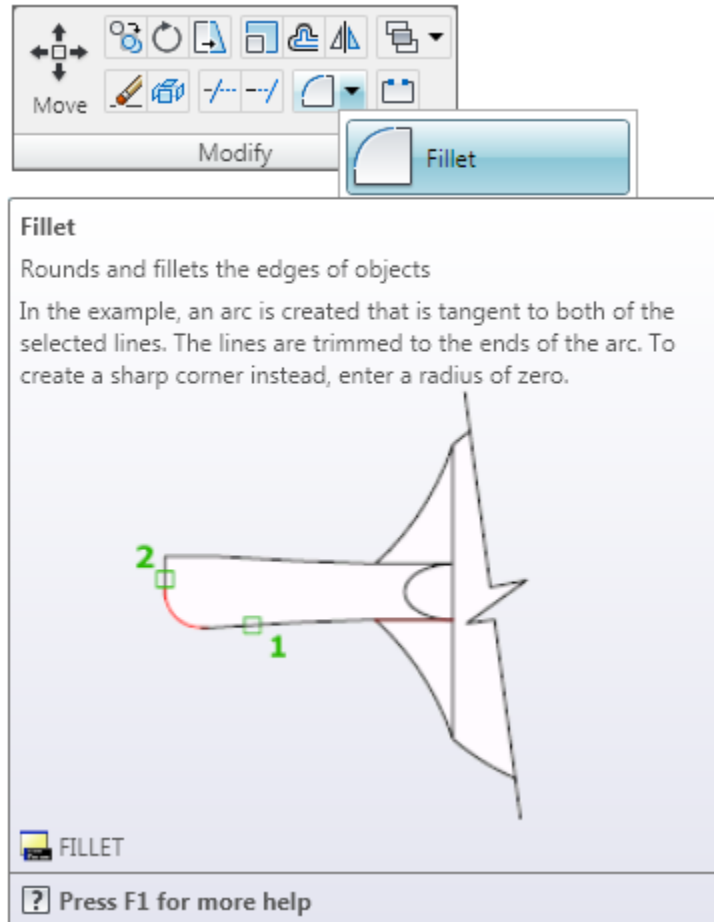
- ▶ Array dialog box appear
- ▶ Select object  Click then **Enter**

- Entry the value Pick Center Point  (X:260, Y:140), Total number of items: **8**, Angle to fill: **360** then **OK**



Fillet Command : used to connect two objects with an arc that is tangent to the objects and has a specified radius.

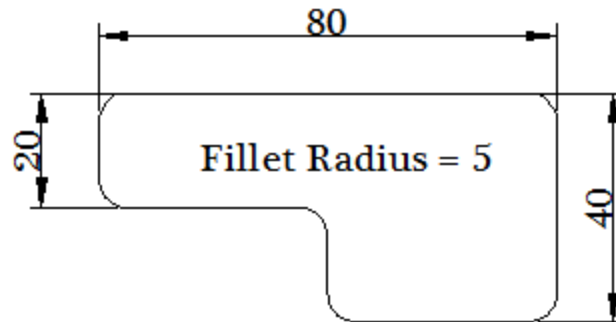
- ▶ Select **Fillet** from modify panel **OR**
- ▶ In command line write **fillet** or **f** then Enter



□ **Fillet Tool:**

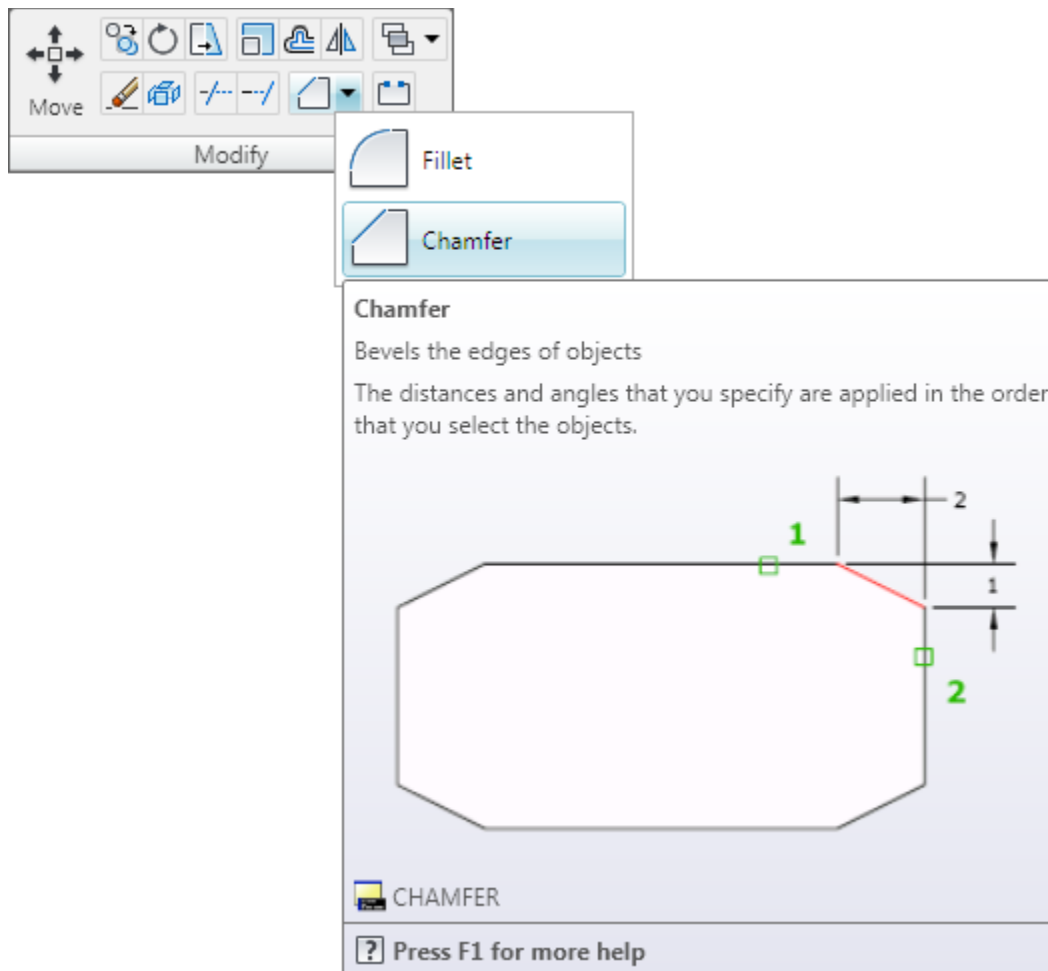
Example: click on fillet tool icon and follow the step below:

- ▶ Command: **_fillet** Current settings: Mode = TRIM, Radius = 0 Select first object or [Undo / Polyline / Radius / Trim / Multiple]: **r** then **Enter**
- ▶ Specify fillet radius <0>: **5** then **Enter**
- ▶ Select first object or [Undo / Polyline / Radius / Trim/ Multiple]: **pick**
- ▶ Select second object or shift – select to apply corner: **pick**
- ▶ Repeated same step for other corners



Chamfer Command : used to connect two objects to meet in a flattened or beveled corner (with an angled line).

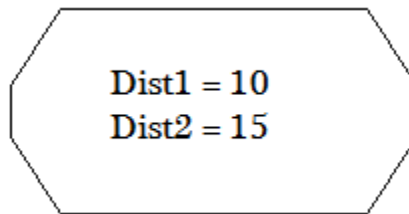
- Select **Chamfer** from modify panel **OR**
- In command line write **chamfer** or **cha** then Enter



□ Chamfer Tool:

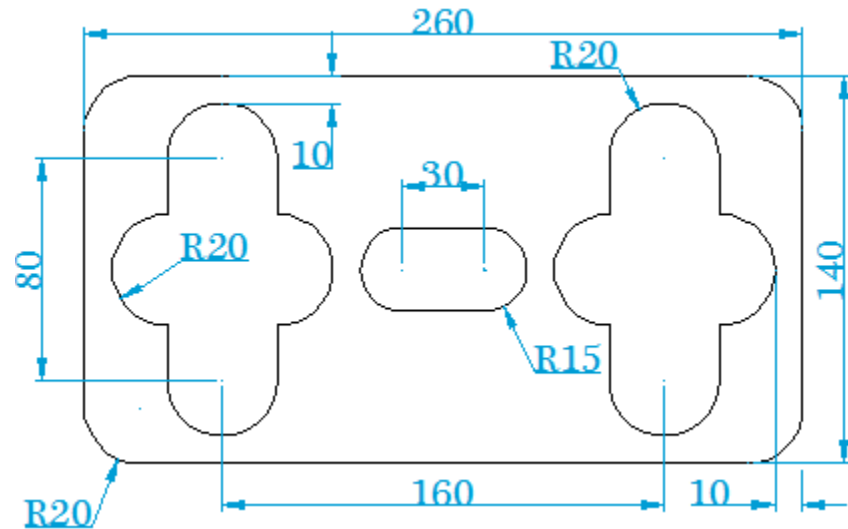
Example: click on chamfer tool icon and follow the step below:

- ▶ Command: `_chamfer` (TRIM mode) Current chamfer Dist1 = 0, Dist2 = 0 Select first line or [Undo / Polyline / Distance / Angle/ Trim / mEthod / Multiple]: **d** (Distance) then **Enter**
- ▶ Specify first chamfer distance <0>: **10** then **Enter**
- ▶ Specify second chamfer distance <10>: **15** then **Enter**
- ▶ Select first line or [Undo / Polyline / Distance / Angle/ Trim / mEthod / Multiple]: **pick**
- ▶ Select second line or shift – select to apply corner: **pick**

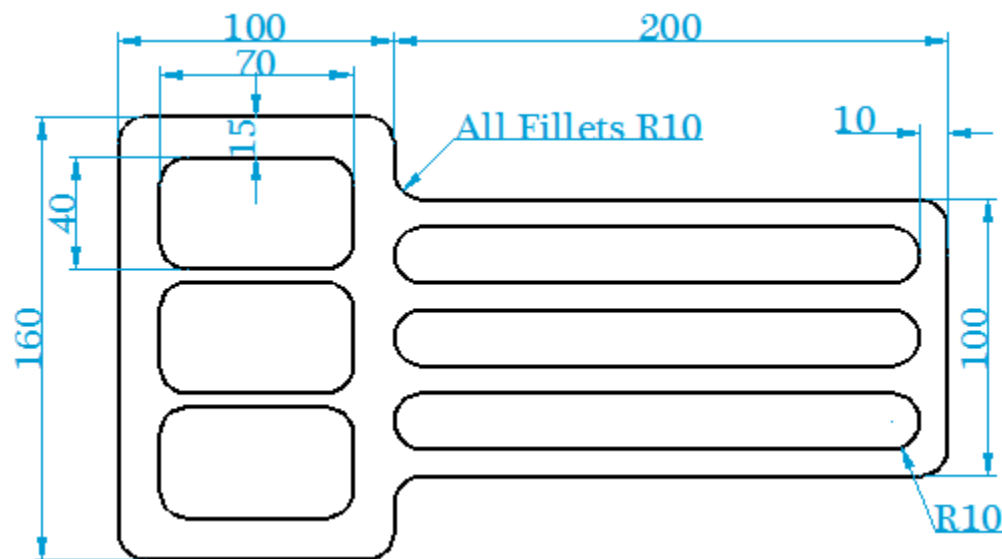


Exercise

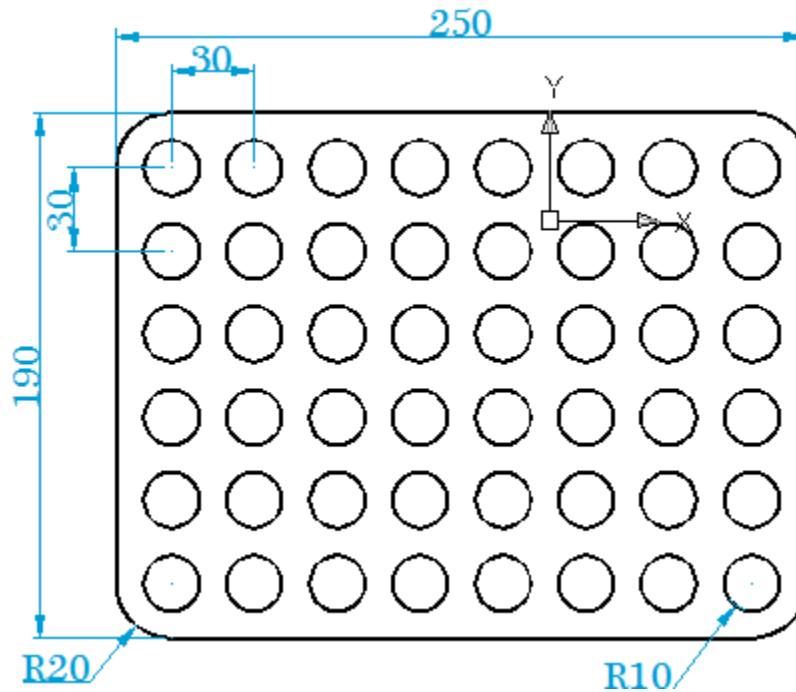
1. Using the tools **Polyline**, **Circle**, **Trim**, **Polyline Edit**, **Mirror** and **Fillet** construct the drawing given below.



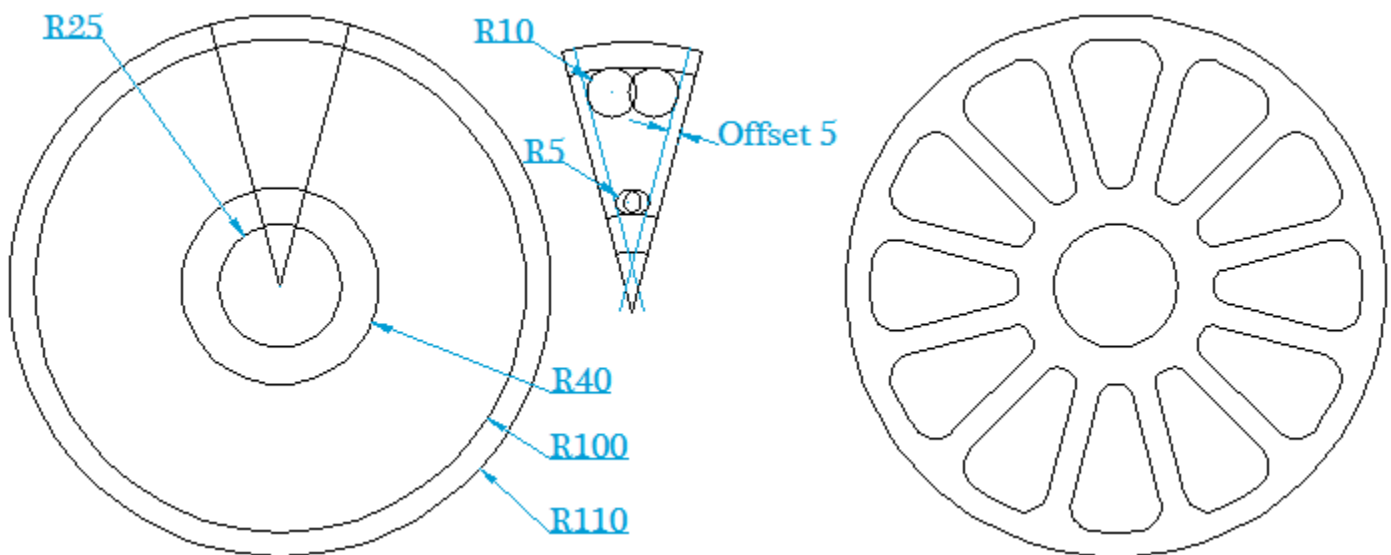
2. Construct the drawing given below. All parts are plines of width = 0.7 with corners filleted R10. The long strips have been constructed using **Circle**, **Polyline**, **Trim** and **Polyline Edit**. Construct one strip and then copy it using **Copy**.



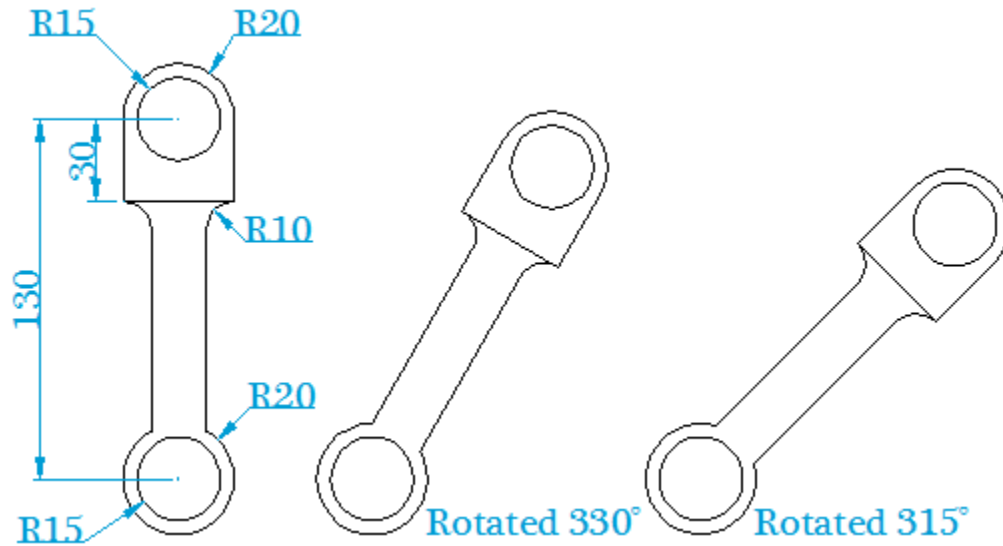
3. Construct the drawing given below. All parts of the drawing are plines of width = 0.7. The setting in the **Array** dialog is to be **180** in the **Angle of array** field.



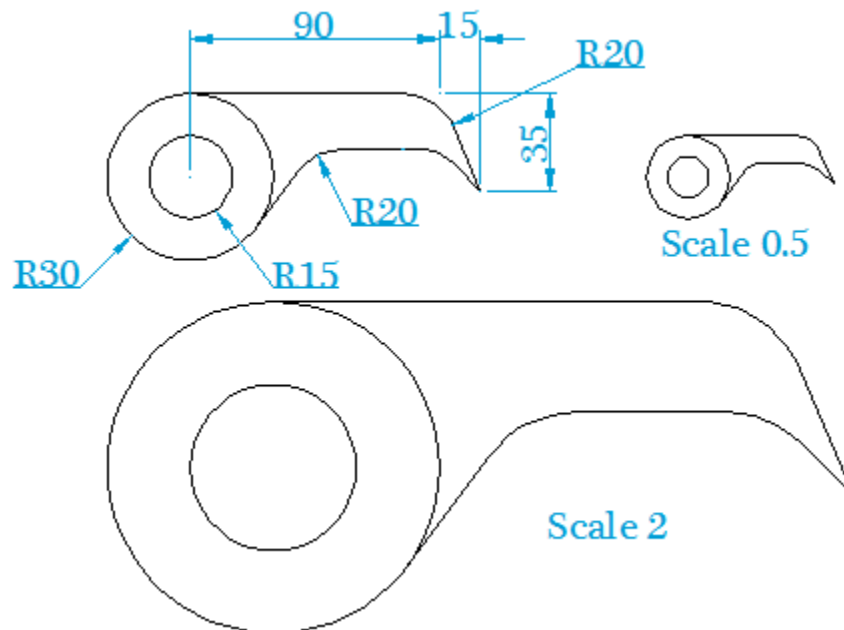
4. Construct the circles and lines shown below. Using **Off set** and the **Ttr** prompt of the **Circle** tool followed by **Trim**, construct one of the outlines arrayed within the outer circle. Then, with **Polyline Edit**, change the lines and arcs into a pline of width = 0.3. Finally, array the outline twelve times around the center of the circles.



5. Construct the drawing shown below and make two copies using **Copy**. With **Rotate** rotated each of the copies to the angles as shown.



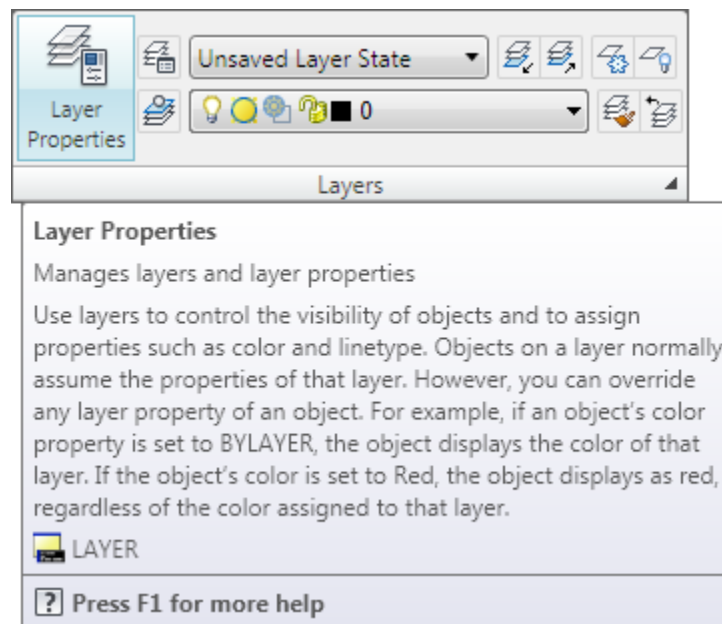
6. Construct the dimensioned drawing of Figure below. With **Copy** copy the drawing. Then with **Scale** scale the drawing to a scale of 0.5 and 1.



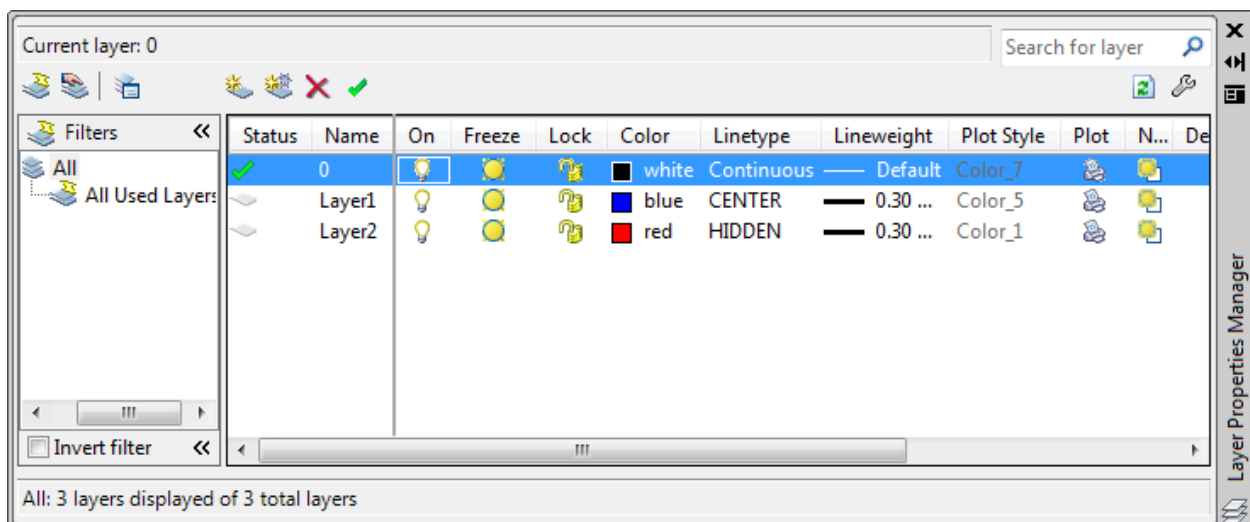
LAYERS COMMAND

Layer Properties : used to control the visibility of objects and to assign properties such as color and linetype to objects. Layers can be locked to prevent objects from being modified.

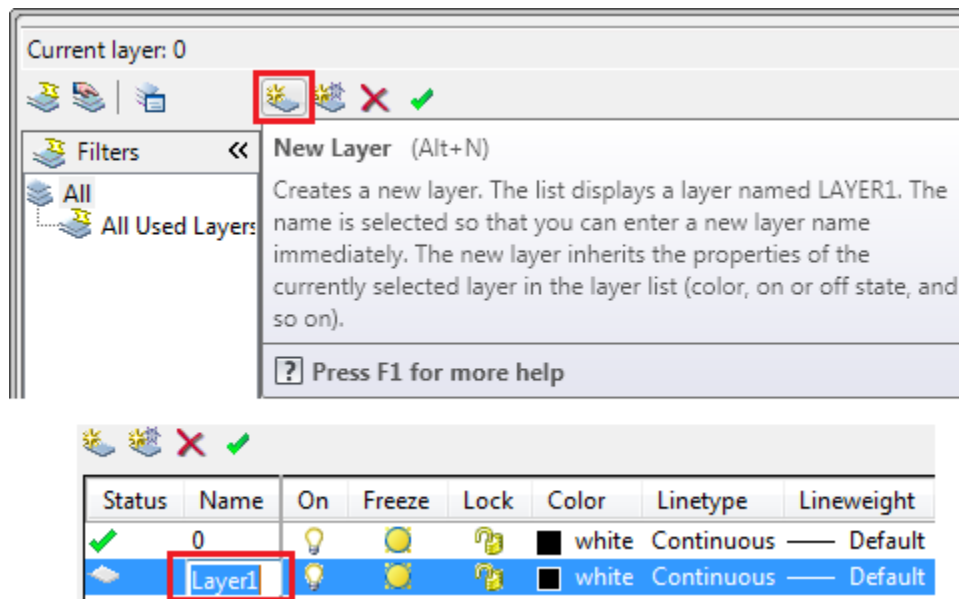
- Select **Layer Properties** from layers panel **OR**
- In command line write **layer** or **la** then Enter



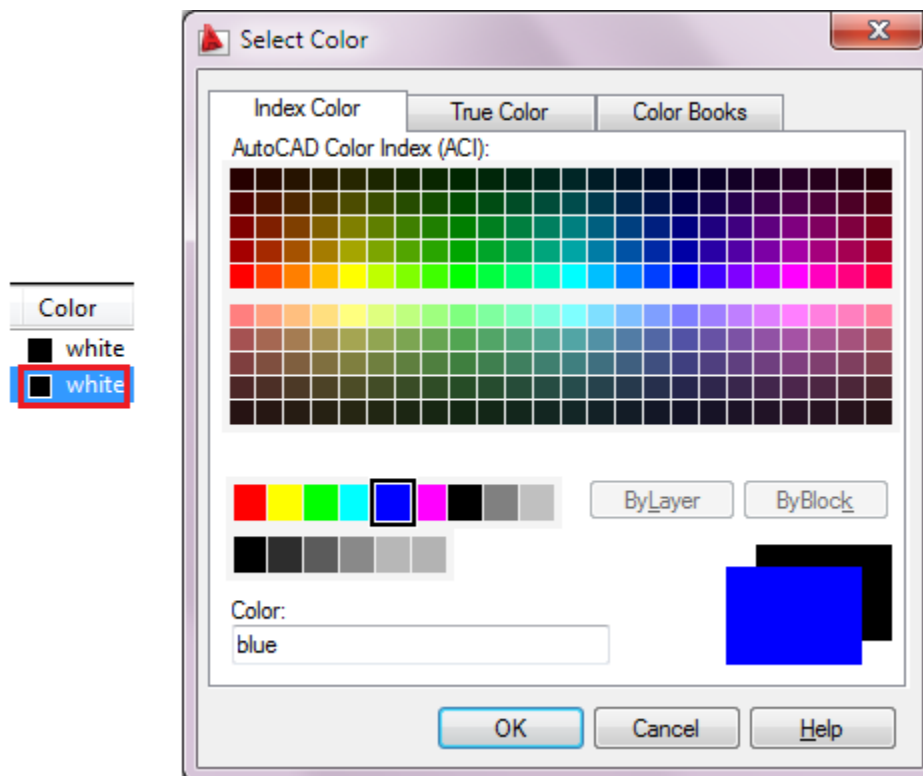
- **Layer Properties Manager** dialog box appear



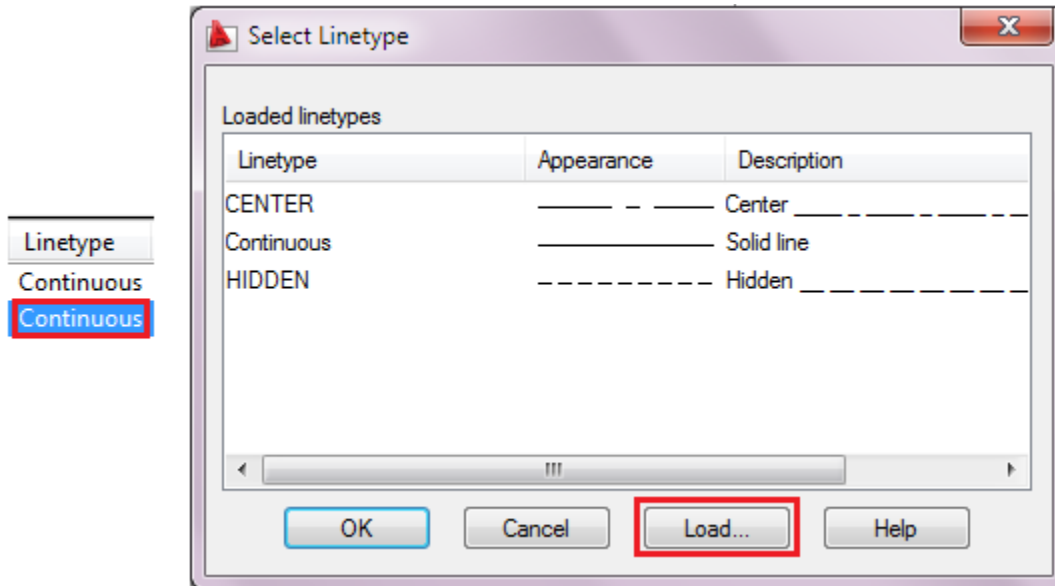
- Click on **New Layer**  icon. Layer1 is appearing by default and you can change the name.



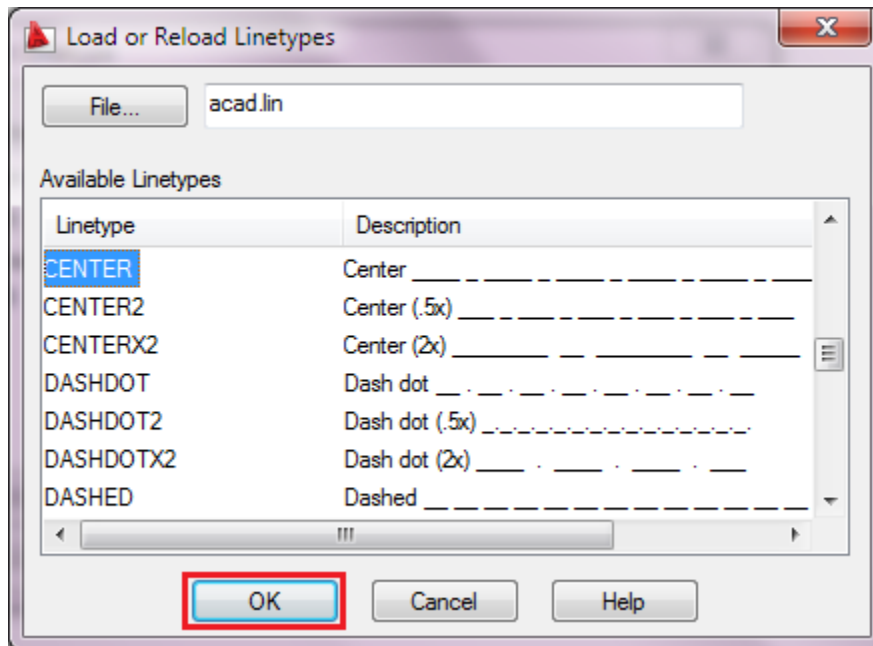
- Click on the color name (**White**) and **Select Color** dialog box appear then OK.



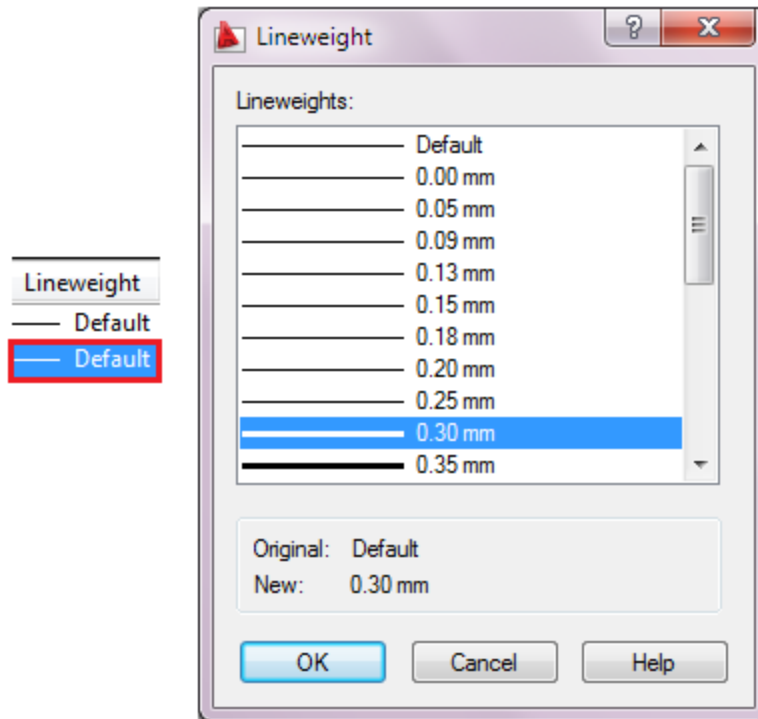
- Click on the linetype (**Continuous**) and **Select Linetype** dialog box appear



- Select **Load...** then Load or Reload Linetypes dialog box appear, selected linetype then **OK**.



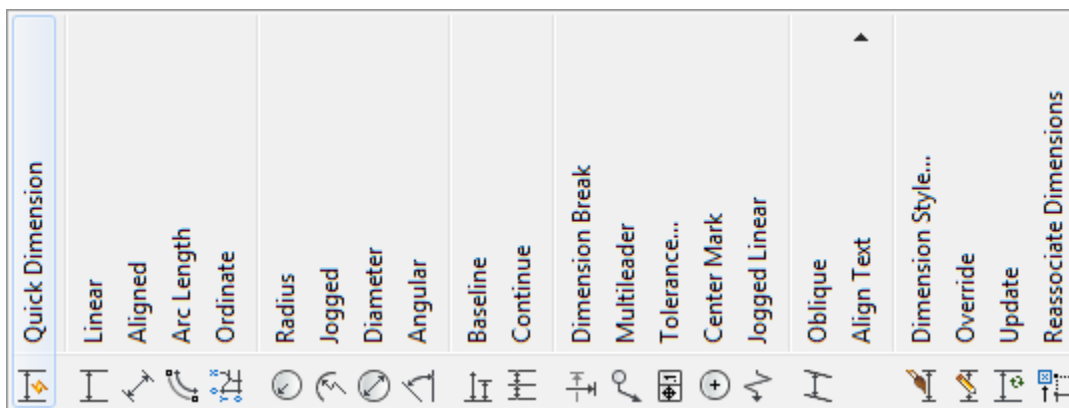
- Click on the any of the **Lineweight** then Lineweight dialog box appear. Select the line weight the value then **OK**.



- Repeat the same for all other the layers. Then Click the Close button of the Layer Properties Manager.

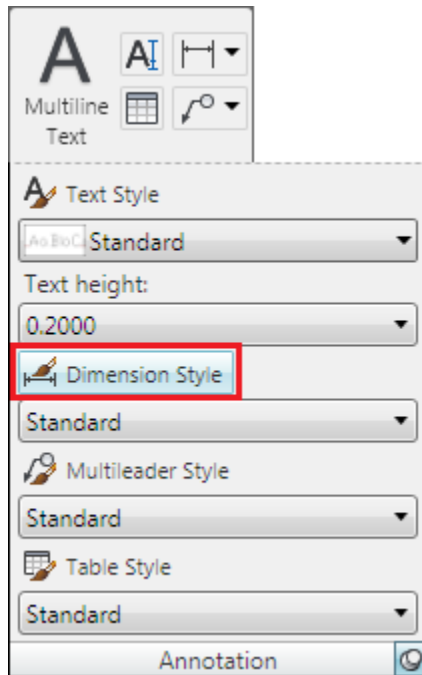
ANNOTATION

Dimensions: the process of adding measurement annotation to a drawing. You can create dimensions for a variety of object types in many orientations. The basic types of dimensioning are: Linear, Radial (radius, diameter and jogged), Angular, Ordinate and Arc Length.

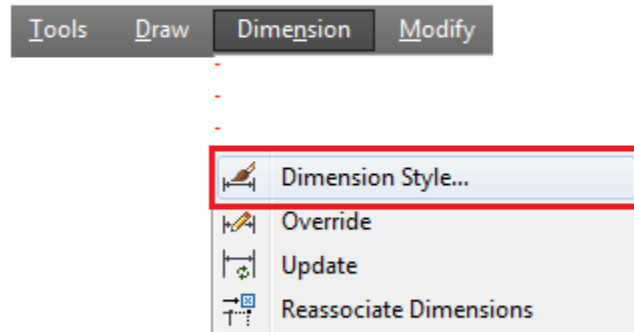


 **Dimension Style:** named collection of dimension settings that controls the appearance of dimensions, such as arrowhead style, text location, and lateral tolerances.

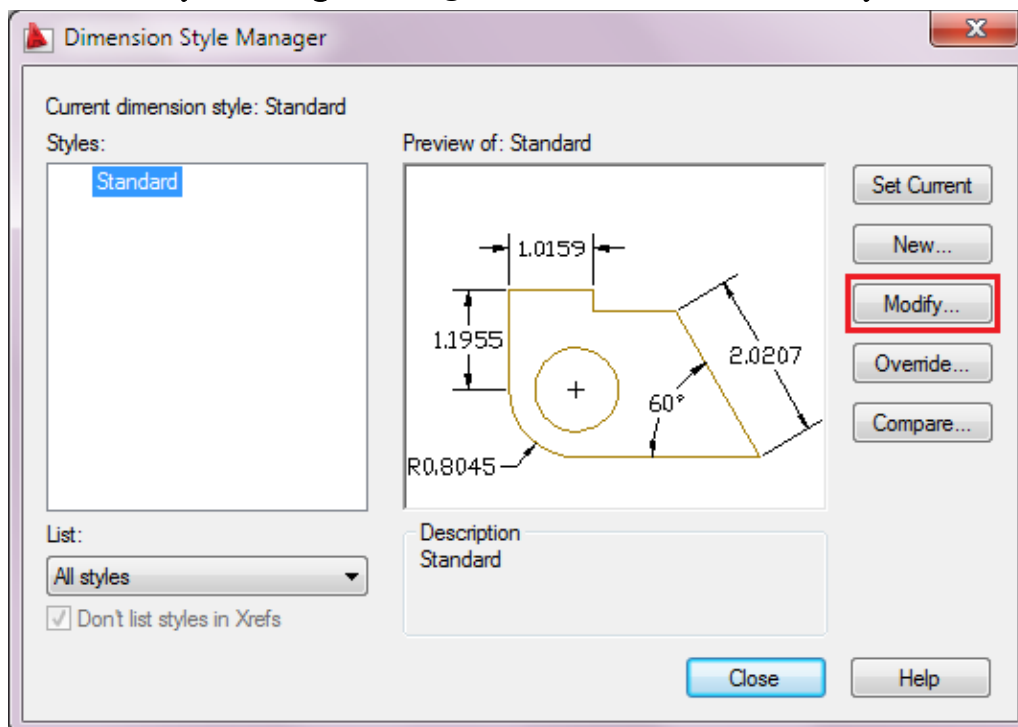
- ▶ Select **Dimension Tool** from annotation panel **OR** Select **Dimension** (Menu Bar) → **Dimension Style...**
- ▶ In command line write **dimstyle** or **d** then Enter



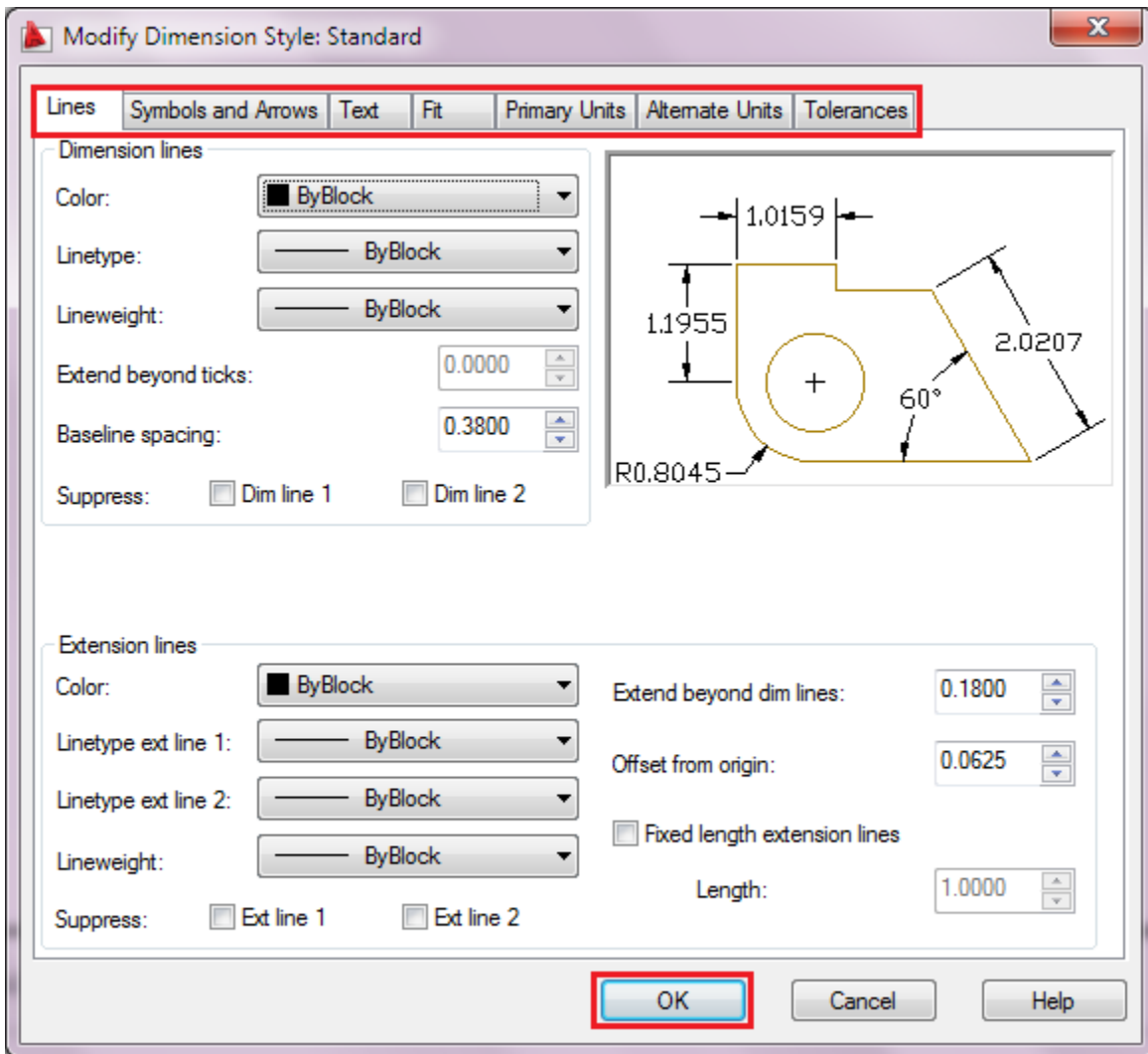
OR



- ▶ **Dimension Style Manager** Dialog box appear then click **Modify...**



- **Modify Dimension Style: Standard** Dialog box appear. You can set
 - **Lines:** Dimension lines (color, linetype, lineweight) and Extension lines (color, extend beyond dim lines, offset from origin)
 - **Symbols and Arrows:** Arrow heads and Arrow size
 - **Text:** Text appearance (Text style, color, height), Text placement (vertical, horizontal, center, above) and Text alignment (horizontal, Aligned with dimension line, ISO standard)
 - **Primary Units:** Linear dimensions (Unit format and Precision)

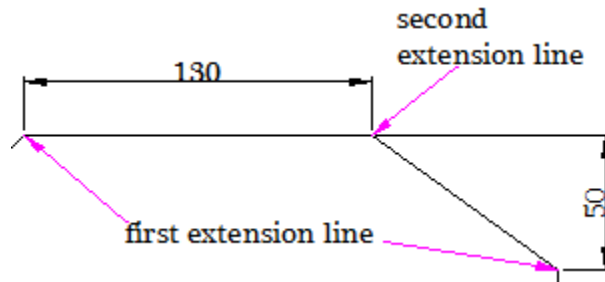


□ Dimension Tool: Example:



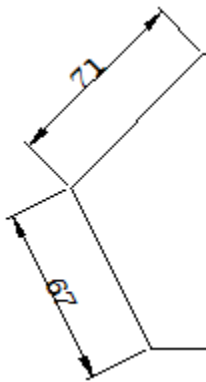
Linear Dimension: Click on linear tool icon or write dimlinear on command line and follow the step below:

- ▶ Command: `_dimlinear` Specify first extension line origin or <select object>: **pick**
- ▶ Specify second extension line origin: **pick**
- ▶ Specify dimension line location or [Mtext / Text / Angle / Horizontal / Vertical / Rotated]: **pick**
- ▶ Dimension text = **130**



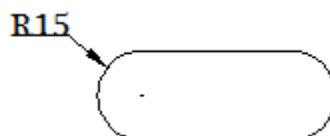
 **Aligned Dimension:** Click on aligned tool icon or write `dimaligned` on command line and follow the step below:

- ▶ Command: `_dimaligned` Specify first extension line origin or <select object>: **pick**
- ▶ Specify second extension line origin: **pick**
- ▶ Specify dimension line location or [Mtext / Text / Angle]: **pick**
- ▶ Dimension text = **71**



Radius Dimension: Click on radius tool icon or write `dimradius` on command line and follow the step below:

- ▶ Command: `_dimradius` Select arc or circle: **pick**
- ▶ Dimension text = **15**
- ▶ Specify dimension line location or [Mtext / Text / Angle]: **pick**



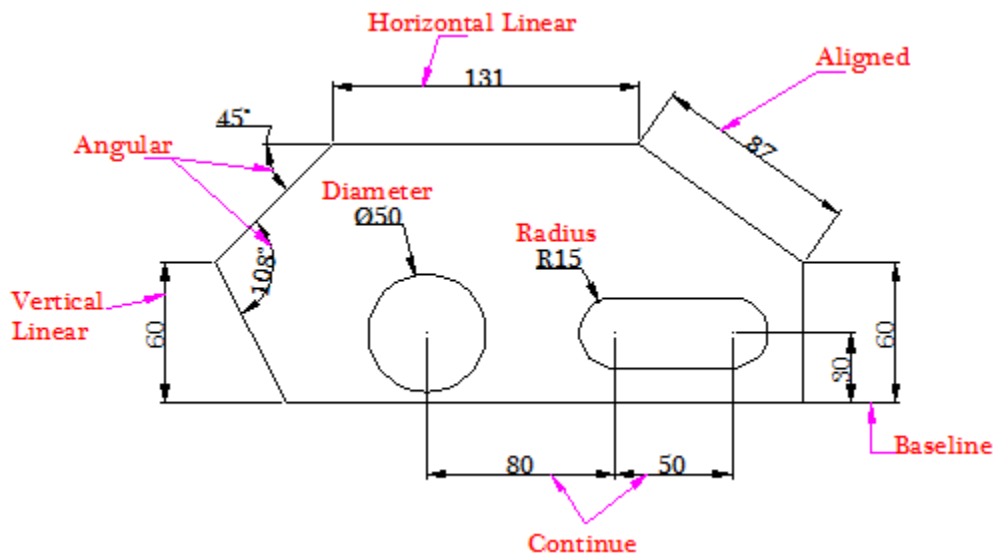


Angular Dimension: Click on angular tool icon or write dimangular on command line and follow the step below:

- ▶ Command: _dimangular Select arc, circle, line, or <specify vertex>: **pick**
- ▶ Select second line: **pick**
- ▶ Specify dimension arc line location or [Mtext / Text / Angle / Quadrant]: **pick**
- ▶ Dimension text = **135**



- ▶ Result

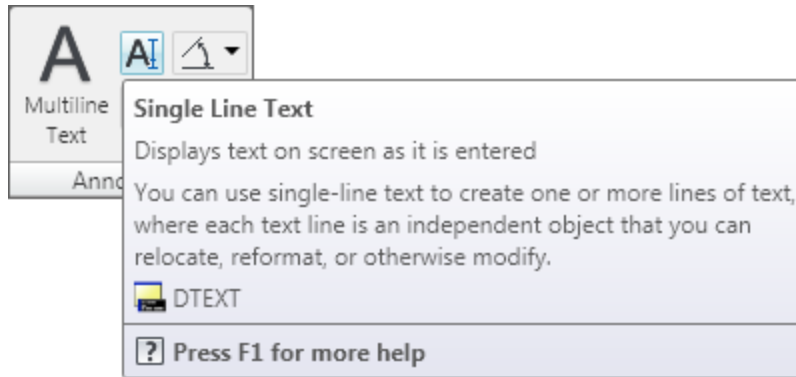


Text: there are two main methods of adding text to drawings: **Single Line Text** and **Multiline Text**.



Single line Text: used to create one or more lines of text.

- ▶ Select **Single line text** from annotation panel **OR**
- ▶ In command line write **text** or **dt** then Enter



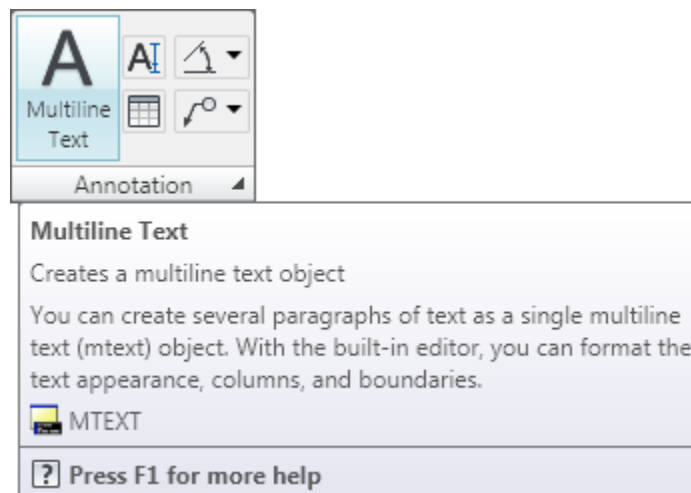
Example: click on single line text tool icon and follow the step below:

- ▶ Command: `_dtext` Current text style: “Standard” Text height: 8 Annotative: No
- ▶ Specify start point of text or [Justify / Style]: **pick**
- ▶ Specify rotation angle of text <0>: **pick** or **enter**
- ▶ After finish the text press **Enter** key twice or **Esc**

AUCTOCAD 2009 TEXT

A Multiline Text: used to create one or more paragraphs of multiline text.

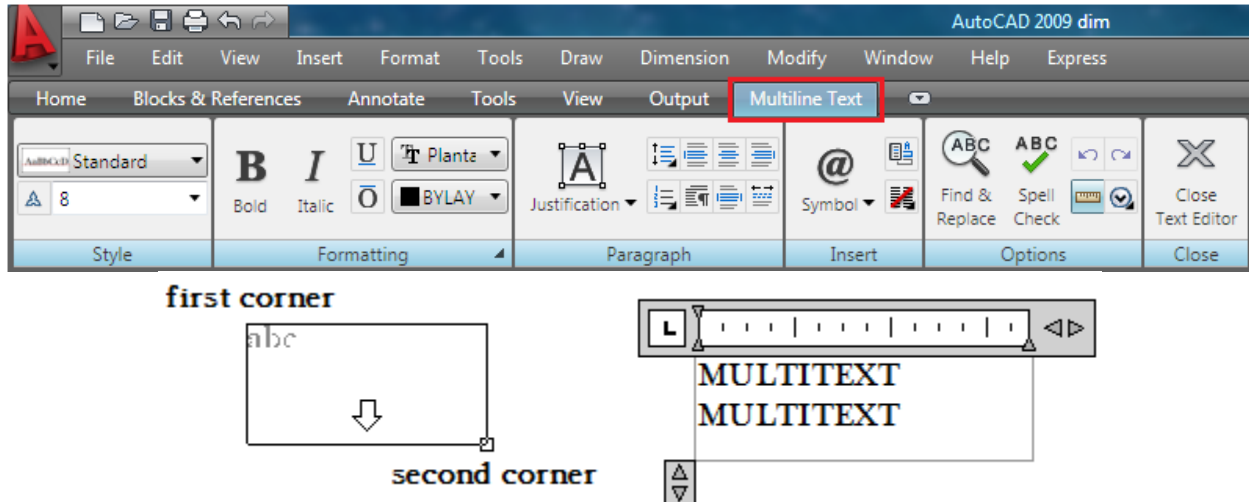
- ▶ Select **Multiline text** from annotation panel **OR**
- ▶ In command line write **mtext** or **t** then Enter



Example: click on multiline text tool icon and follow the step below:

- ▶ Command: `_mtext` Current text style: “Standard” Text height: 8 Annotative No

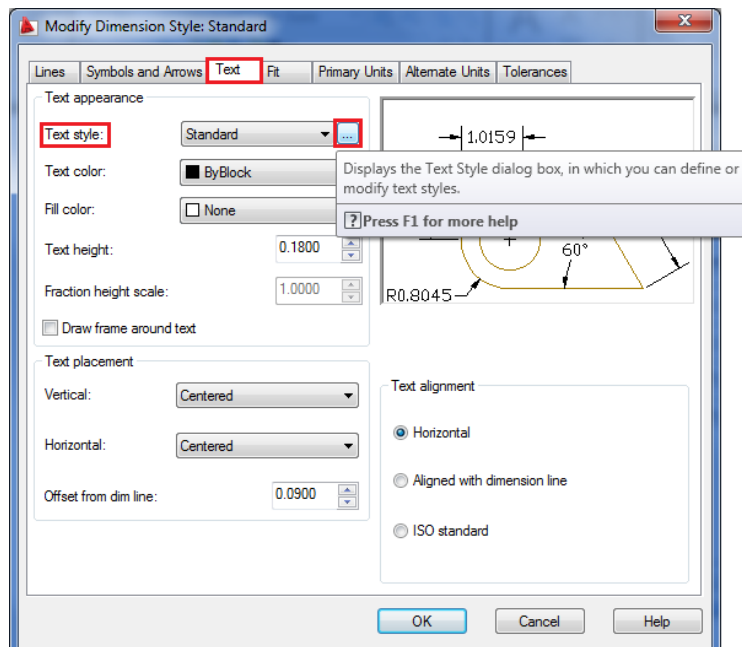
- Specify first corner: **pick**
- Specify opposite corner or [Height / Justify / Line spacing / Rotation / Style / Width / Columns]: **pick** or **enter**



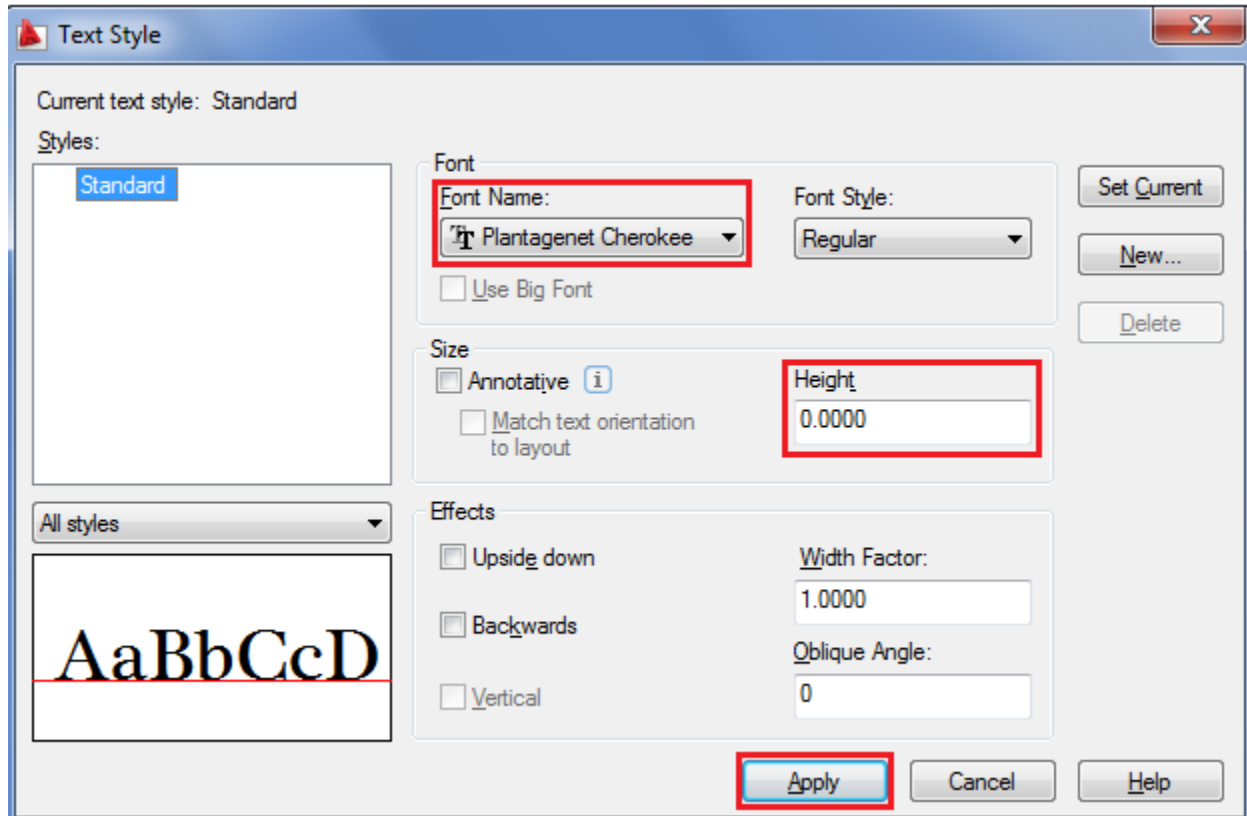
- In Mtext Robbin, you can set text style, formatting, paragraph and insert symbol.
- ❑ Common special characters during text have to be added by entering letters and figures as part of a dimension.
 - To obtain $\emptyset$ enter %%c
 - To obtain $^{\circ}$ enter %%d
 - To obtain $\pm$ enter %%p

Text Setting

- Select **Dimension** (Menu Bar) → **Dimension Style...** → **Dimension Style Manager** Dialog box appear then click **Modify...** → **Modify Dimension Style: Standard** Dialog box appear then click on display the **Text Style** Dialog box. **OR**
- In command line write **st** then Enter



- ▶ **Text Style** Dialog box appearing then set **Font** (Font Name, Font Style) and **Size** (Height).
- ▶ Click on **Apply** button you finish the set
- ▶ Finally click on **Close** button to exit from the dialog box.



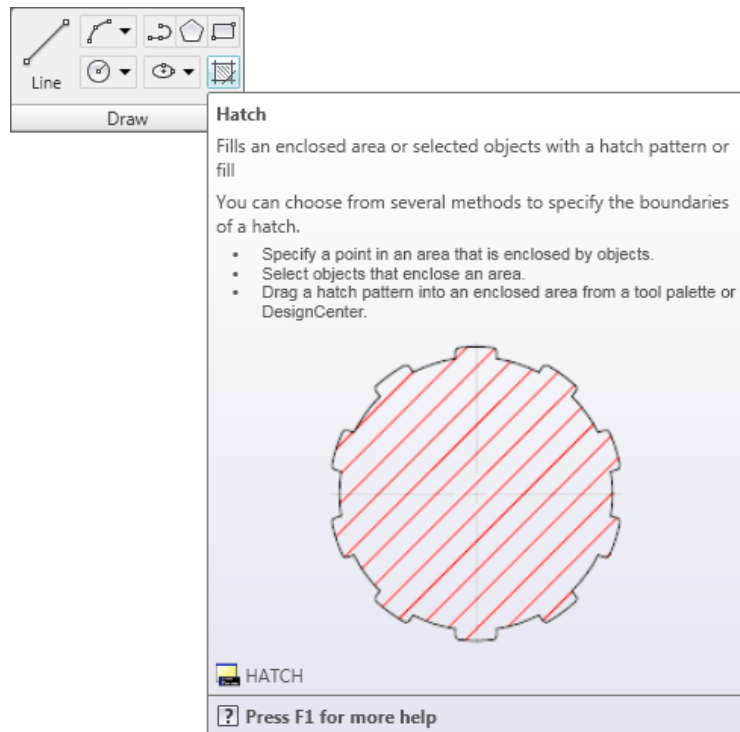
HATCHING

Hatch Command : In order to represent sectional surface into sectional view, it is used to fill an enclosed area or selected objects with a hatch pattern, methods to specify the boundaries of a hatch.

- Specify a point in an area that is enclosed by objects.
- Select objects that enclose an area.
- Drag a hatch pattern into an enclosed area from a tool palette or Design Center.

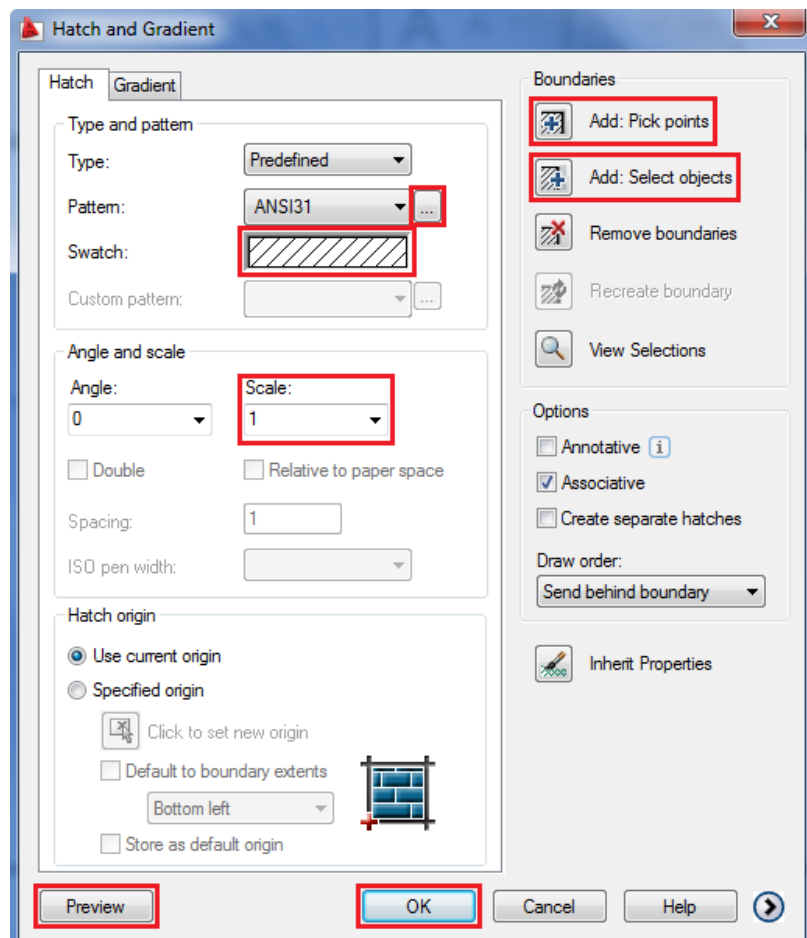
Hatch creates:

- ▶ Select **Hatch** from draw panel **OR**
- ▶ In command line write **hatch** or **h** then Enter

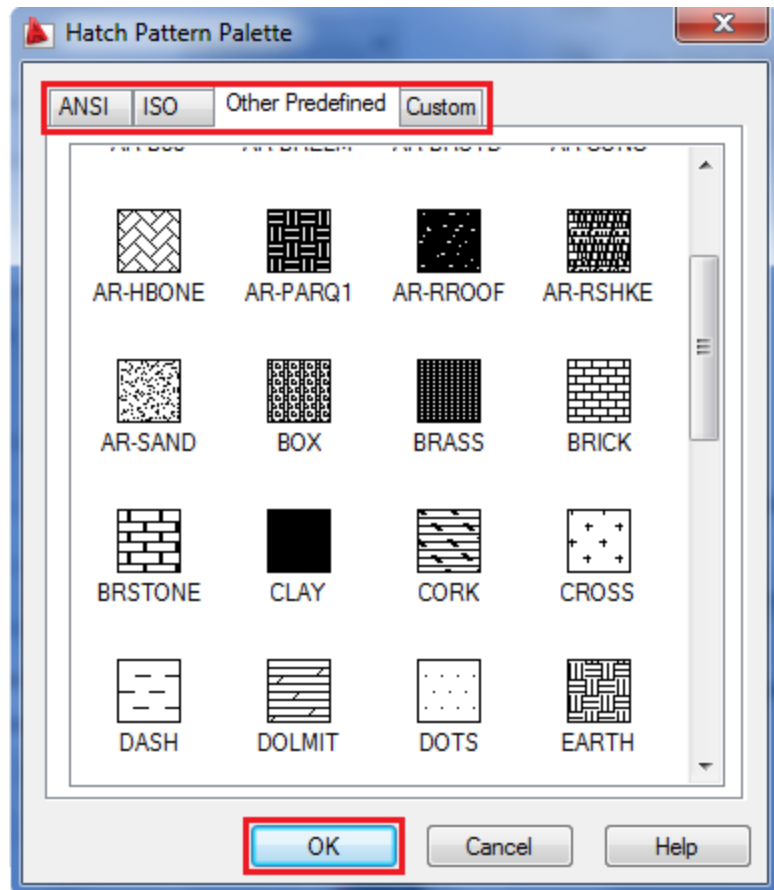


Example: click on hatch tool icon and follow the step below:

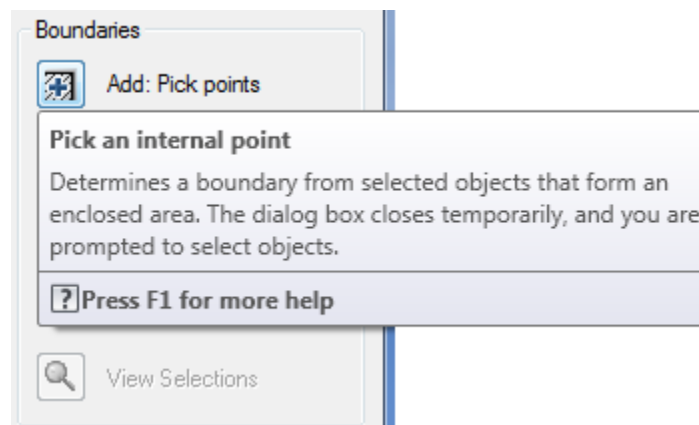
► **Hatch and Gradient**
Dialog box appears



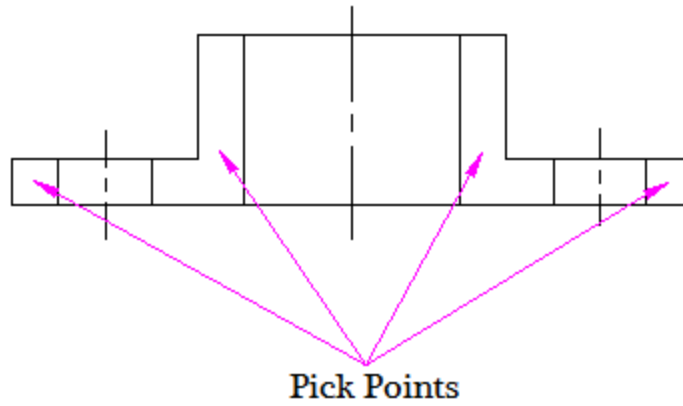
- Click in the **Swatch** field then **Hatch Pattern Palette** Dialog box appear. Select hatch pattern from ANSI, ISO, Other Predefined then click **OK** button.



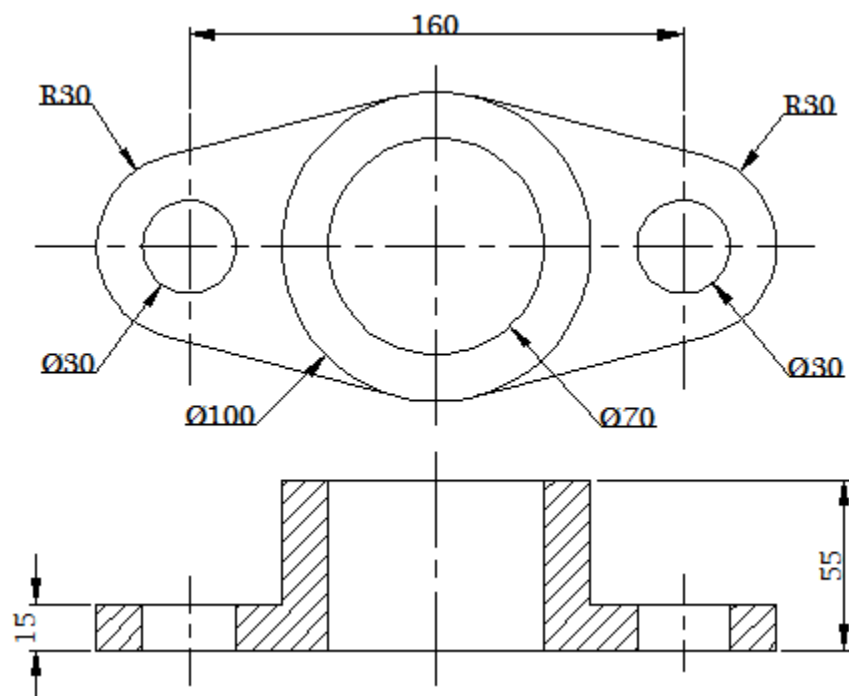
- Click on the **Pick an internal point** button from the hatch and gradient dialog box the dialog box disappear.



- Pick area that hatch pattern enclosed then press **Enter** button. Click the **Preview** button of the dialog and in the sectional view which reappears; check whether the **hatching** and **scale** is satisfactory.

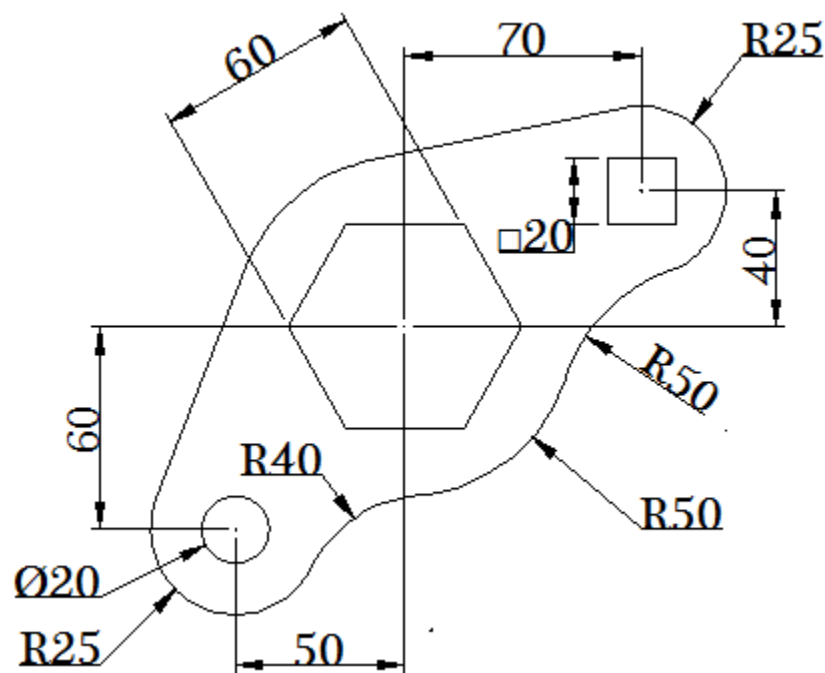
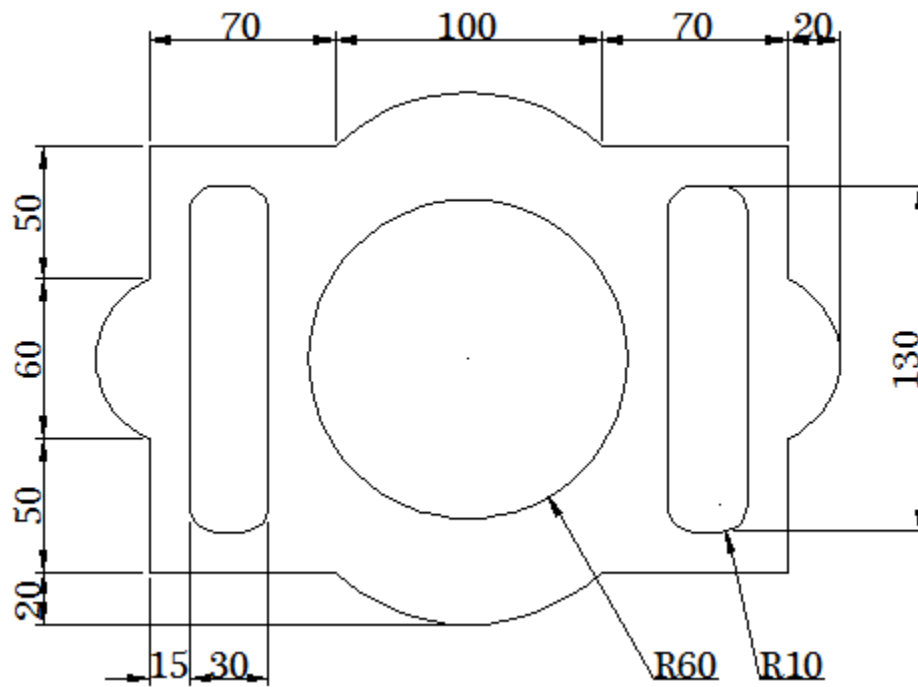


- Click on **OK** to finish then Hatching Result

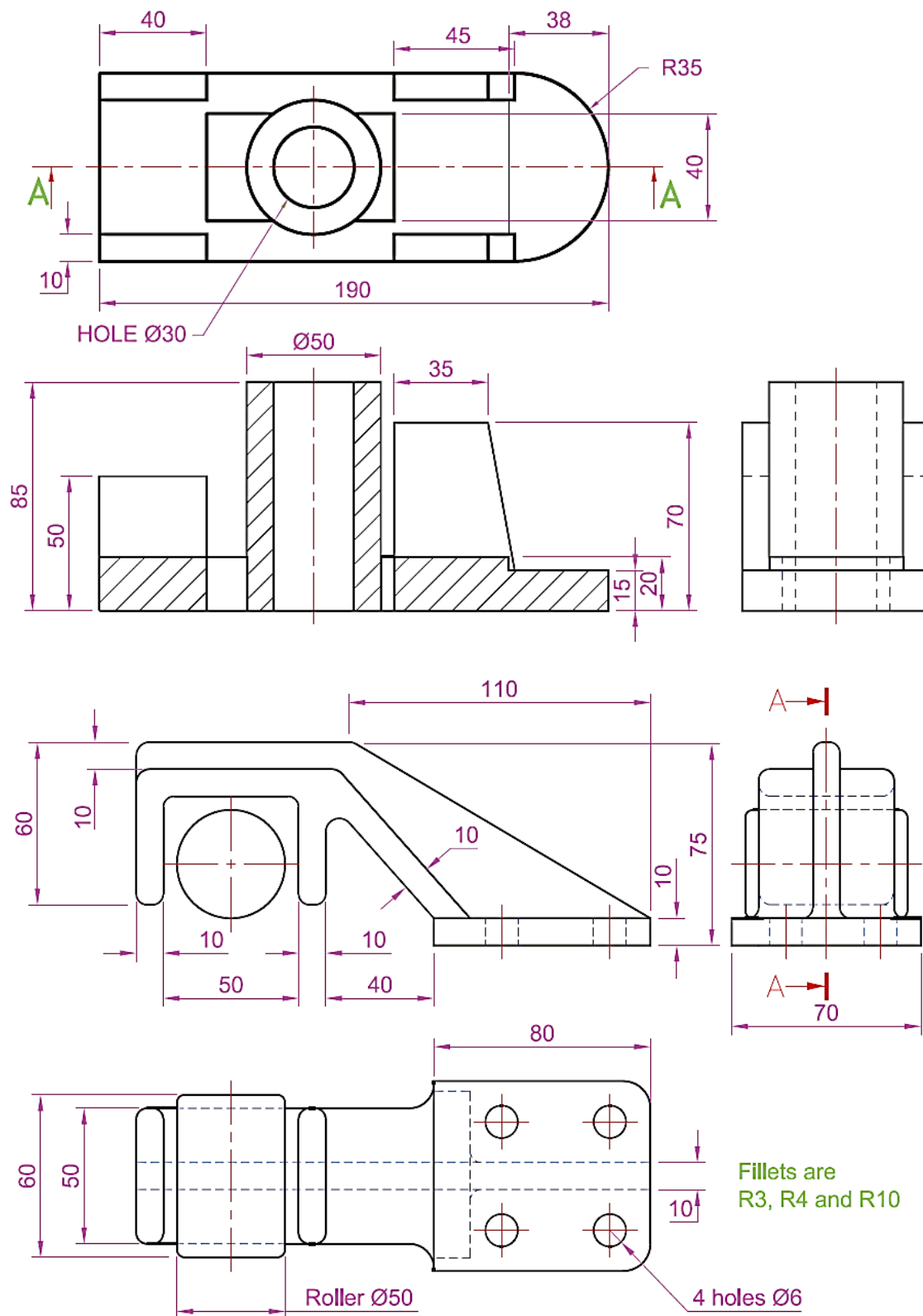


Exercises

Construct and dimension the drawing



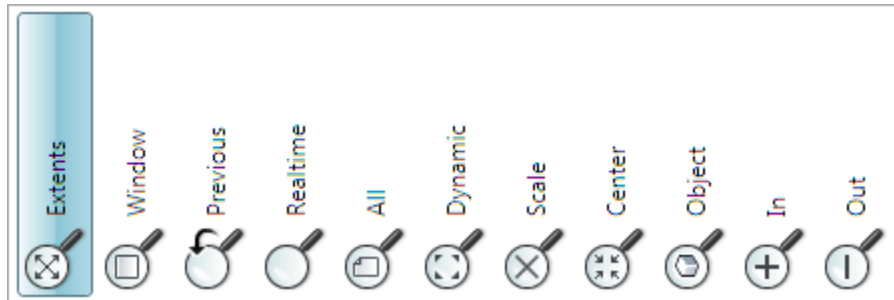
Construct the three views but with the front view as a sectional view based on the section plane A – A .



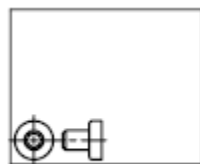
UTILITIES

Zooms

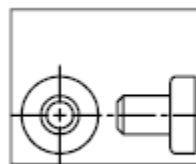
- ▶ Select **Zoom** from utilities panel **OR**
- ▶ In command line write **zoom** or **z** then Enter
- ▶ Command: '_zoom Specify corner of window, enter a scale factor (nX or nXP), or [All / Center / Dynamic / Extents / Previous / Scale / Window / Object] <real time>:



Extents Zooms: to display the drawing extents and results in the largest possible display of all the objects.



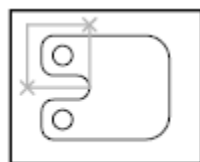
before ZOOM Extents



after ZOOM Extents



Window Zooms: to display an area specified by two opposite corners of a rectangular window.



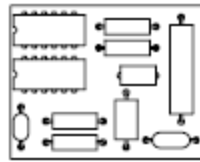
before ZOOM Window



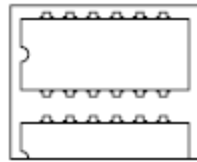
after ZOOM Window



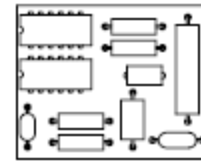
Previous Zooms: to display the previous view. You can restore up to 10 previous views.



original view



current view



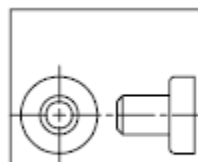
after ZOOM Previous



Realtime Zooms: to increase or decrease apparent size of objects in the current viewport.



All Zooms: to display the entire drawing in the current viewport.



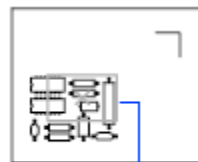
before ZOOM All



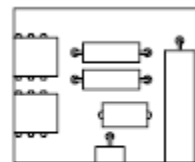
after ZOOM All



Dynamic Zooms: to display the generated portion of the drawing with a view box.



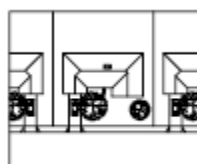
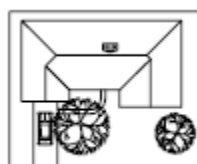
view box



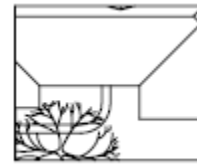
new view



Scale Zooms: to display at a specified scale factor.



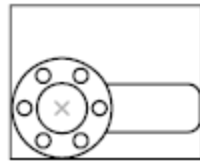
ZOOM .5x



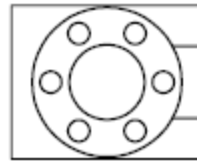
ZOOM 2



Center Zooms: to display a window defined by a center point and a magnification value or height.



before ZOOM center

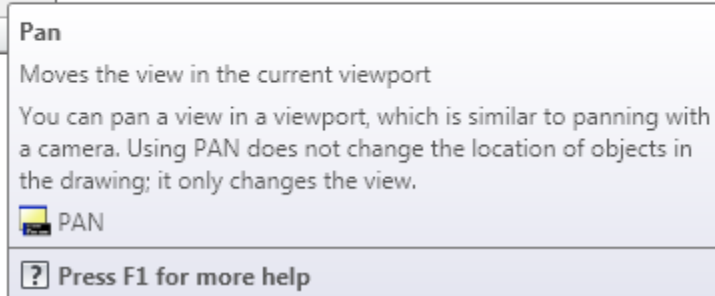
after ZOOM center,
magnification increased

Object Zooms: to display one or more selected objects as large as possible and in the center of the drawing area.



Pan: Moves the view in the current viewport

- ▶ Select **Pan** from utilities panel **OR**
- ▶ In command line write **pan** or **p** then Enter
- ▶ Command: '_pan Press ESC or ENTER or exit, or right – click to display shortcut menu.

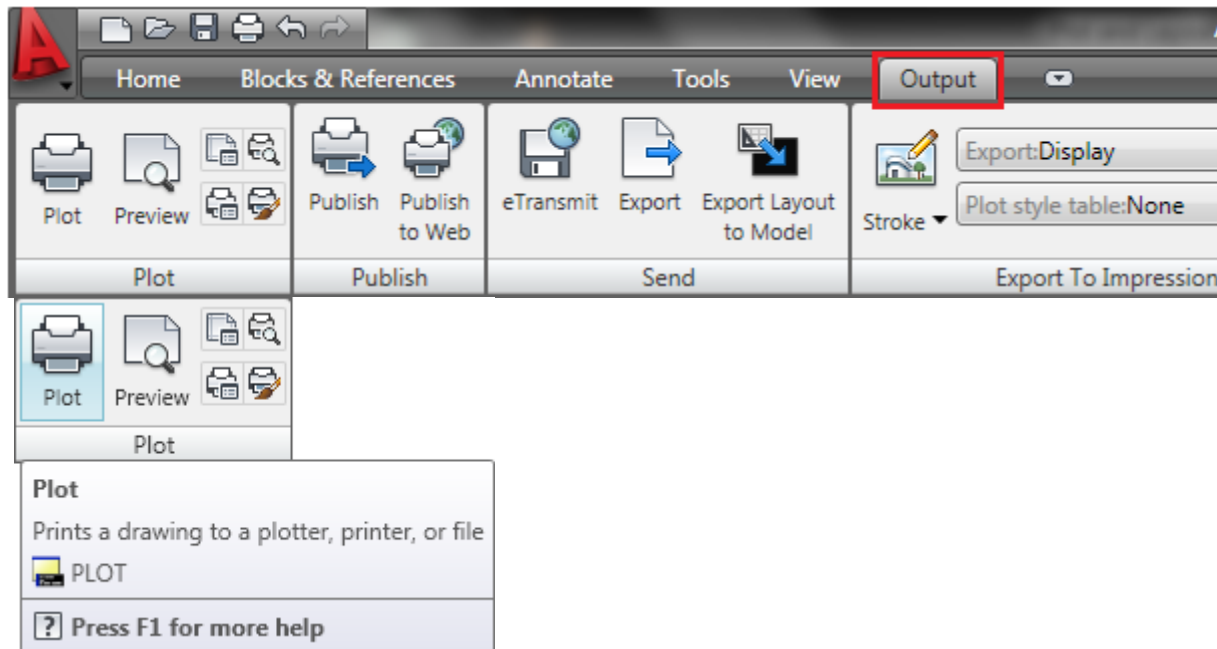


PLOT AND LAYOUT

Plot : Getting hardcopy output from AutoCAD is something of an art. You'll need to be intimately familiar with both your output device and the settings available in AutoCAD.

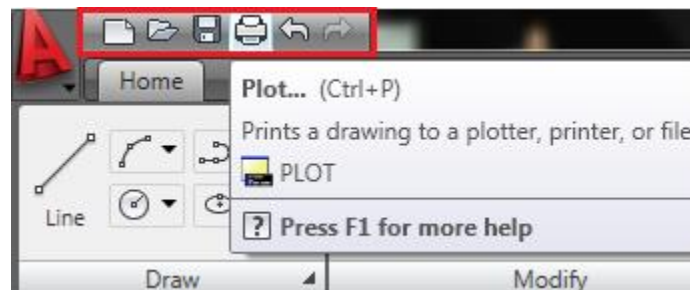
To Plot a Drawing

- **Ribbon:** Output → Plot Panel → Plot



OR

- **Toolbar:** Standard (On Quick Access)

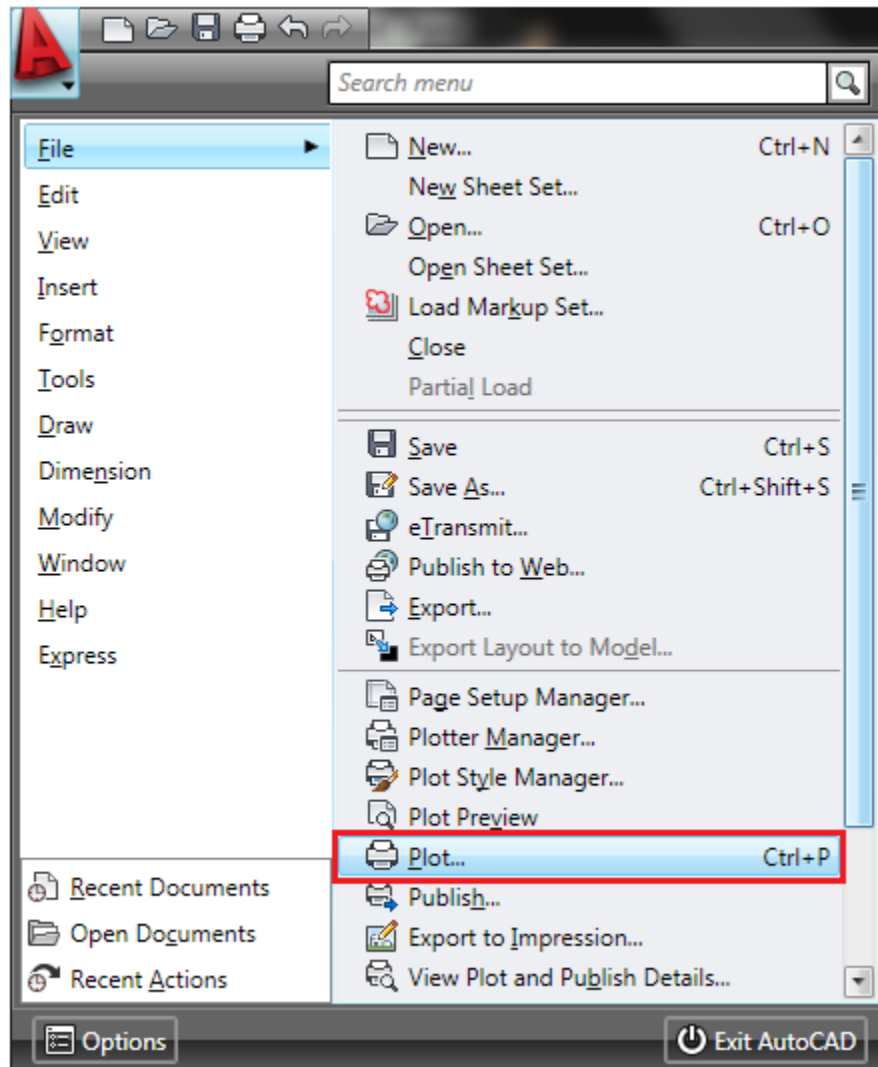


OR

- **Ctrl+P**

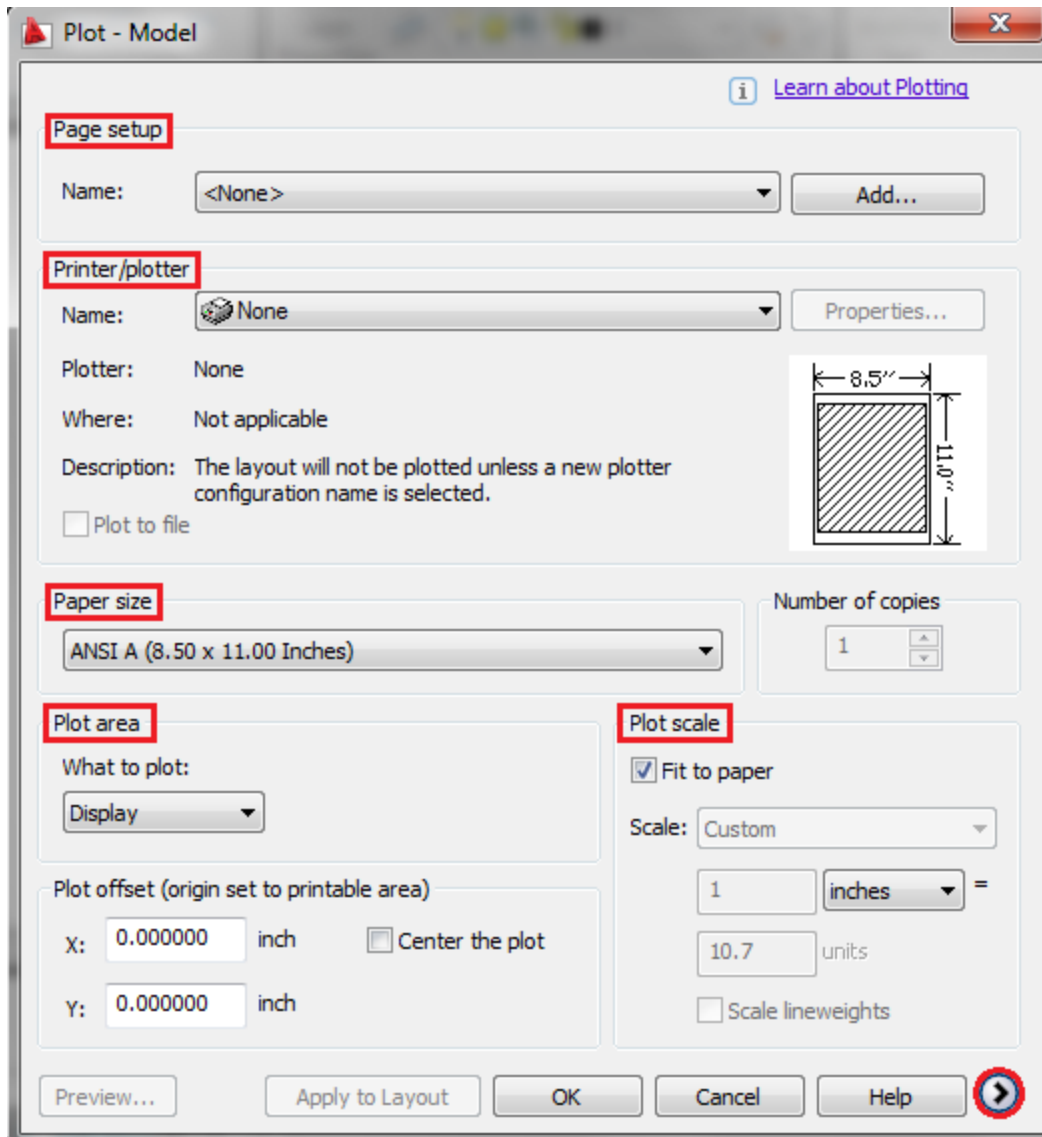
OR

- Menu Browser → File → Plot...

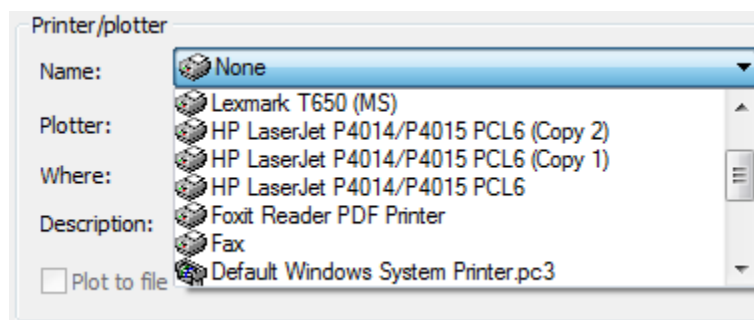


The Plot – Model dialog box is appearing. In this dialog box set the following:

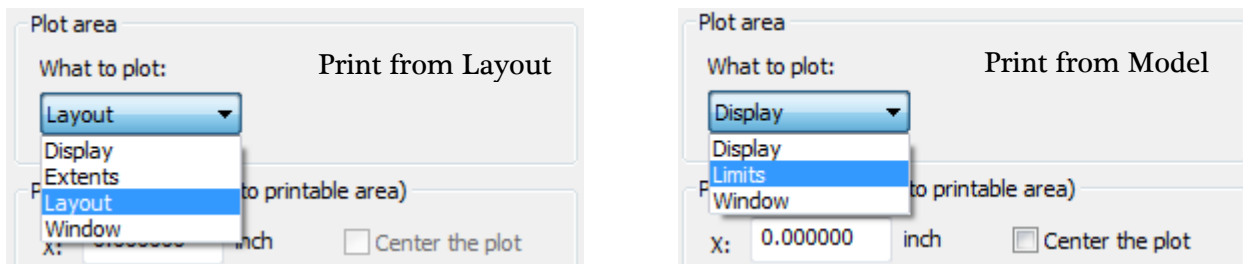
- ☐ Printer/Plotter
- ☐ Plot Area
- ☐ Paper Size
- ☐ Plot Scale
- ☐ Drawing Orientation (in More Options)



Printer/Plotter: If the name option in the Printer/Plotter group shows None, click the drop-down arrow and select your current Windows system printer.

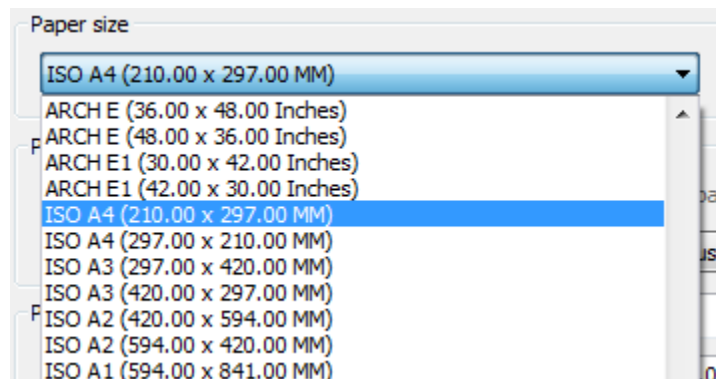


Plot Area: When plotting a drawing, you must specify the area of the drawing to plot. The Plot dialog box provides the following options under Plot Area.

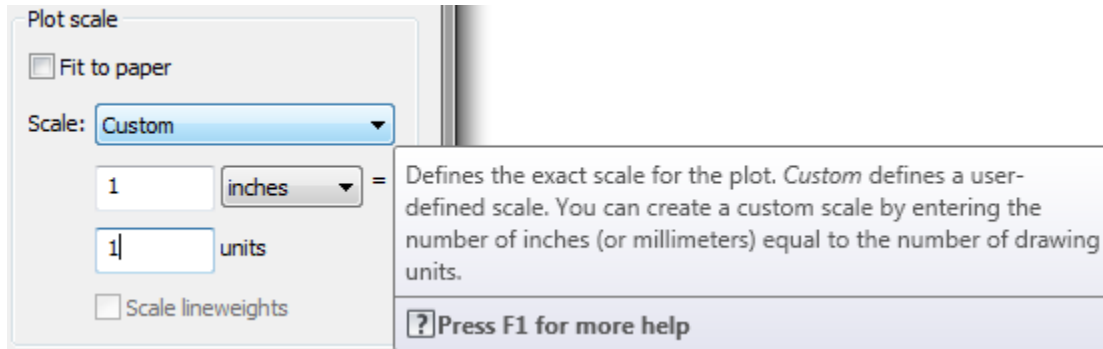


- **Layout or Limits:** When plotting a layout, plots everything within the printable area of the specified paper size, with the origin calculated from 0, 0 in the layout. When plotting the Model tab, plots the entire drawing area defined by the grid limits. If the current viewport does not display a plan view, this option has the same effect as the Extents option.
- **Extents:** Plots the portion of the current space of the drawing that contains objects. All geometry in the current space is plotted. The drawing might be regenerated to recalculate the extents before plotting.
- **Display:** Plots the view in the current viewport in the Model tab or the current paper space view in a layout tab.
- **Window:** Plots any portion of the drawing you specify. Click the Window button to use a pointing device to specify opposite corners of the area to be plotted, or enter coordinate values.

Paper Size: You can select a paper size from a standard list, or you can add custom paper sizes using the Plotter Configuration Editor. The paper sizes available in the list are determined by the plot device that is currently selected for the layout.

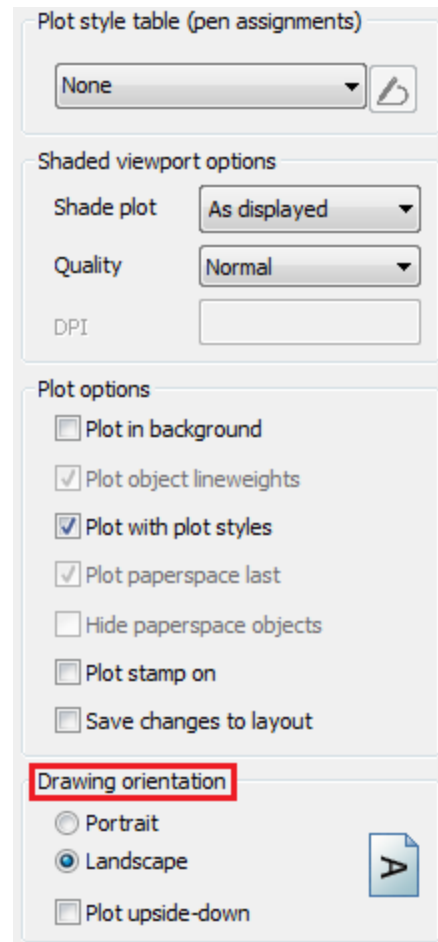


Plot Scale: When you specify a scale to output your drawing, you can choose from a list of real-world scales, enter your own scale (Custom), or select Fit to Paper to scale the drawing to fit onto the selected paper size.

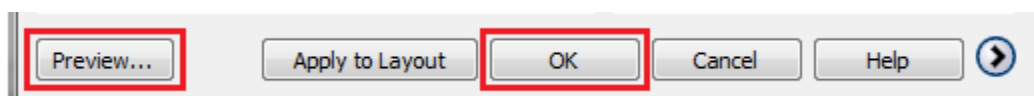


Drawing Orientation: You can specify the orientation of the drawing on the paper using the Landscape and Portrait settings. Landscape orients the drawing on the paper so that the long edge of the paper is horizontal, and Portrait orients the paper so that the short edge is horizontal.

- ▶ Click the More Options button 
- ▶ Select the orientation of your drawing

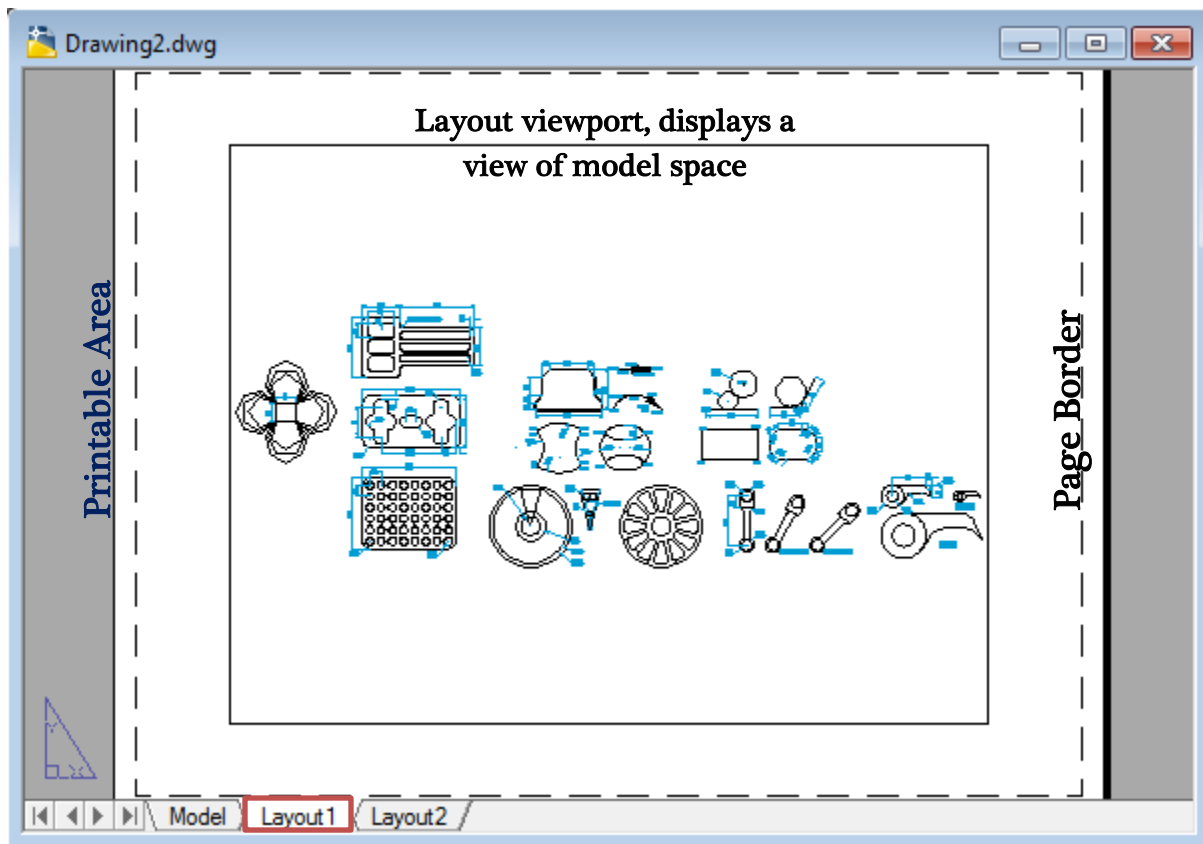


Click the **Preview** button in the lower-left corner of the dialog box. If you want to view of how your drawing will appear when printed. Finally click **OK**.



Layouts : A layout represents a plotted page. You can create as many layouts as you need. Each layout is saved on its own layout tab and can be associated with a different page setup. Elements that appear only on a plotted page, such as title blocks and notes, are drawn in paper space in a layout. The objects in the drawing are created in model space on the Model tab. To view these objects in the layout, you create layout viewports. Layout viewports are areas, similar to picture frames or windows, through which you can see your model. You scale the views in viewports by zooming in or out. You can display objects differently by setting property overrides for color, linetype, lineweight, and plot style and apply them to individual layout viewports.

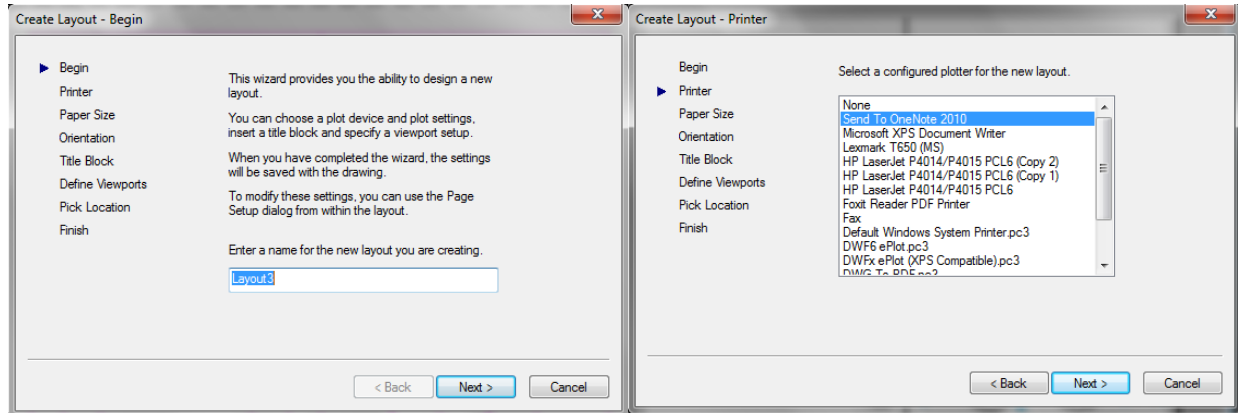
Paper space is a sheet layout environment where you can specify the size of your sheet, add a title block, display multiple views of your model, and create dimensions and notes for your drawing.



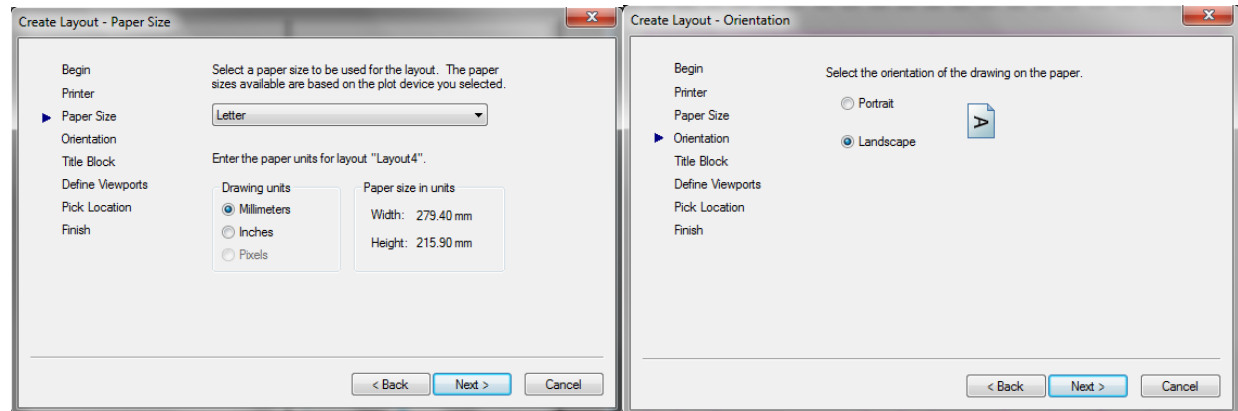
To Create Layout

- ▶ Click Menu Browser → Tools → Wizards → Create Layout...
- ▶ Follow the steps in the wizard to create a layout.
- ▶ Create Layout – **Begin**: You can enter a name for new layout. Click Next.

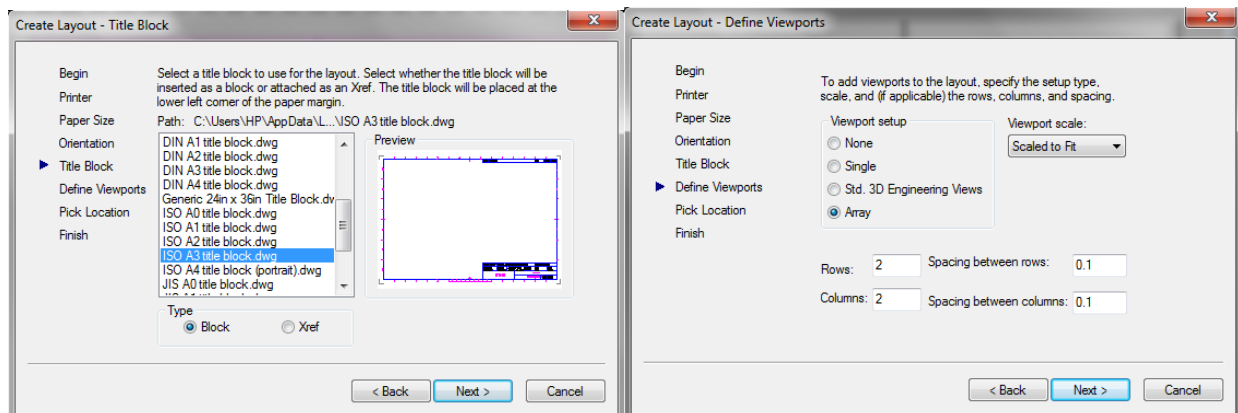
- **Create Layout – Printer:** Select a configured Plotter. Click Next.



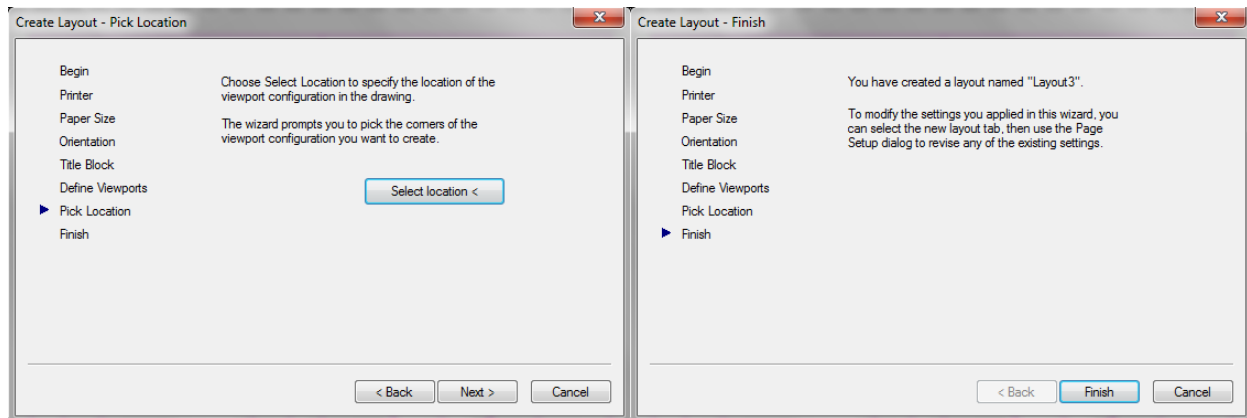
- **Create Layout – Paper Size:** Select a paper size to be used for the layout.
- **Create Layout – Orientation:** Select the orientation of the drawing on the paper.



- **Create Layout – Title Block:** Select a title block to use for the layout. Click Next.
- **Create Layout – Define Viewports:** To add viewports to the layout. Click Next.

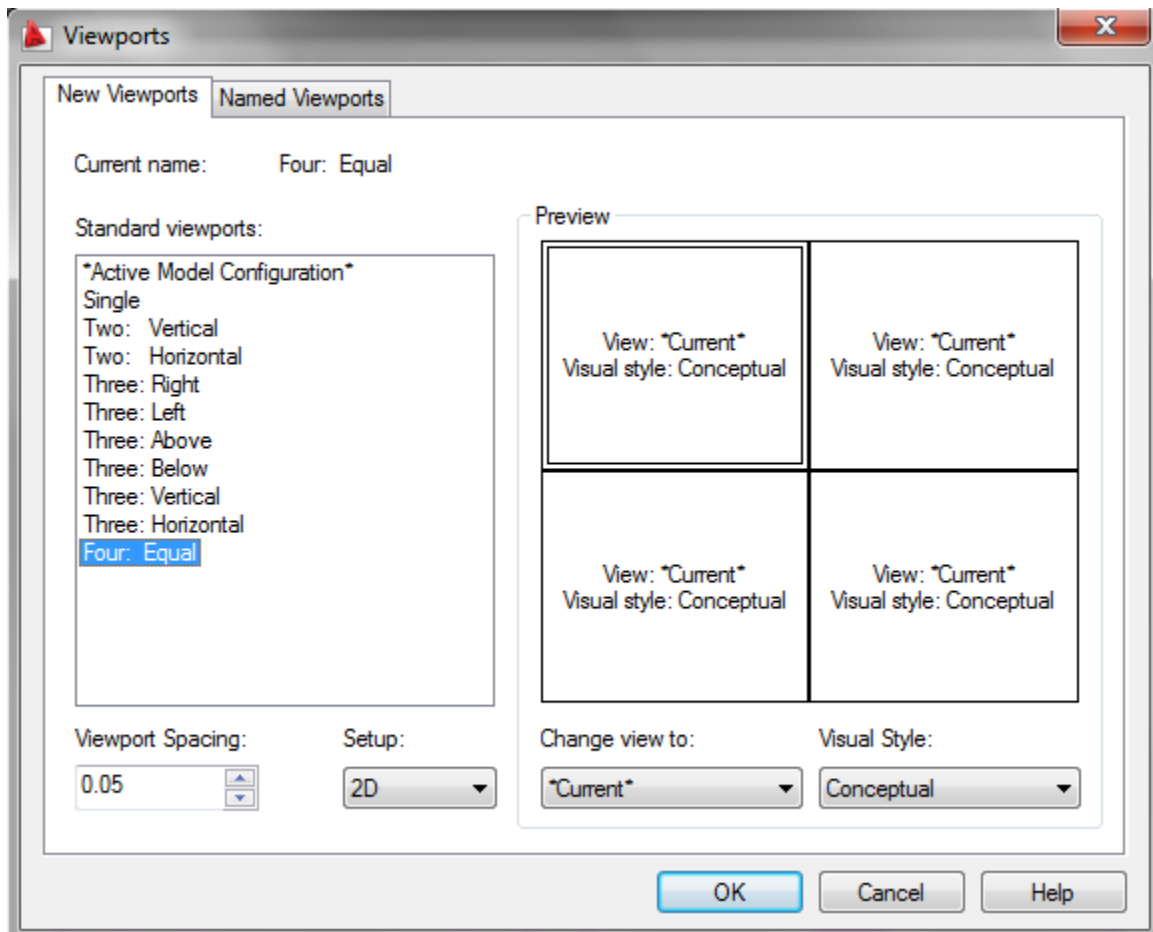


- Create Layout – **Pick Location:** Choose select location to specify the location of the viewport configuration in the drawing. Click Next.
- Create Layout – **Finish:** You have created a layout named “Layout3”.



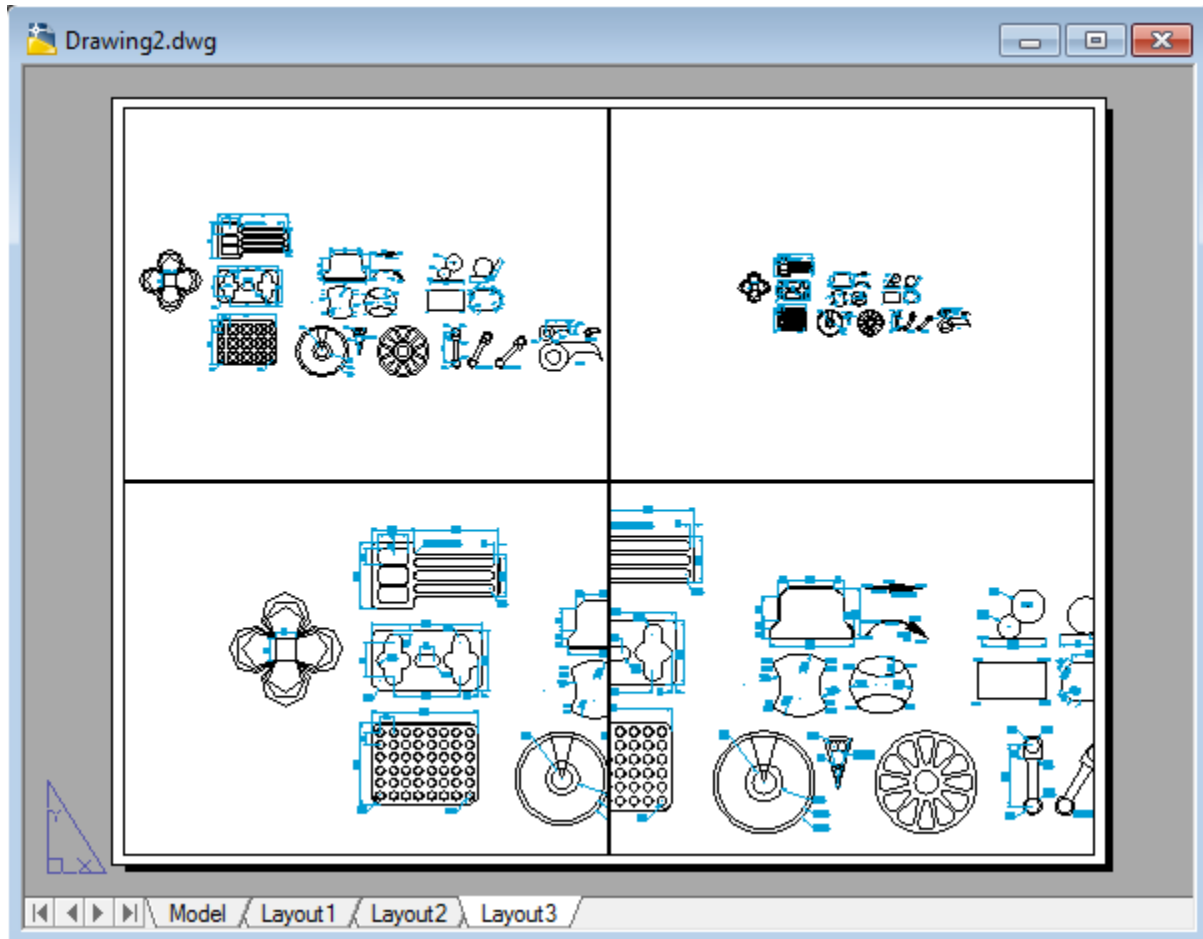
To Create Viewport

- Click Menu Browser → View → Viewports → New Viewports...
- You can select standard viewports



OR

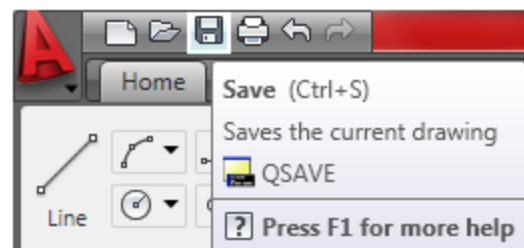
- Click Menu Browser → View → Viewports → 4 Viewports.



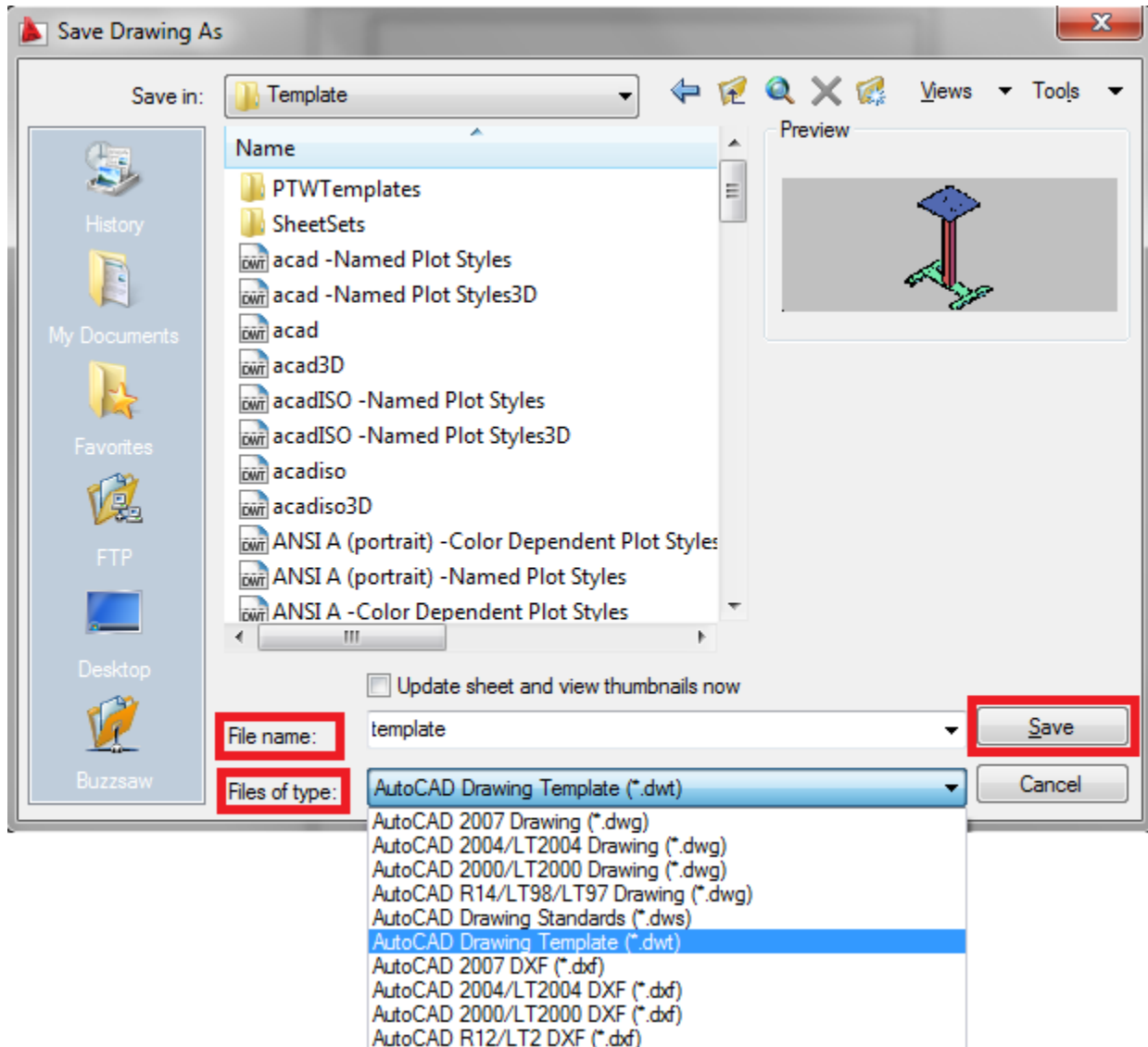
To Create Template file

Template file contains predefined settings, standards, and definitions that will save you significant setup time. Template files include settings and basic drawing elements that you will use often, such as: Unit type and precision; Tool settings and preferences; Layer organization; Title blocks, borders, and logos; Dimension styles; Text styles; Plot styles; Linetypes and lineweights.

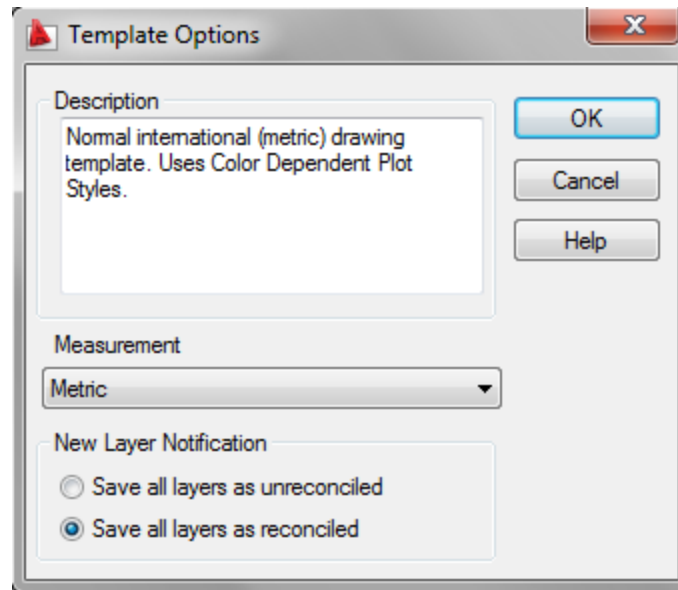
- Menu Browser → File → Save as...
- OR
- Click the Save icon in the Quick Access toolbar.



- In the Save Drawing As dialog box, save to the default template folder (or select the folder in which you want to store the template file). Enter file name in the File name box.



- Select **AutoCAD Drawing Template (*.dwt)** in Files of type.
- Pick **Save** in the Save Drawing As dialog box to close the dialog box.
- In the Template Options dialog box, you can set the Description, Measurement and New Layer Notification.



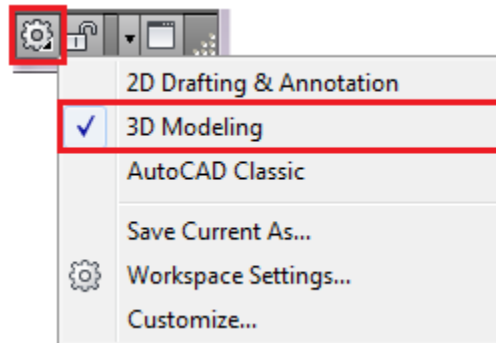
- Pick **OK** to close the dialog box and save the template file.

AUTOCAD 2009

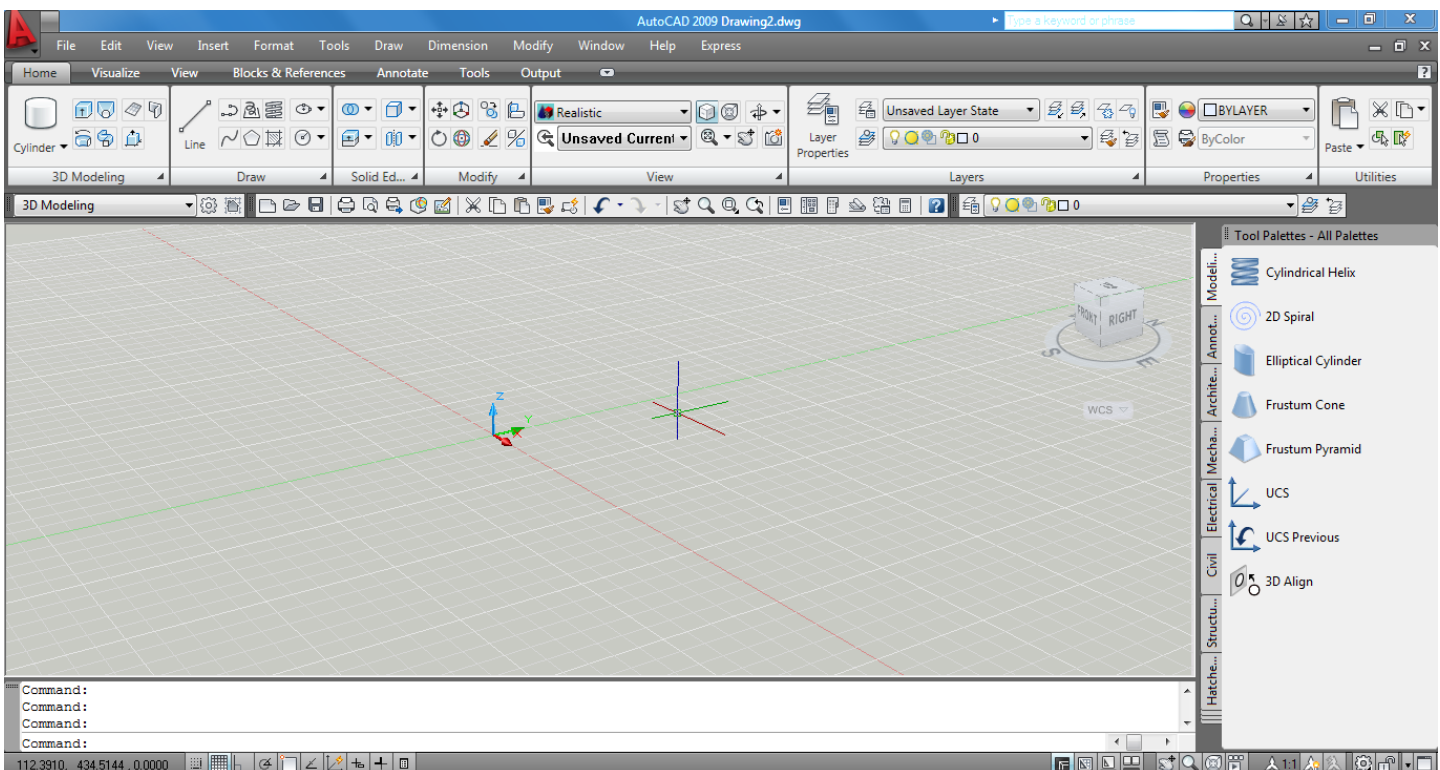
3D DESIGN

THE 3D MODELING WORKSPACE

- ▶ It used to construct 3D modeling drawings
- ▶ To set workspaces Click the Menu Browser → Tools → Workspaces → 3D Modeling OR
- ▶ To set workspaces click on the **Workspace Switching** icon from the status bar.

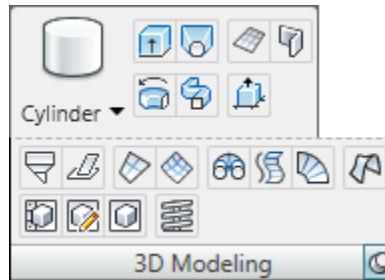


- ▶ The 3D Modeling workspace appears.



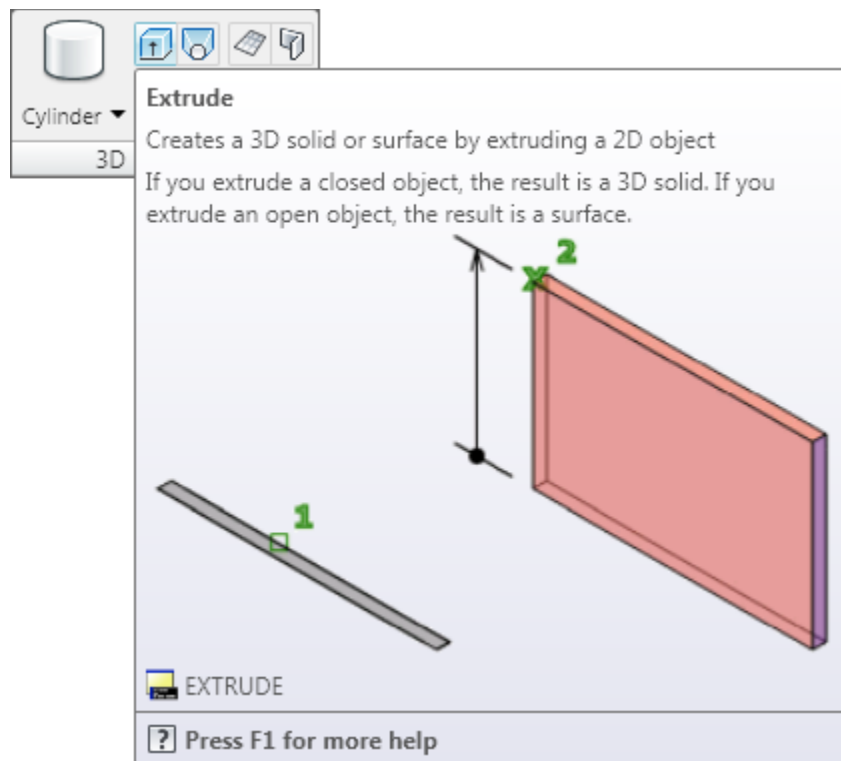
3D MODELING COMMAND

- ❑ **3D Modeling:** used to create 3D solids and surfaces from scratch or from existing objects. It contains different tools such as extrude, loft, revolve, polysolid, sweep, press/pull, helix, cylinder, box, sphere, cone, wedge and pyramid.

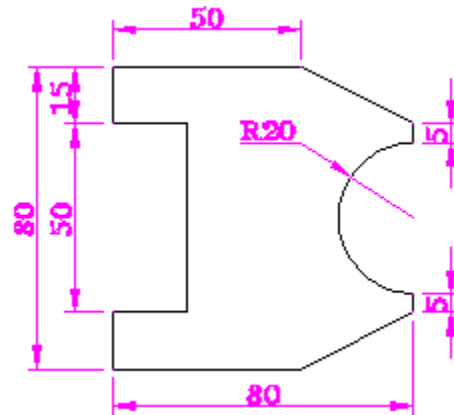


Extrude Tool : used to create solids and surfaces by extruding selected objects. If the extrude object is closed, the resulting object is a solid otherwise the object is a surface.

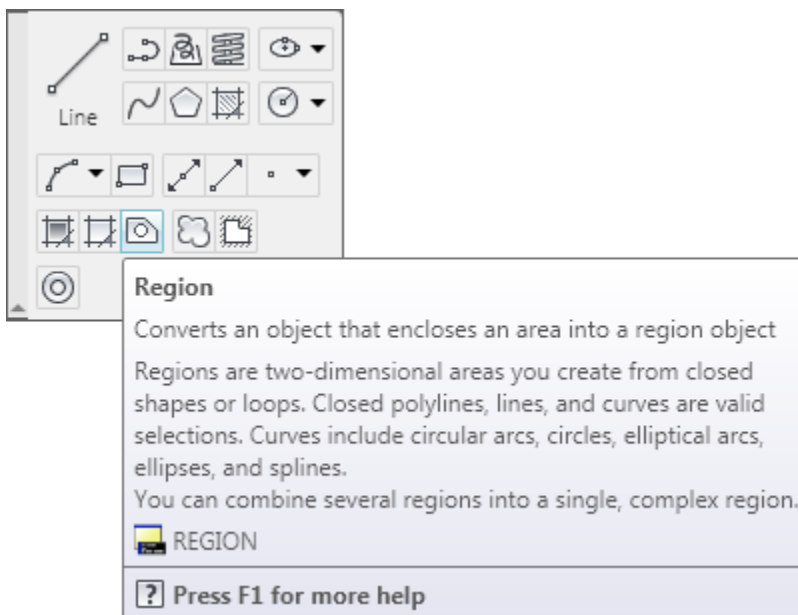
- ▶ Select **Extrude** from 3D modeling panel **OR**
- ▶ In command line write **extrude** or **ext** then Enter



Example: Direct Extrude:

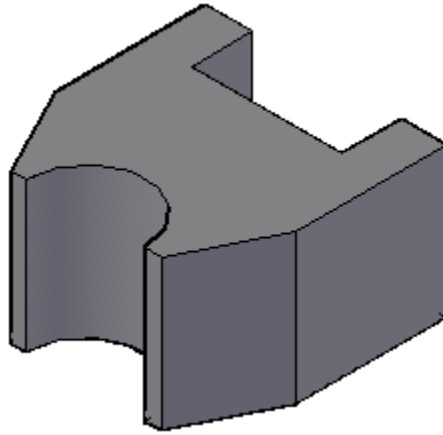


- ▶ First construct the above drawing using different necessary commands in the given dimension then
- ▶ Click the **Region** tool  from the draw panel **OR** write **region** or **reg** on the command line then press **Enter**.



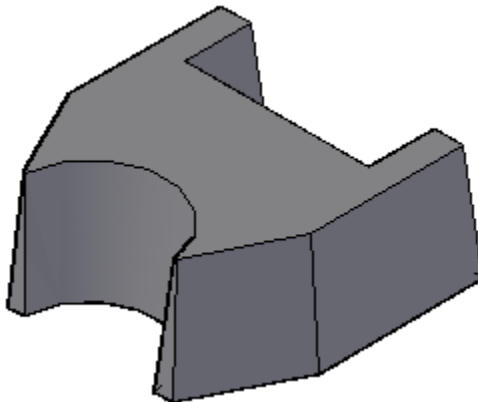
- ▶ Command: `_region`
Select objects: Specify opposite corner: 12 found
Select objects: press **Enter** or **Right Click**.
1 Region created.
- ▶ Click on **Extrude** tool
- ▶ Command: `_extrude` Current wire frame density: ISOLINES = 4
- ▶ Select objects to extrude: **pick** 1 found

- ▶ Select objects to extrude: press **Enter** or **Right Click**.
- ▶ Specify height of extrusion or [Direction / Path / Taper angle]: **50** then **Enter**.

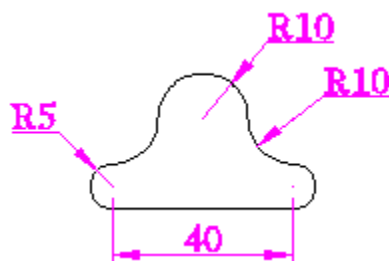


Example: Taper angle Extrude:

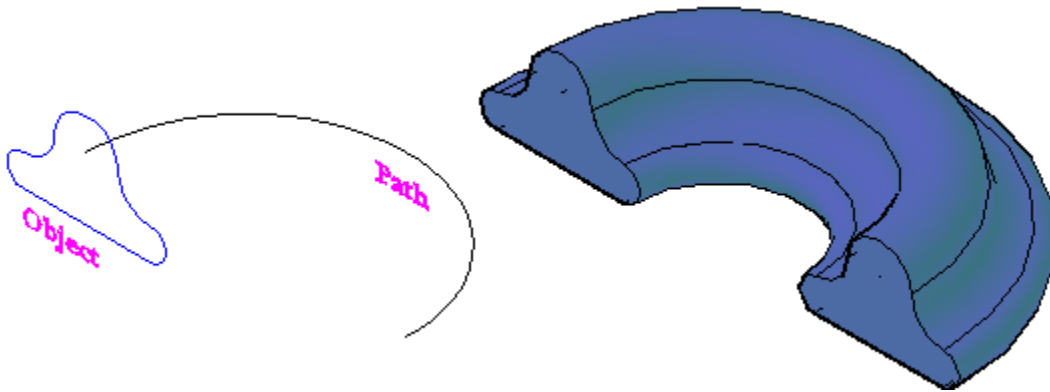
- ▶ Command: `_extrude` Current wire frame density: ISOLINES = 4
- ▶ Select objects to extrude: **pick 1** found
- ▶ Select objects to extrude: press **Enter** or **Right Click**.
- ▶ Specify height of extrusion or [Direction / Path / Taper angle]: **t** then **Enter**.
- ▶ Specify angle of taper for extrusion <0>: **5** then **Enter**
- ▶ Specify height of extrusion or [Direction / Path / Taper angle]: **35** then **Enter**.



Example: Path Extrude:



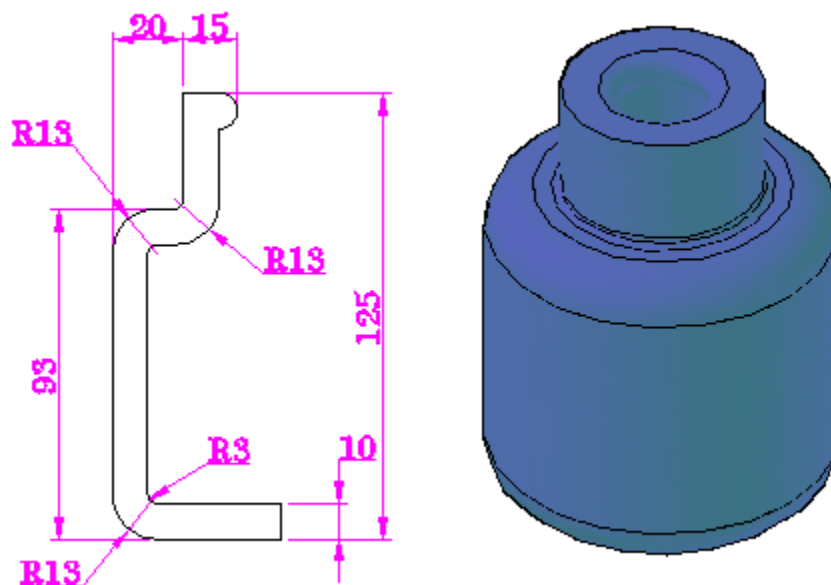
- ▶ First construct the above drawing using different necessary commands in the given dimension then Region the object.
- ▶ Click on **Extrude** tool
- ▶ Command: `_extrude` Current wire frame density: ISOLINES = 4
- ▶ Select objects to extrude: **pick 1 found**
- ▶ Select objects to extrude: press **Enter** or **Right Click**.
- ▶ Specify height of extrusion or [Direction / Path / Taper angle]: **p** then **Enter**.
- ▶ Select extrusion path or [Taper angle]: **pick the path**.



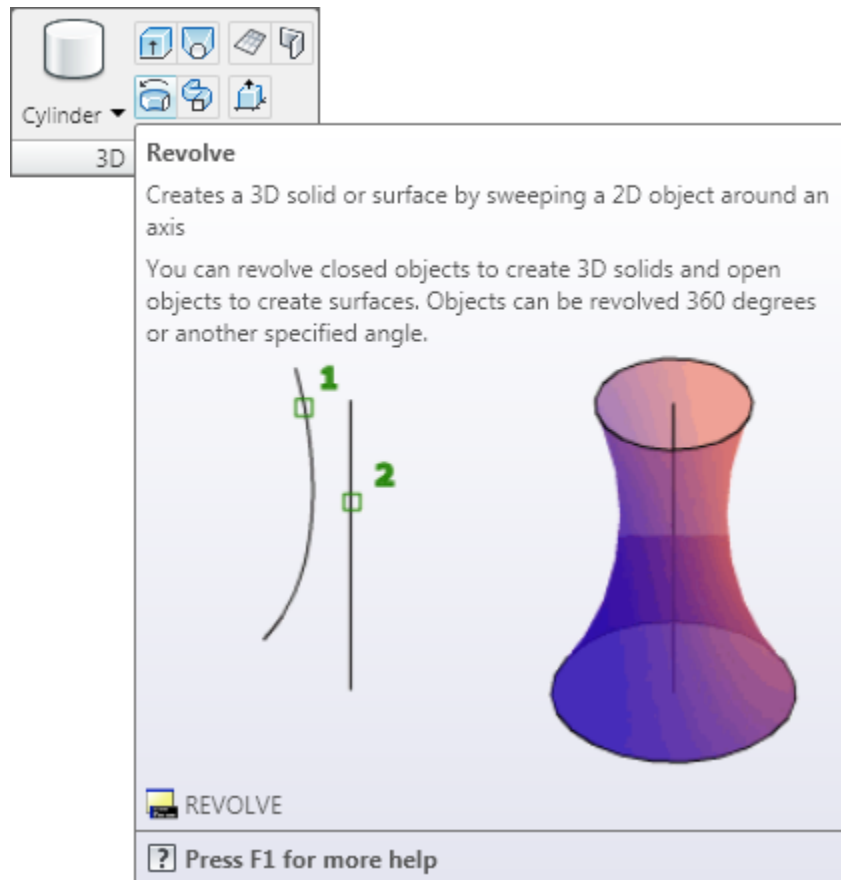
Revolve Tool : used to create a solid or surface by revolving or sweeping open or closed objects about an axis.

- ▶ Select **Revolve** from 3D modeling panel **OR**
- ▶ In command line write **revolve** or **rev** then Enter

Example: click on revolve tool icon and follow the step below:

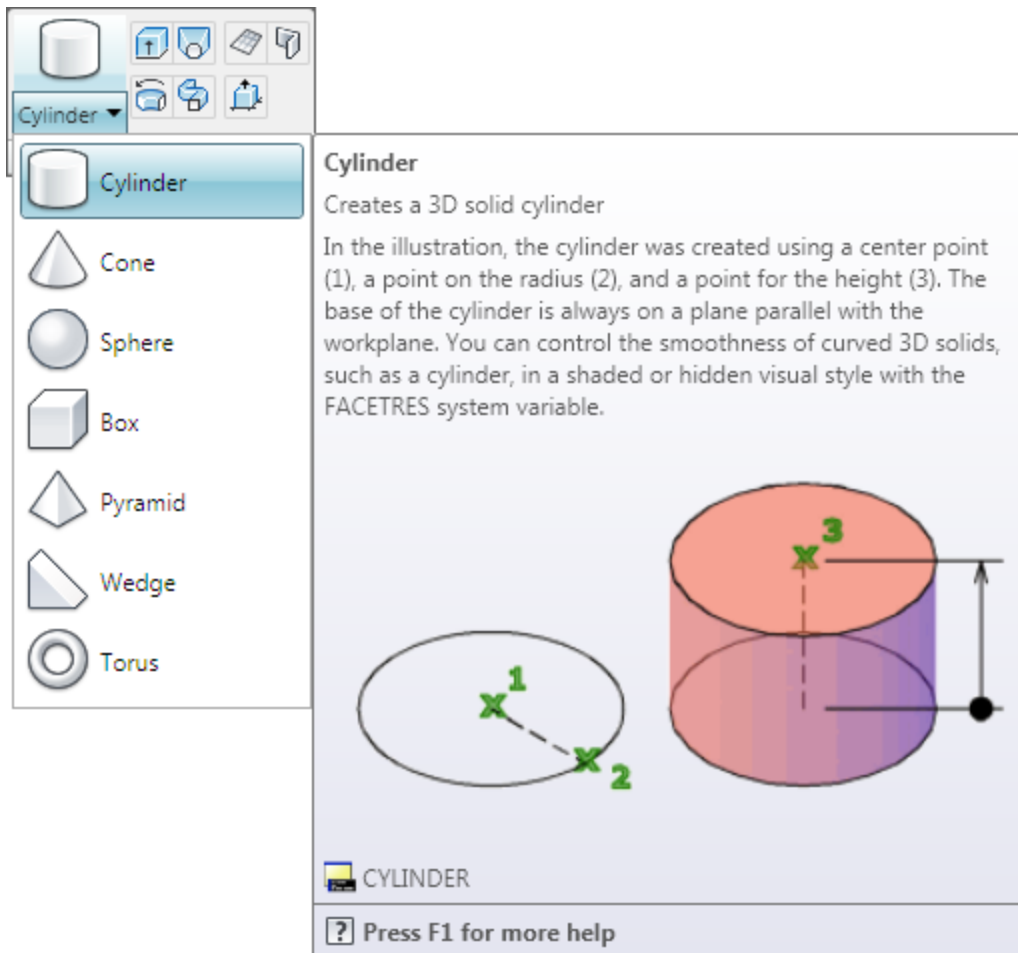


- ▶ First construct the above drawing using different necessary commands in the given dimension then
- ▶ Command: `_revolve` Current wire frame density: ISOLINES = 4
- ▶ Select objects to revolve: Specify opposite corner: 17 found
- ▶ Select objects to revolve: press **Enter**
- ▶ Specify axis start point or define axis by [Object / X / Y / Z /] <objects>: **pick**
- ▶ Specify axis endpoint: **pick**
- ▶ Specify angle of revolution or [Start angle] <360>: press **Enter**.



Cylinder : used to create a solid cylinder with a circular or an elliptical base.

- ▶ Select **Cylinder** from 3D modeling panel **OR**
- ▶ In command line write **cylinder** or **cyl** then Enter



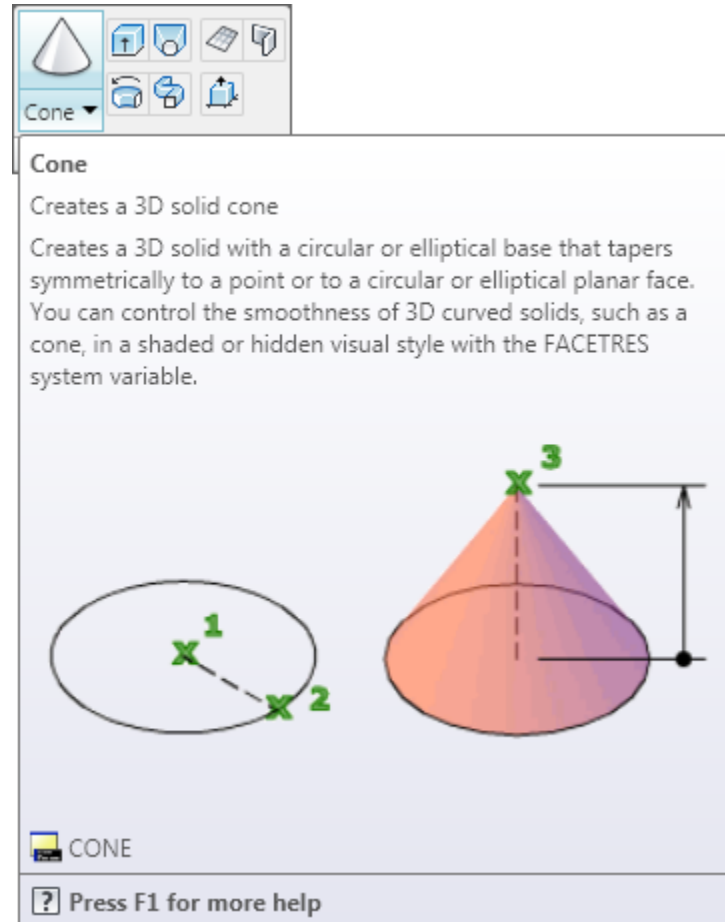
Example: click on cylinder tool icon and follow the step below:



- ▶ Command: `_cylinder` Specify center point of base or [3P / 2P / Ttr / Elliptical]: **pick**
- ▶ Specify base radius or [Diameter]: **5** then **Enter**
- ▶ Specify height or [2Point / Axis endpoint]: **40** then **Enter**

Cone : create a solid cone with a circular or an elliptical base tapering to a point and also create a cone frustum, which tapers to a circular or elliptical planar face that is parallel to its base.

- ▶ Select **Cone** from 3D modeling panel **OR**
- ▶ In command line write **cone** then Enter



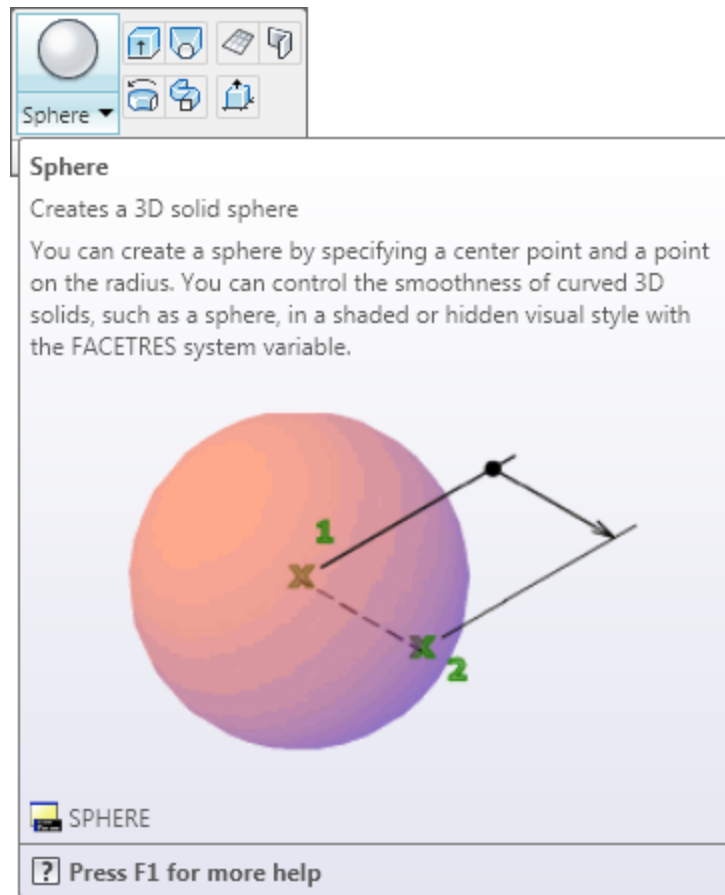
Example: click on cone tool icon and follow the step below:



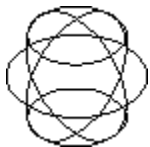
- ▶ Command: `_cone` Specify center point of base or [3P / 2P / Ttr / Elliptical]: **pick**
- ▶ Specify base radius or [Diameter]: **25** then **Enter**
- ▶ Specify height or [2Point / Axis endpoint / Top radius]: **t** then **Enter**
- ▶ Specify top radius: **5** then **Enter**
- ▶ Specify height or [2Point / Axis endpoint / Top radius]: **-40** then **Enter**

Sphere : used to create a solid sphere.

- ▶ Select **Sphere** from 3D modeling panel **OR**
- ▶ In command line write **sphere** then Enter



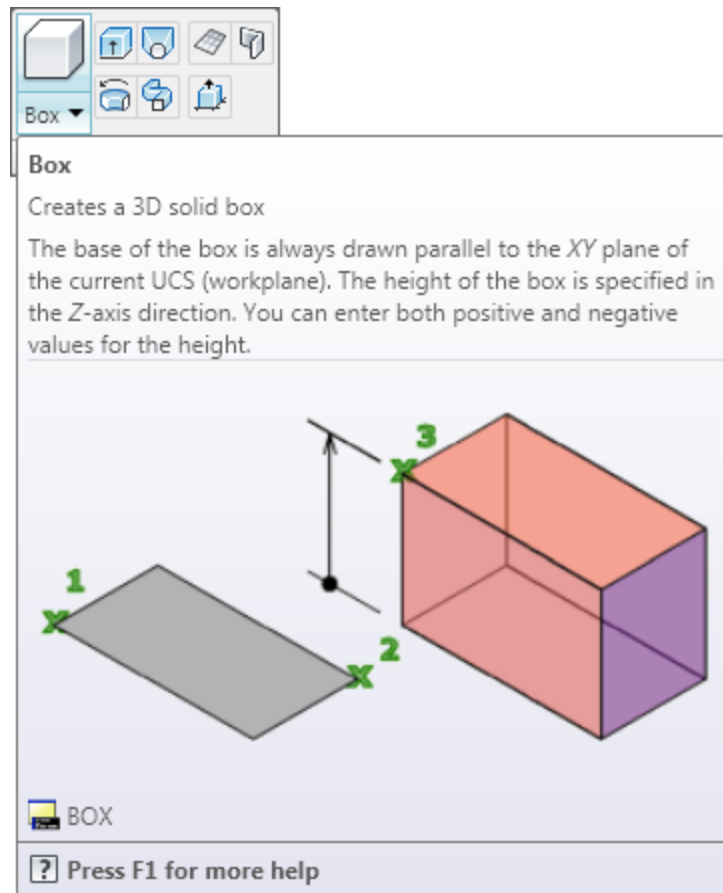
Example: click on sphere tool icon and follow the step below:



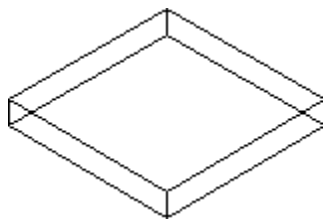
- ▶ Command: `_sphere` Specify center point or [3P / 2P / Ttr]: **pick**
- ▶ Specify radius or [Diameter]: **22** then **Enter**

Box : used to create a solid box by using cube or length option. The base of the box is always drawn parallel to the XY plane of the current UCS (workplane).

- ▶ Select **Box** from 3D modeling panel **OR**
- ▶ In command line write **box** then Enter



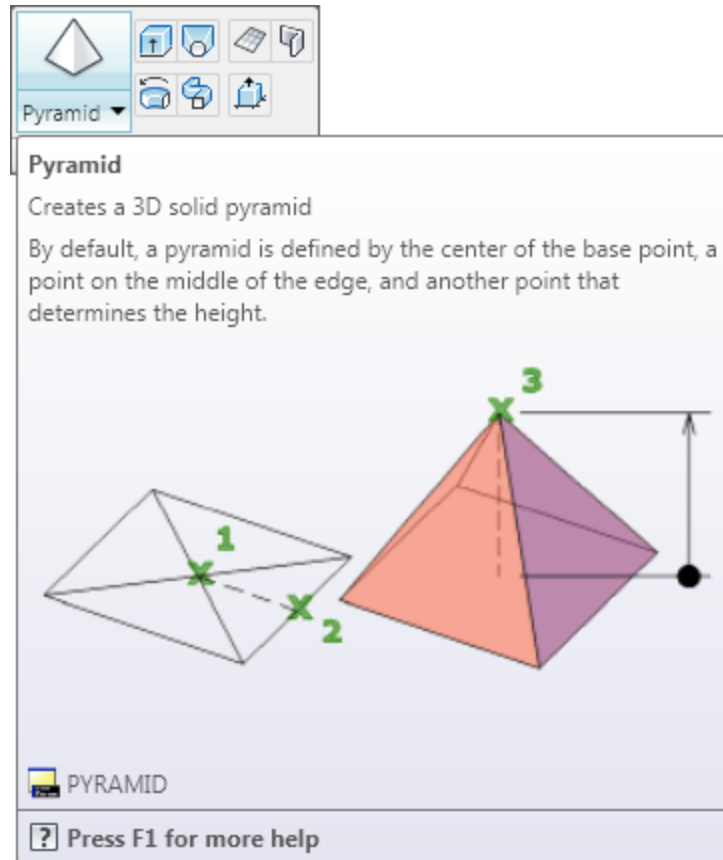
Example: click on box tool icon and follow the step below:



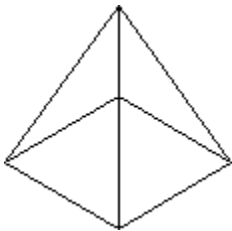
- ▶ Command: `_box` Specify first corner or [Center]: **pick**
- ▶ Specify other corner or [Cube / Length]: **1** then **Enter**
- ▶ Specify length: **50** then **Enter**
- ▶ Specify width: **50** then **Enter**
- ▶ Specify height or [2Point]: **8** then **Enter**

Pyramid : used to create a solid pyramid and can define the number of sides for a pyramid from 3 to 32.

- ▶ Select **Pyramid** from 3D modeling panel **OR**
- ▶ In command line write **pyramid** and **pyr** then Enter



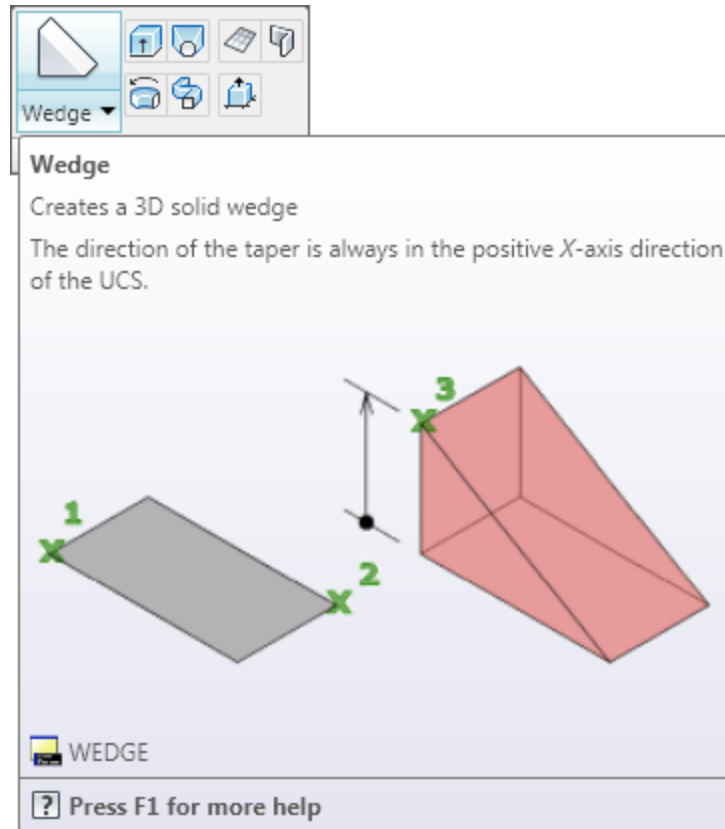
Example: click on pyramid tool icon and follow the step below:



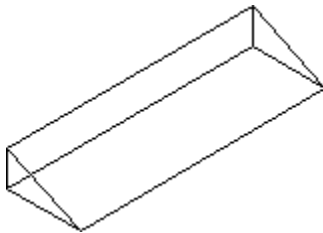
- ▶ Command: `_pyramid` 4 sides Circumscribed
- ▶ Specify center point of base or [Edge / Sides]: **pick**
- ▶ Specify base radius or [Inscribed]: **25** then **Enter**
- ▶ Specify height or [2Point / Axis endpoint / Top radius]: **60** then **Enter**

Wedge : used to create a solid wedge. The base of the wedge is drawn parallel to the *XY* plane of the current UCS with the sloped face opposite the first corner. The height of the wedge is parallel to the *Z* axis.

- ▶ Select **Wedge** from 3D modeling panel **OR**
- ▶ In command line write **wedge** and **we** then Enter



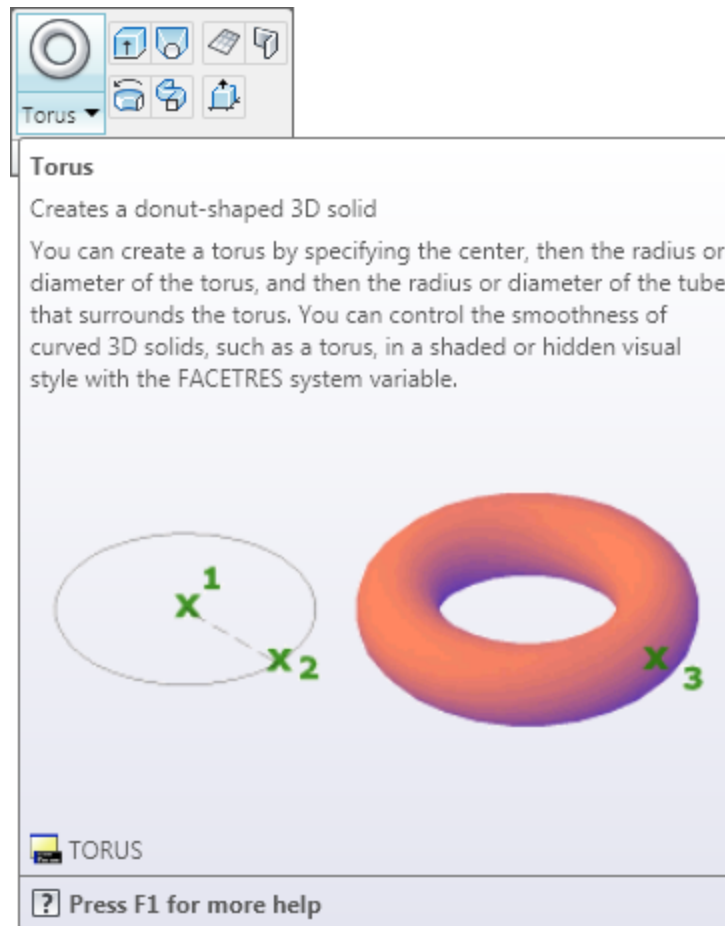
Example: click on wedge tool icon and follow the step below:



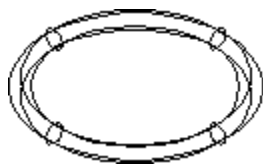
- ▶ Command: `_wedge` Specify first corner or [Center]: **pick**
- ▶ Specify other corner or [Cube / Length]: **1** then **Enter**
- ▶ Specify length: **15** then **Enter**
- ▶ Specify width: **50** then **Enter**
- ▶ Specify height or [2Point]: **8** then **Enter**

Torus : used to create a ring-shaped solid similar to the inner tube of a tire. A torus is defined by two radius values, one for the tube and the other for the distance from the center of the torus to the center of the tube.

- ▶ Select **Torus** from 3D modeling panel **OR**
- ▶ In command line write **torus** and **tor** then Enter

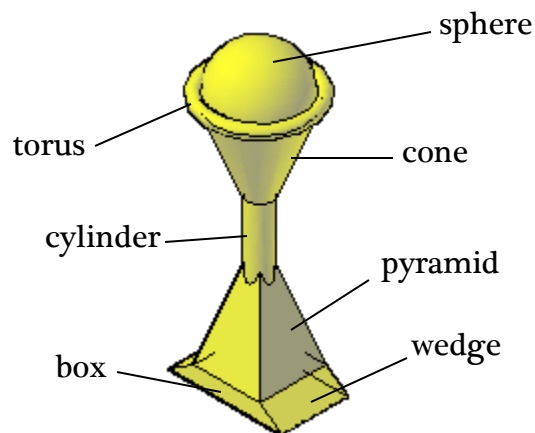


Example: click on wedge tool icon and follow the step below:



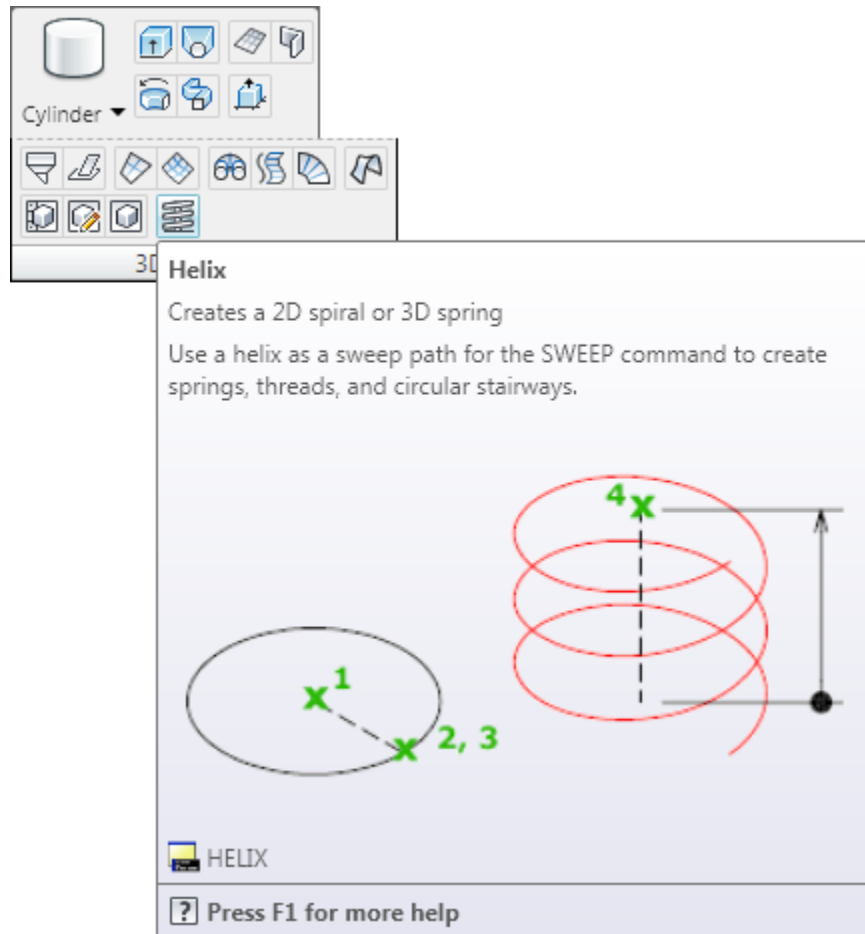
- ▶ Command: `_torus` Specify center point or [3P / 2P / Ttr]: **pick**
- ▶ Specify radius or [Diameter]: **25** then **Enter**
- ▶ Specify tube radius or [2Point / Diameter]: **2.5** then **Enter**

Combination Result of the above 3D Modeling

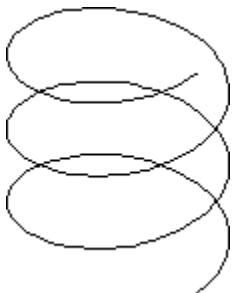


Helix : used to create a 2D spiral or 3D spring.

- ▶ Select **Helix** from 3D modeling panel **OR**
- ▶ In command line write **helix** then Enter



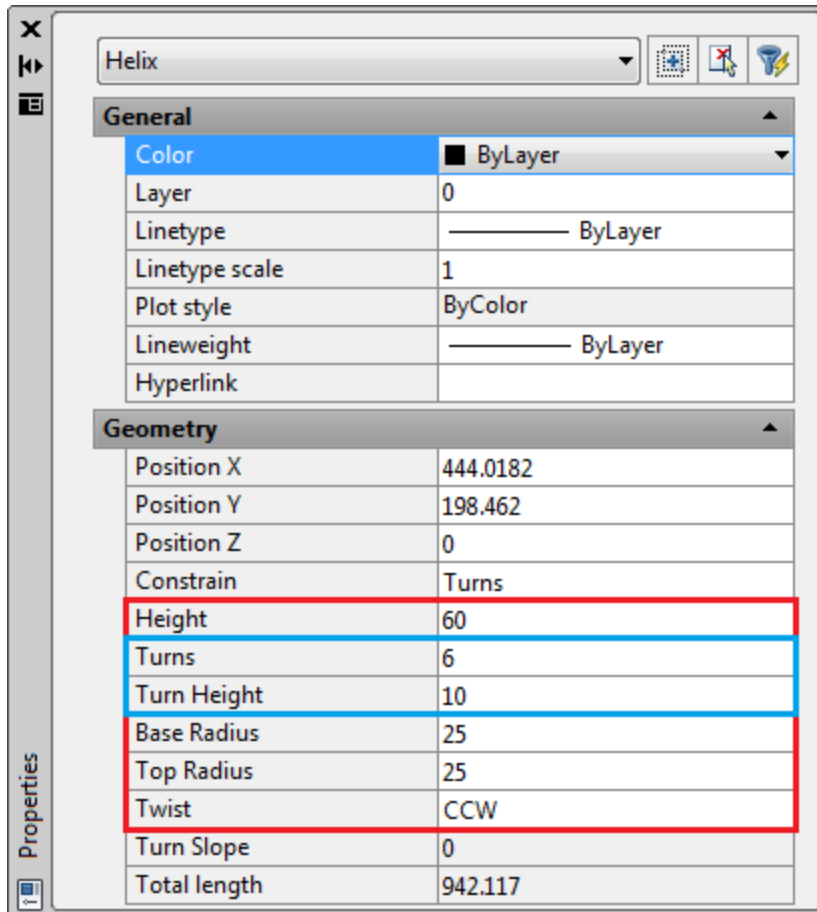
Example: click on helix tool icon and follow the step below:



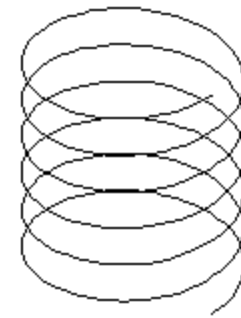
- ▶ Command: **_Helix** Number of turns = 3 Twist=CCW
- ▶ Specify center point of base: **pick**
- ▶ Specify base radius or [Diameter]: **25** then **Enter**
- ▶ Specify top radius or [Diameter] **<25>**: press **Enter**
- ▶ Specify helix height or [Axis endpoint / Turns / turn Height / tWist]: **60** then **Enter**

You can adjust the helix turn, turn height, base radius, top radius, twist and height.

- ▶ Double click on the helix then the properties dialog box appearing

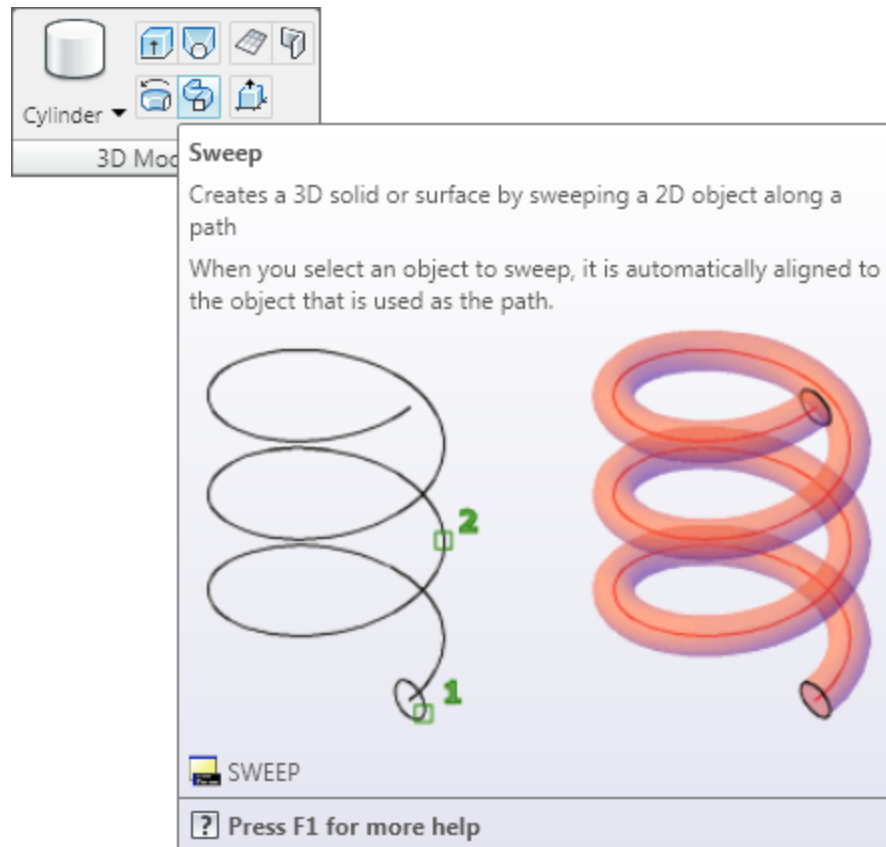


Adjusted Helix



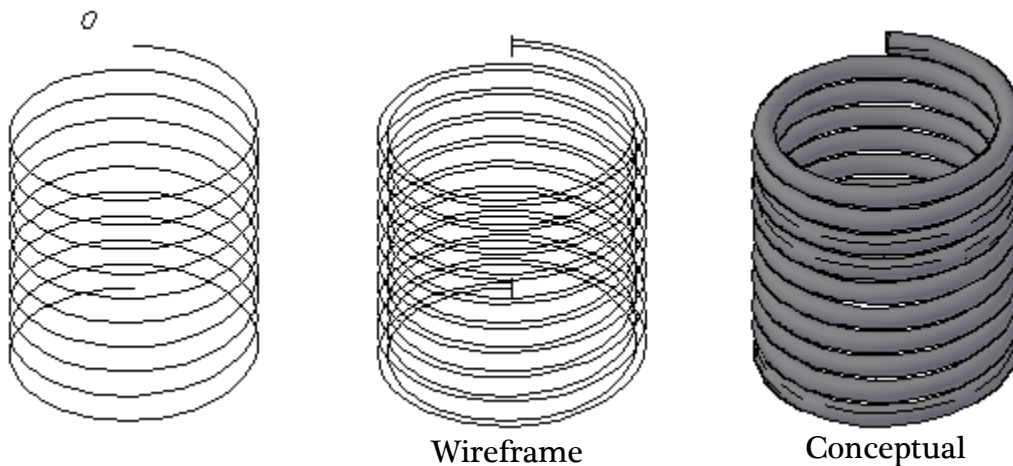
Sweep : used to create a solid or surface in the shape of the specified profile (the swept object) along the specified 2D or 3D path. If you sweep a closed curve along a path the result is a solid. If you sweep an open curve along a path, the result is a surface. Sweeping is different from extruding. When you sweep a profile along a path, the profile is moved and aligned normal (perpendicular) to the path. Then the profile is swept along the path.

- ▶ Select **Sweep** from 3D modeling panel **OR**
- ▶ In command line write **sweep** then Enter



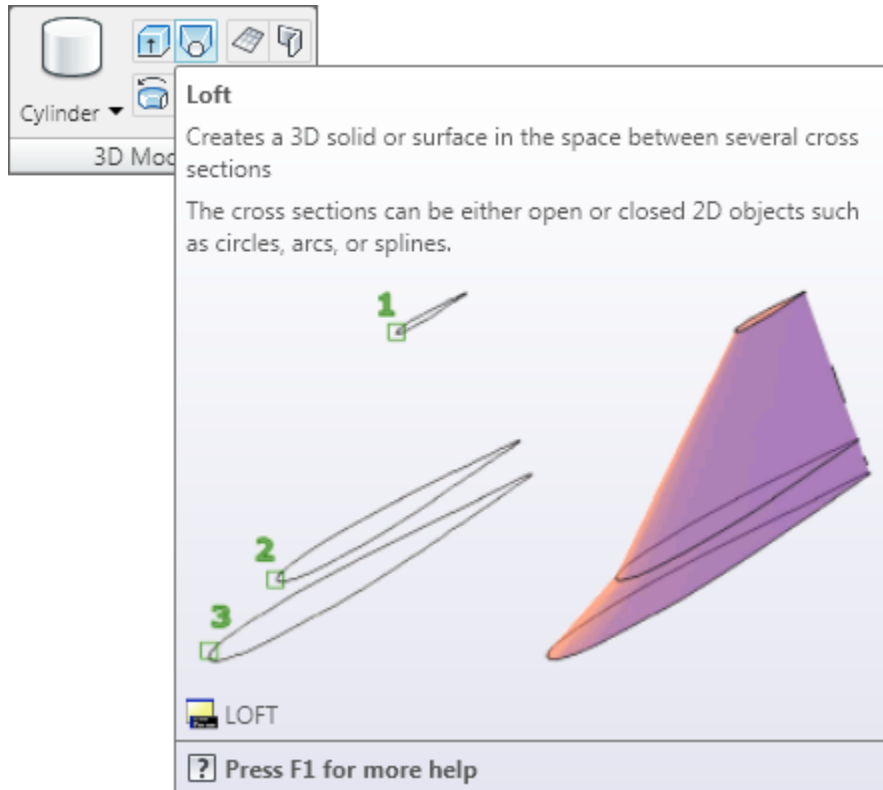
Example: click on sweep tool icon and follow the step below:

- ▶ Command: `_sweep` Current wire frame density: ISOLINES=4
- ▶ Select objects to sweep: **pick** (circle R = 2), 1 found
- ▶ Select objects to sweep: press **Enter**
- ▶ Select sweep path or [Alignment / Base point / Scale / Twist]: **pick** (helix)



Loft : used to create a 3D solid or surface by lofting (drawing a solid or surface) through a set of two or more cross-section curves.

- ▶ Select **Loft** from 3D modeling panel **OR**
- ▶ In command line write **loft** then Enter



Example: click on loft tool icon and follow the step below:

- ▶ Command: `_loft` Select cross sections in lofting order: pick, 3 total
- ▶ Select cross section in lofting order: press Enter or Right Click
- ▶ Enter an option [Guides / Path / Cross sections only] <Cross sections only>: press **Enter**
- ▶ Loft setting dialog box appearing then select **Normal to:** and click **OK**

Circle:

- Radius = 10

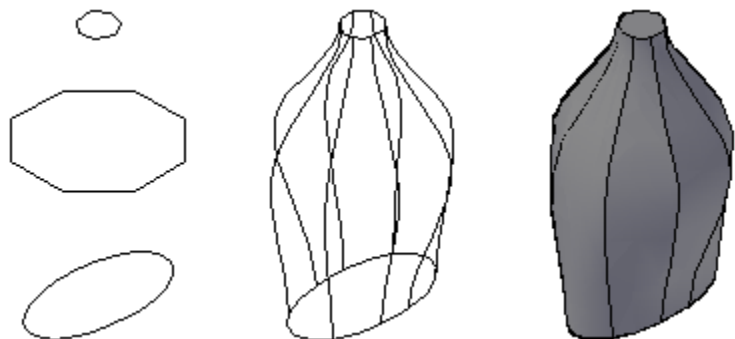
Octagon:

- Inscribed in Circle R = 40

Ellipse:

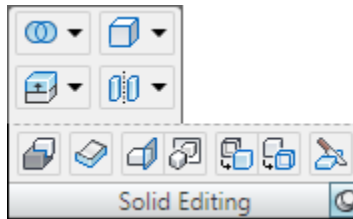
- Major Axis = 80
- Minor Axis = 40

Distance between each object = 70



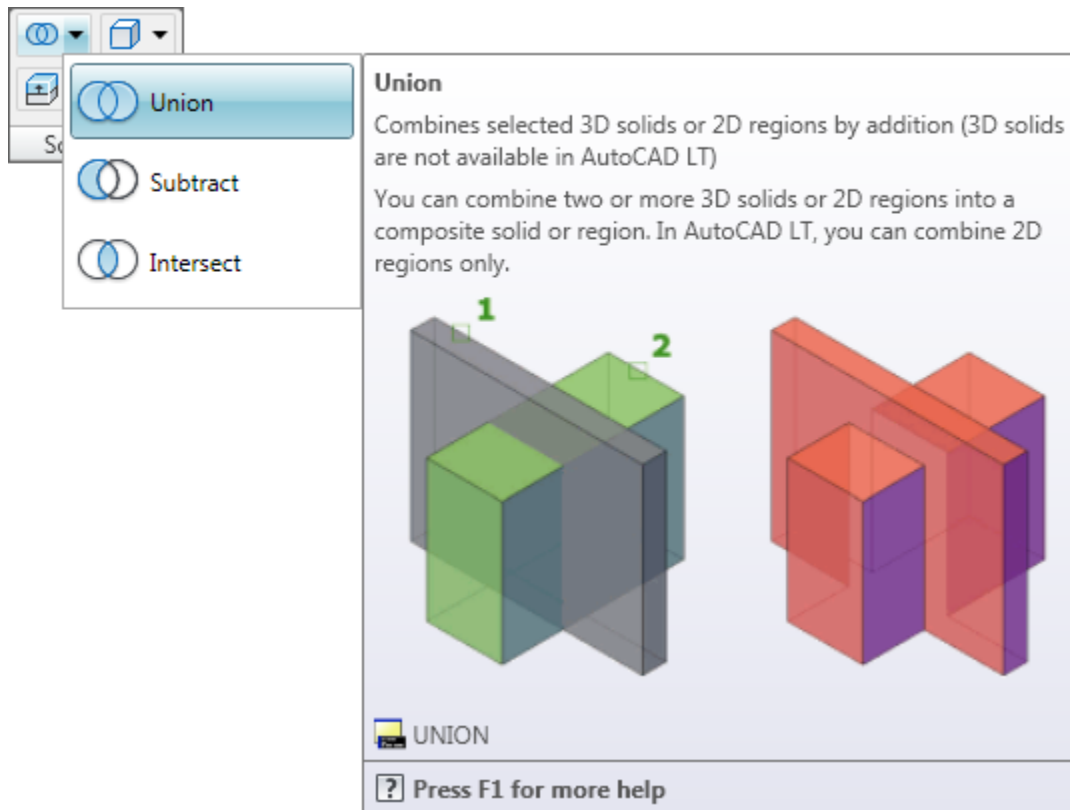
SOLID EDITING COMMAND

□ **Solid Editing:** contain tools such as union, subtract and intersect.



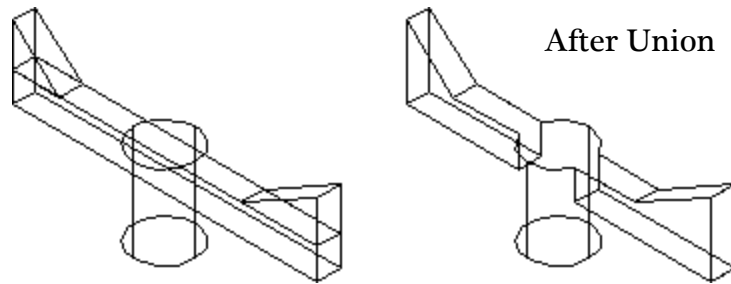
Union : used to combine the total volume of two or more solids or two or more regions into a composite object.

- ▶ Select **Union** from solid editing panel **OR**
- ▶ In command line write **union** and **uni** then Enter



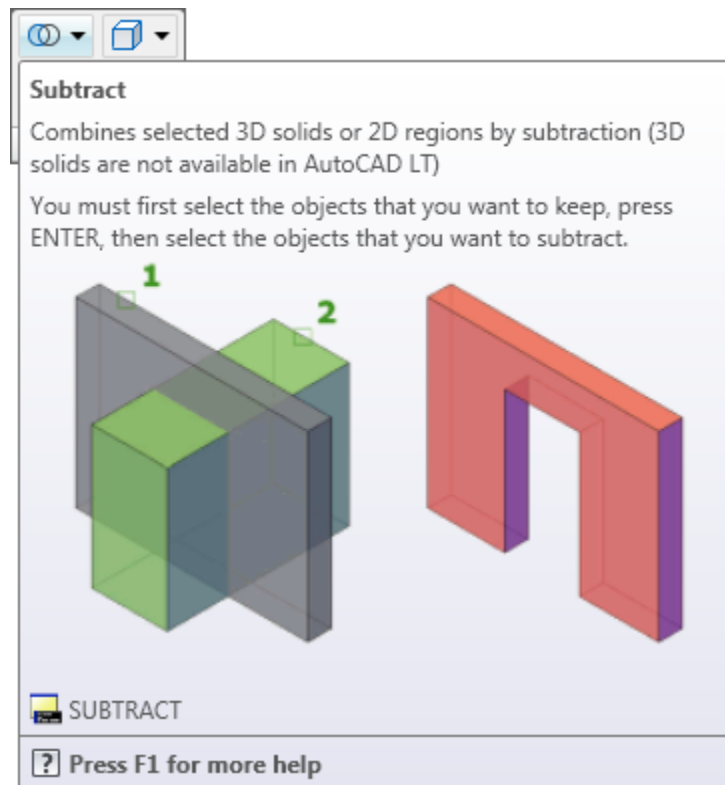
Example: click on union tool icon and follow the step below:

- ▶ Command: **_union**
- ▶ Select objects: **pick** Specify opposite corner: 4 found
- ▶ Select objects: press **Enter** or **Right Click**



Subtract : used to remove the common area of one set of solids from another. For example, you can use subtract to add holes to a mechanical part by subtracting cylinders from the object.

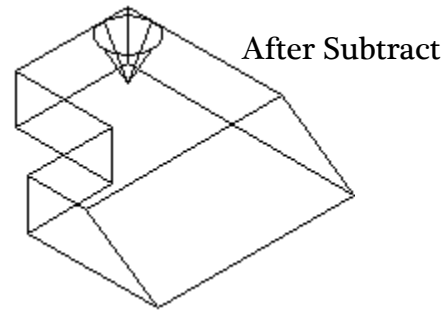
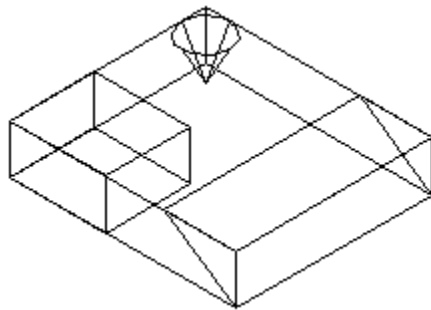
- ▶ Select **Subtract** from solid editing panel **OR**
- ▶ In command line write **subtract** and **su** then Enter



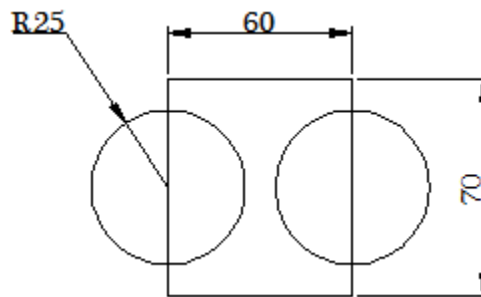
Example: click on subtract tool icon and follow the step below:

- ▶ Command: `_subtract` Select solids and regions to subtract from . .
- ▶ Select objects: **pick** (box), 1 found then press **Enter**
- ▶ Select solids and regions to subtract . .
- ▶ Select objects: **pick** (cone box and wedge) 3 total

- Select objects: press **Enter**



Example: Union and Subtract tool in Regions



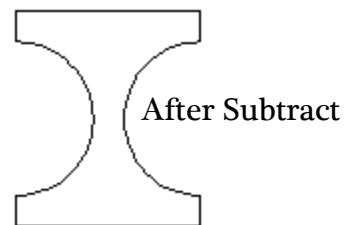
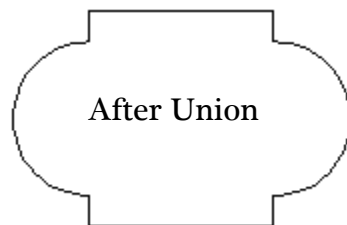
- Draw the above drawing in the given dimension and make region.

UNION

- Command: `_union`
- Select objects: **pick**, 3 total
- Select objects: press **Enter** or **Right Click**

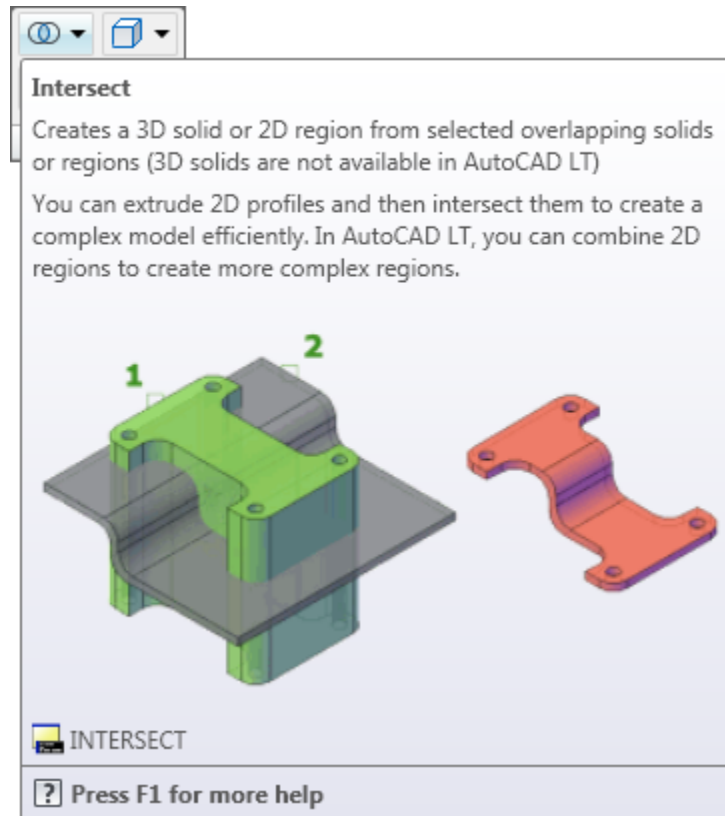
SUBTRACT

- Command: `_subtract` Select solids and regions to subtract from . .
- Select objects: **pick** (rectangle), 1 found then press **Enter**
- Select solids and regions to subtract . .
- Select objects: **pick** (circles) 2 total
- Select objects: press **Enter**



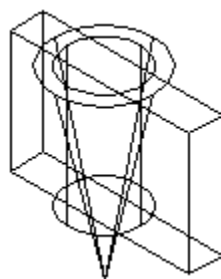
Intersect : used to create a composite solid from the common volume of two or more overlapping solids.

- ▶ Select **Intersect** from solid editing panel **OR**
- ▶ In command line write **intersect** and **in** then Enter



Example: click on intersect tool icon and follow the step below:

- ▶ Command: `_intersect`
- ▶ Select objects: **pick**, 3 found
- ▶ Select objects: press **Enter** or **Right Click**

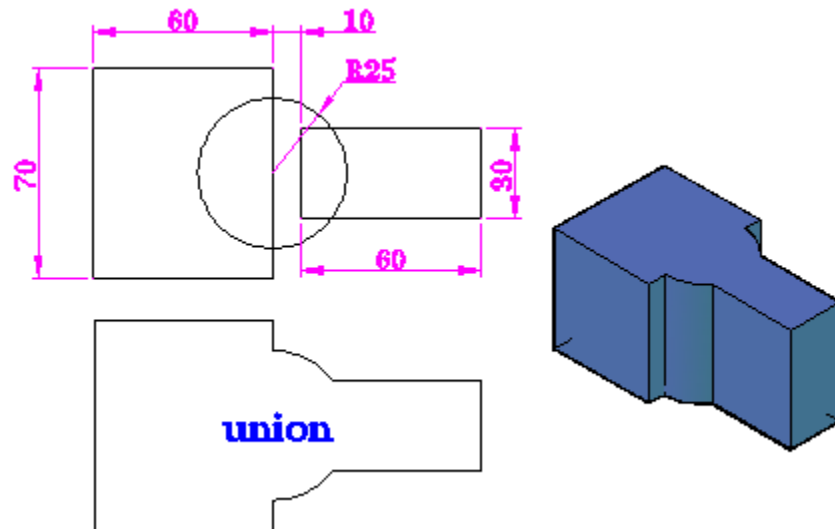


After intersect

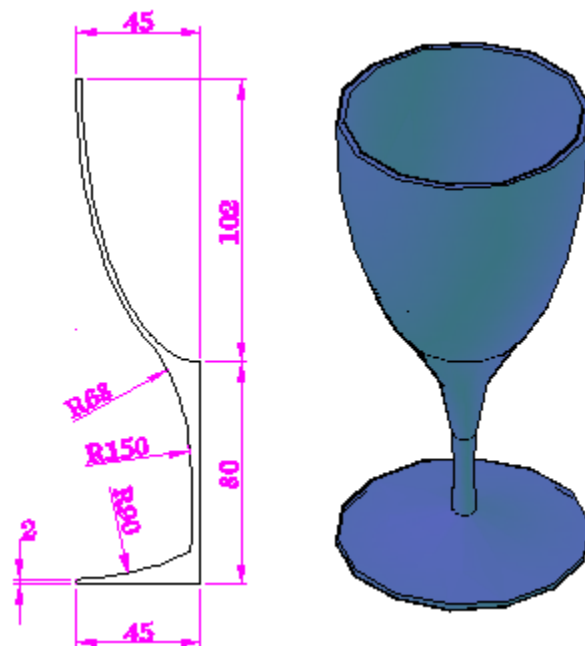


Exercise

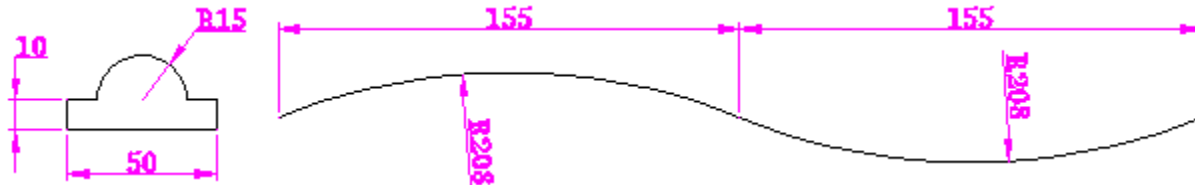
1. Construct 3D model using extrude tool as show below. First construct the outline in the given dimensions, change it into region and union the objects. Use extrusion height 65.



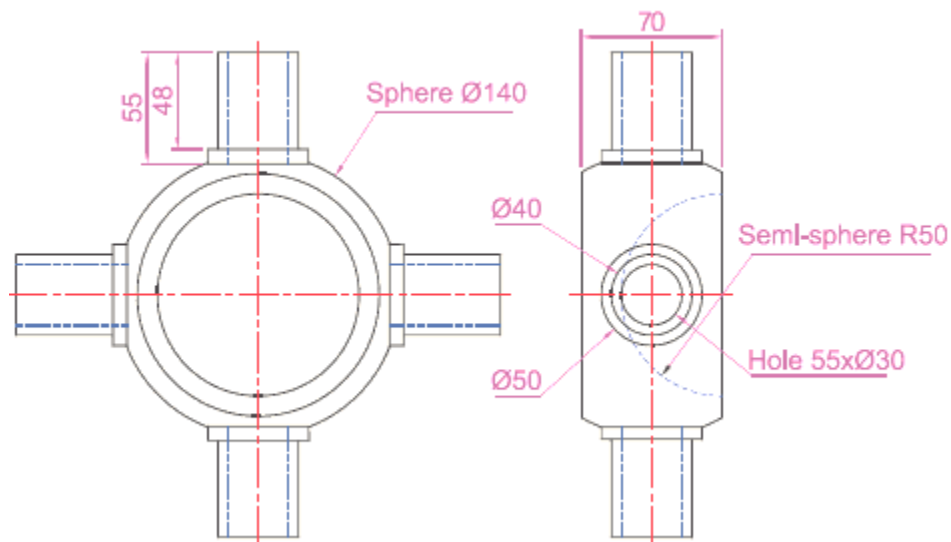
2. Construct the 3D model of a wine glass as shown below, working to the dimensions given in the outline drawing. You will need to construct the outline and change it into a region before being able to change the outline into a solid of revolution using the Revolve tool from the Home/3D Modeling panel.



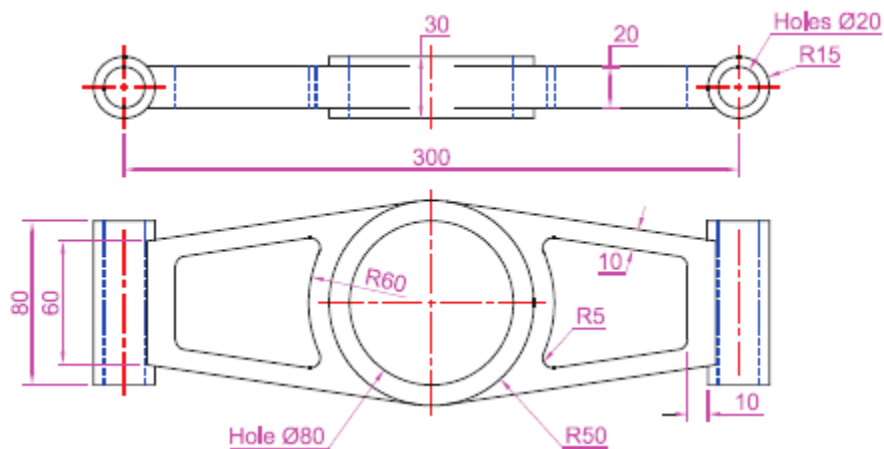
3. Construct 3D model using sweep tool, dimensions is given in the outline drawing as shown below.



4. Using the tools Sphere, Box, Union and Subtract and working to the dimensions given in below, construct the 3D solid model.



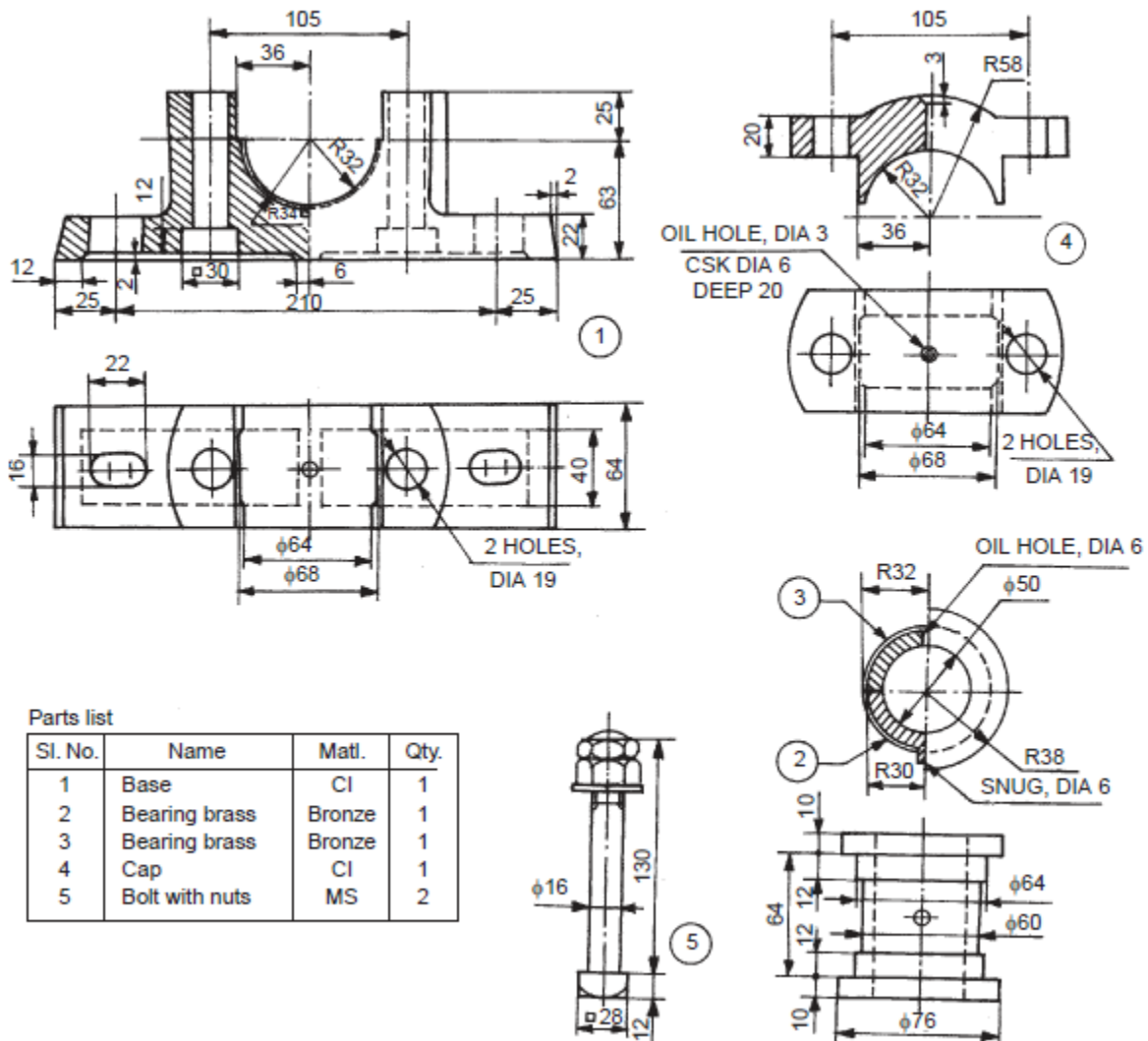
5. Construct the 3D solid model using the given dimensions below.



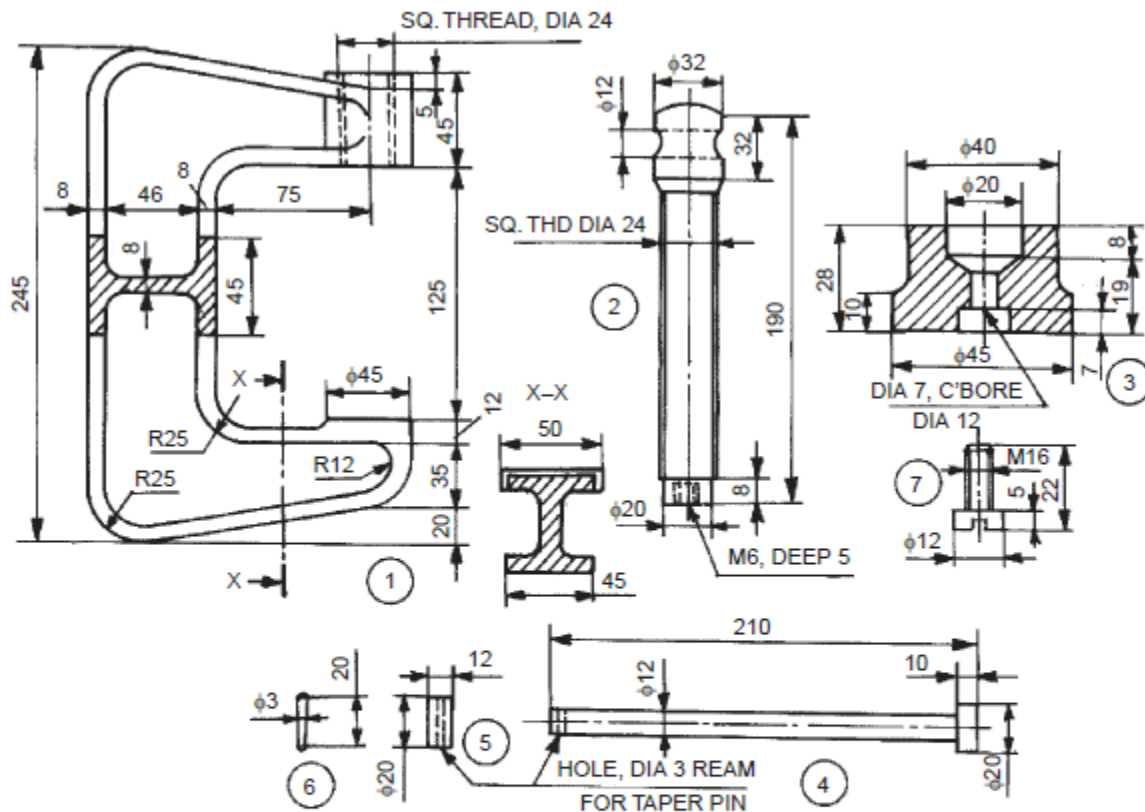
PROJECT

Assemble the parts and Draw the three view orthographic projection in an angle of your own choice and 3D Model of the Machine by using a suitable scale and appropriate sectional view.

Plummer Block



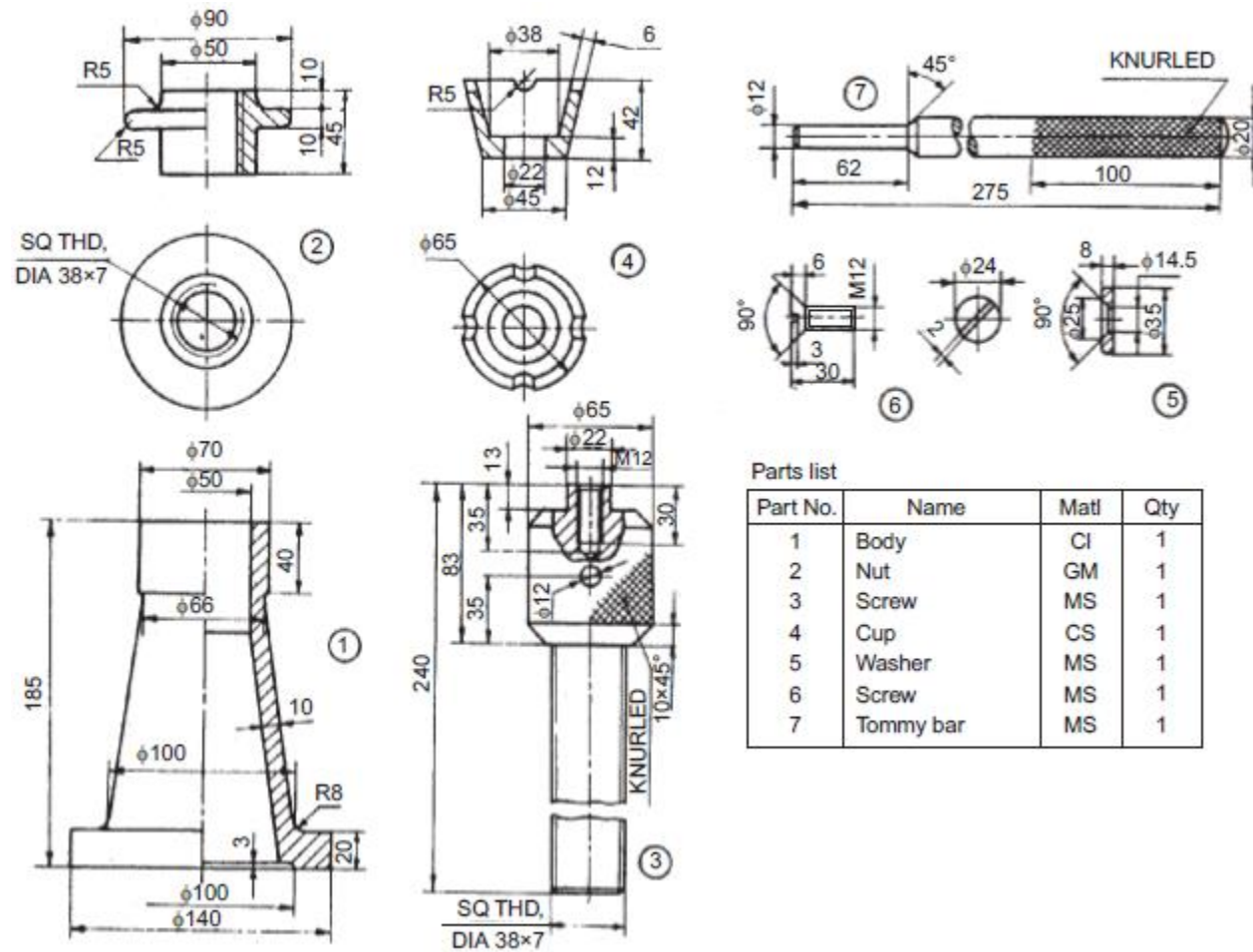
C- Clamp



Parts list

Sl. No.	Name	Matl.	Qty.
1	Frame	CI	1
2	Screw	MS	1
3	Pad	MS	1
4	Tommy bar	MS	1
5	Collar	MS	1
6	Pin	MS	1
7	Cap screw	MS	1

Screw Jack



Parts list

Part No.	Name	Matl	Qty
1	Body	CI	1
2	Nut	GM	1
3	Screw	MS	1
4	Cup	CS	1
5	Washer	MS	1
6	Screw	MS	1
7	Tommy bar	MS	1

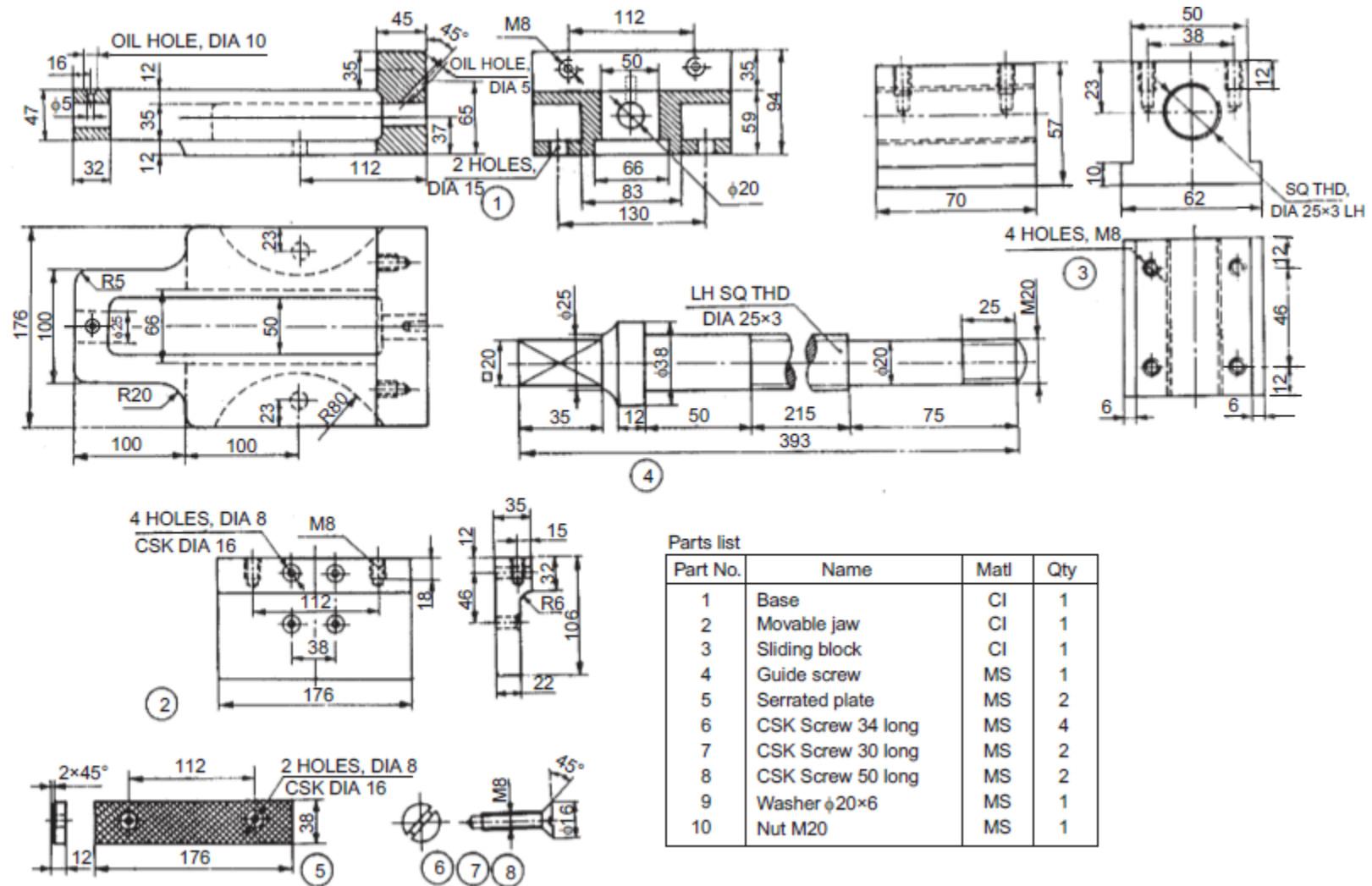
Technical drawing of a mechanical assembly with 9 numbered parts. The drawing includes various cross-sections (X-X, Y-Y) and detailed views of the assembly. The parts are:

- Tool holder
- Base plate
- Stud
- Clamping nut
- Handle
- Knob
- Set screw
- Grub screw
- Spring

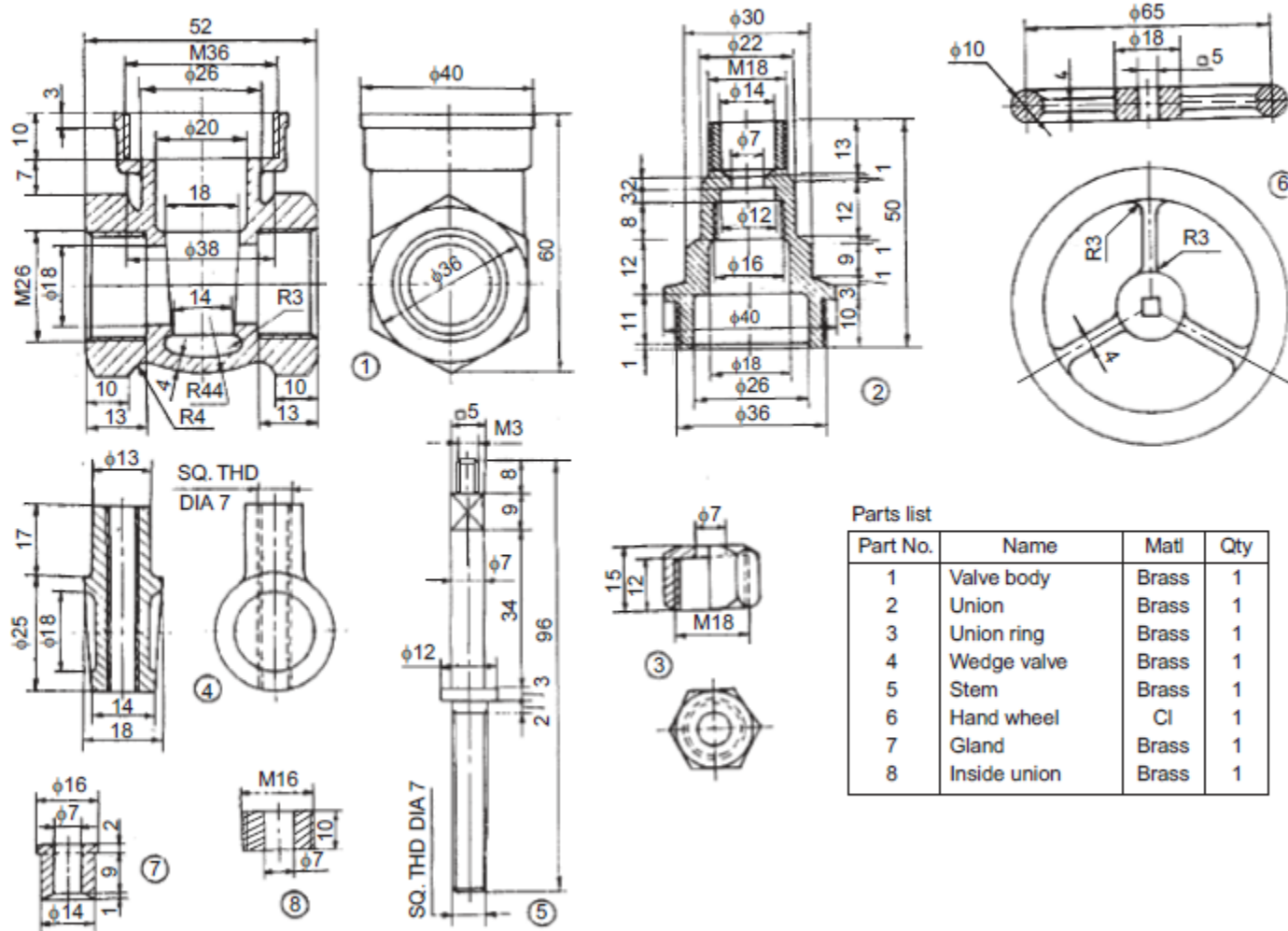
The drawing includes various dimensions and specifications for each part, such as diameters, lengths, and hole sizes.

[illegible]

Machine Vice



Gate Valve



REFERENCE

1. Alf Yarwood, Introduction to AutoCAD 2009, 2008
2. George Omura; Mastering AutoCAD 2009 and AutoCAD LT 2009; 2008
3. K.L. Narayana, P. Kannaiah, K. Venkata Reddy; Machine Drawing, 2006
4. Kristen S. Kurland; AutoCAD 2009 2D Training Manual; 2008
5. Randy H. Shih; Auto CAD 2010 Tutorial; 2009.